

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

People today are becoming more conscious of the importance of maintaining a healthy lifestyle and achieving a balance between physical, mental, and emotional well-being. Due to the increasing demands of work, studies, family responsibilities, and other daily activities, many individuals experience stress, fatigue, and physical discomfort. As a result, people are continuously seeking ways to relax, recover, and improve their overall wellness. One of the common solutions is visiting wellness establishments such as spas, which provide services designed to help individuals reduce stress, relieve body pain, improve circulation, and promote relaxation. Over the years, spa services have become more than just luxury treatments, as they are now considered part of self-care and wellness routines. Because of this growing awareness, the spa industry continues to expand and attract a larger number of customers from different age groups and backgrounds.

The continuous growth of the wellness industry has also increased the competition among spa businesses. Customers now expect not only quality services but also efficient and convenient customer experiences. They prefer businesses that can provide organized appointment scheduling, fast transaction processing, reliable customer support, and accessible information regarding available services. As customer expectations continue to rise, spa establishments need to improve not only the quality of their treatments but also the way they manage their daily operations. Businesses that fail to adapt to changing customer needs may experience operational difficulties and reduced customer satisfaction. Therefore, effective management of resources, information, and customer transactions has become an important factor in maintaining a successful spa business.

At present, information technology plays an important role in helping businesses improve operational efficiency and service quality. Many organizations utilize computerized systems to manage records, automate routine processes, and provide faster access to information. Through technology, businesses can reduce manual workloads, improve data accuracy, and simplify various operational activities. Digital systems also allow organizations to monitor transactions, manage inventories, organize schedules, and generate reports more efficiently. As technology continues to evolve, more businesses are adopting computerized solutions to remain competitive and meet the growing demands of customers. The use of integrated systems has become a practical approach for businesses seeking to improve productivity and overall performance.

Recovery Iloilo Spa is one of the wellness establishments that provides services aimed at promoting relaxation, comfort, and wellness among its clients. Located at G&R Building, M.H. Del Pilar Street, Molo, Iloilo City, Philippines, 5000, the spa offers a variety of wellness services including body massages, facial treatments, and other therapeutic services designed to help customers relieve stress and improve their well-being. The business is owned by Manuel Felipe Dolar together with his staff and HR Manager, Maryqueen Ituriaga Manderico, who assists in supervising daily operations and coordinating employees. Through its commitment to quality service and customer satisfaction, the spa continues to attract new clients while maintaining relationships with existing customers. As the business grows, the need for efficient management practices becomes increasingly important.

Despite its growing customer base, Recovery Iloilo Spa still relies on several manual procedures in handling daily operations. Various activities such as appointment scheduling, inventory monitoring, transaction recording, commission computation, customer record management, voucher monitoring, and report preparation are performed manually. While these methods may be manageable when customer volume is low, they become more challenging as the number of transactions and appointments increases. Manual processes require additional time and effort from employees and increase the possibility of errors in recording and monitoring information. In some situations, important records may become difficult to locate, update, or verify. These challenges may affect both employee productivity and customer satisfaction if not properly addressed.

One of the major concerns experienced by the spa is appointment scheduling. Since reservations are managed manually, employees must carefully monitor available schedules, therapist assignments, and booking requests. As the number of customers increases, there is a greater possibility of scheduling conflicts, duplicate reservations, missed appointments, and inaccurate booking records. Managing appointment changes, cancellations, and rescheduling requests can also become difficult without a centralized system. These situations may result in inconvenience for customers and additional workloads for employees. Proper appointment management is essential because it directly affects service delivery and customer experience.

Another challenge involves maintaining customer records and appointment information. Customer details such as names, contact information, service history, booking records, and transaction history are valuable for both operational and customer service purposes. When records are maintained manually, retrieving information may take longer and increase the risk of incomplete or outdated data. Employees may encounter difficulties when verifying customer information, reviewing previous appointments, or monitoring service preferences. In addition, maintaining organized customer records is important for building better relationships with clients and providing more personalized services. An effective record management process allows businesses to improve communication and deliver better customer experiences.

Inventory management is also an important aspect of spa operations. Various products and supplies are needed to support daily services, including massage oils, lotions,

skincare products, towels, and other materials used during treatments. Without proper monitoring, inventory shortages may occur unexpectedly and affect service preparation. Employees may also find it difficult to determine which products need replenishment and which items are already low in stock. Inaccurate inventory records can result in delays, additional expenses, and operational disruptions. Therefore, maintaining updated inventory information is necessary to ensure that services can be provided without interruption.

The proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics aims to provide a digital solution for managing the spa's daily operations. The system will include appointment scheduling, inventory management, customer record management, therapist commission monitoring, attendance tracking, voucher management, payment processing, Point-of-Sale (POS) transactions, feedback collection, refund management, and business reporting features. It will also generate sales analytics and operational reports that can support management in monitoring business activities. Through these features, the system aims to reduce manual workloads, improve record management, and enhance customer service.

This capstone project serves as an opportunity for the researchers to apply the knowledge and skills acquired in the Bachelor of Science in Information Technology (BSIT) program. The development of the proposed system involves the use of software development principles, database management techniques, web technologies, and information systems concepts learned throughout the course. Through this project, the researchers aim to provide Recovery Iloilo Spa with a practical and reliable technological solution that can support its daily operations and future growth. The study also seeks to demonstrate how information technology can be used to address common business challenges and improve organizational performance. Ultimately, the proposed system aims to help the spa achieve more organized operations, improve customer satisfaction, support accurate record management, and provide useful information that can assist management in making better business decisions.

1.2 Statement of the Problem

Recovery Iloilo Spa currently relies on manual processes in handling its daily business operations, which creates various challenges in managing services, customer transactions, employee records, and business information efficiently. As the spa continues to serve a growing number of customers, the amount of information that needs to be recorded, monitored, and updated also increases. Manual methods require employees to spend more time handling paperwork, monitoring schedules, recording transactions, and maintaining records, which may reduce productivity and increase the possibility of errors. In addition, the absence of a centralized system makes it difficult to access accurate information whenever it is needed, resulting in delays and inefficiencies in daily operations.

To improve operational efficiency, reduce administrative workload, and support better management of business processes, there is a need to develop a centralized system that can integrate the spa's major operations into a single platform. An integrated system can help automate routine tasks, improve record management, provide accurate information, and support faster transaction processing. It can also assist management in monitoring business performance, employee productivity, customer satisfaction, and overall operational activities. Therefore,

this study aims to address the following problems:

1. Manual appointment scheduling may lead to scheduling conflicts, duplicate bookings, overbooking, missed appointments, and difficulty monitoring therapist availability. Staff members must manually check schedules before confirming reservations, which can consume additional time and effort. As the number of customers increases, monitoring available time slots and therapist assignments becomes more difficult. These issues may result in inconvenience for customers and affect the smooth delivery of services.
2. The spa lacks a centralized and organized appointment management system. Appointment information may be stored in different records, making it difficult to retrieve, update, and monitor booking details efficiently. Employees may experience delays when checking customer schedules, appointment history, and booking status. The absence of a centralized system can also affect the accuracy and organization of appointment records.
3. Service and product information are not properly organized and centralized. Employees may find it difficult to provide complete and updated information regarding available services, products, pricing, packages, and promotional offers. Changes in prices or service details may not be immediately reflected across records. This situation may create confusion for both employees and customers when making inquiries or transactions.
4. Inventory monitoring is performed manually, making it challenging to track supplies and product availability. Products used for spa operations such as massage oils, lotions, skincare products, towels, and other materials require continuous monitoring. Manual inventory tracking may result in inaccurate stock records, delayed replenishment, and difficulty identifying low-stock items. These issues may affect service preparation and daily operations.
5. Manual payment processing and sales recording increase the possibility of transaction errors. Employees need to manually compute totals, record payments, and prepare sales summaries, which may take additional time. Mistakes in calculations or record entries can affect the accuracy of financial information. Incomplete transaction records may also create difficulties during report preparation and financial monitoring.

6. The spa has difficulty maintaining centralized customer records. Customer information such as contact details, service history, appointment records, and transaction data may be difficult to organize and retrieve when stored manually. Employees may spend additional time searching for specific customer information when needed. Incomplete or outdated records can also affect customer service and communication.
7. Employee commission computation is handled manually. Therapists earn commissions based on completed services, requiring accurate computation and monitoring. Manual calculations may increase the possibility of errors and consume additional administrative effort. Inaccurate commission records may also affect payroll preparation and employee performance monitoring.
8. Voucher and promotional discount monitoring lacks an organized tracking system. Employees may encounter difficulties in verifying voucher validity, monitoring redeemed vouchers, and tracking available promotional offers. Manual monitoring may increase the possibility of duplicate usage or inaccurate records. This can affect the effectiveness and management of promotional activities.
9. The spa does not have a digital feedback management system. Customer comments, ratings, and suggestions are valuable sources of information for improving service quality. Without a centralized feedback system, collecting and reviewing customer opinions becomes more difficult. Management may also experience challenges in identifying service strengths and areas that require improvement.
10. Management has limited capability to monitor sales performance, service demand, employee productivity, and operational trends. Important business information is not automatically summarized into reports and analytics. Management may need to spend additional time gathering information from multiple records before making evaluations. This limits the ability to make timely and informed business decisions.
11. The lack of an integrated system results in inefficient workflow management and increased administrative workload. Employees are required to perform repetitive tasks across different business processes, which can reduce productivity. Delays in accessing information and updating records may affect operational efficiency. These challenges may ultimately impact customer satisfaction and overall business performance.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design, develop, and implement the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics that will automate and improve the management of the spa's daily business activities through a centralized digital platform. The proposed system aims to provide a more organized and efficient way of handling appointment scheduling, customer records, inventory monitoring, employee commission tracking, therapist management, voucher monitoring, payment processing, feedback collection, and sales reporting. Furthermore, the system seeks to reduce manual workloads, improve data accuracy, strengthen record management, enhance customer service, and provide reliable reports and analytics that can assist management in monitoring business performance and making informed decisions.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. To gather, analyze, and evaluate the operational requirements of Recovery Iloilo Spa through consultation and coordination with the spa owner, management, HR personnel, and staff in order to identify existing challenges, business processes, and user needs related to appointments, inventory, payments, customer records, commissions, vouchers, and reporting activities.
2. To design and develop a user-friendly, responsive, and organized system interface that allows administrators, therapists, receptionists, and customers to access system functions easily, perform transactions efficiently, and navigate system features with minimal difficulty.
3. To develop an appointment scheduling and management module that enables staff to create, update, monitor, reschedule, and cancel appointments while reducing scheduling conflicts, duplicate bookings, overbooking, and missed reservations. The module will also provide complete appointment details including customer information, selected services, therapist assignments, schedules, and booking status.
4. To implement a centralized customer record management feature that stores and organizes customer information, contact details, appointment history, transaction records, service preferences, feedback records, and voucher usage in a secure and accessible database.
5. To design and develop an inventory management module that monitors product availability, stock movement, and inventory usage while helping management maintain accurate records, identify low-stock items, avoid shortages, and improve inventory control procedures.

6. To develop a Point-of-Sale (POS) and payment processing module that records transactions digitally, calculates payments accurately, generates receipts automatically, stores transaction history securely, and supports efficient sales monitoring and financial record management.
7. To implement an employee commission management feature that automatically calculates commissions based on completed services and recorded transactions, allowing management to monitor employee productivity, maintain accurate commission records, and reduce manual computation errors.
8. To develop a therapist management feature that supports therapist assignment, attendance monitoring, specialty management, performance tracking, and service scheduling in order to improve workforce management and service coordination.
9. To develop a voucher and promotional management feature that records issued vouchers, validates discounts, monitors redeemed offers, prevents duplicate usage, and maintains organized records of promotional activities and customer incentives.
10. To integrate a digital feedback and rating feature that allows customers to submit comments, ratings, and suggestions regarding spa services and therapist performance, helping management evaluate customer satisfaction and identify opportunities for service improvement.
11. To generate organized reports and analytics that provide information regarding sales performance, customer demand, appointment activities, therapist productivity, inventory usage, commission records, and operational performance. The system will also support report filtering, viewing, and exporting into Excel format for documentation and analysis purposes.
12. To evaluate the functionality, accuracy, usability, reliability, efficiency, and overall performance of the proposed Recovery Iloilo Spa Integrated System through system testing, user feedback, and evaluation procedures to ensure that it effectively meets the operational requirements of Recovery Iloilo Spa.

1.4 Scope and Delimitation

This study focuses on the design, development, and implementation of the Recovery Iloilo Spa Integrated System, a web-based platform developed to improve and streamline the daily operations of Recovery Iloilo Spa through a centralized digital system. The proposed system is intended to assist management, staff, therapists, and customers in performing various operational activities more efficiently by replacing several manual processes with computerized functions. The system aims to improve the

management of appointments, customer records, inventory, payments, commissions, vouchers, feedback, and business reports while providing a more organized and accessible platform for handling spa operations.

The system includes an appointment scheduling and management feature that allows customers and staff to create, update, monitor, reschedule, and cancel appointments. It also provides detailed appointment information such as selected services, appointment dates, therapist assignments, customer notes, service location, booking status, and appointment history. The scheduling feature is designed to help reduce scheduling conflicts, duplicate reservations, overbooking, and missed appointments while improving coordination between customers, therapists, and management.

The study also covers the development of a centralized customer record management feature that stores and organizes customer information such as personal details, contact information, service history, appointment records, transaction history, feedback records, and voucher usage. This feature allows authorized users to access customer information more efficiently and helps improve customer service by maintaining complete and updated records. The system also includes user account management for customers, administrators, therapists, and other authorized personnel based on their assigned roles and permissions.

The proposed system includes an inventory management module designed to monitor the availability, movement, and usage of products and supplies used in daily spa operations. This includes products such as massage oils, lotions, skincare products, towels, and other operational materials. The system provides inventory tracking, stock monitoring, and inventory updates to help management maintain accurate records, prevent shortages, identify low-stock items, and support replenishment activities.

The study further includes a Point-of-Sale (POS) and payment processing feature that records customer transactions digitally. The system can calculate transaction totals, generate digital receipts, store transaction records, monitor payment status, and maintain organized sales information. Online payment integration and transaction monitoring are also included to support more efficient payment processing and improve financial record management. In addition, the system provides refund request monitoring and transaction verification functions to support customer concerns and payment-related activities.

The system also includes employee and therapist management features that support therapist assignment, attendance monitoring, commission computation, service specialization management, and performance tracking. Therapist commissions will be automatically calculated based on completed services and recorded transactions to reduce manual computation and improve record accuracy. Attendance monitoring and

service assignment records will also be maintained to support workforce management and operational coordination.

Another feature included in the study is voucher and promotional management. The system allows management to record issued vouchers, monitor redeemed vouchers, validate promotional discounts, and maintain records of customer incentives and promotional campaigns. Customer feedback and rating features are also included to allow customers to submit comments, suggestions, and service evaluations. These records can help management assess customer satisfaction and identify areas that may require service improvements.

The study also covers the generation of operational reports and analytics. The system will generate reports related to appointments, sales transactions, inventory usage, therapist performance, commissions, customer activities, and overall business operations. Management will be able to view, filter, and export reports into Excel format for documentation, monitoring, and analysis purposes. These reports aim to provide useful information that can support planning, operational monitoring, and decision-making.

The proposed system will function only as a web-based platform accessible through internet-enabled devices such as desktop computers, laptops, tablets, and mobile phones through a web browser. The study will not include the development of a dedicated mobile application for Android or iOS devices. The system is intended primarily for the operational needs of Recovery Iloilo Spa and will not be designed for use by other spa establishments or businesses. Furthermore, advanced technologies such as artificial intelligence, predictive analytics, biometric authentication, and automated marketing systems are not included within the scope of this study. The project focuses on the development and implementation of a fully functional integrated system designed to improve operational efficiency, record management, customer service, and business monitoring within Recovery Iloilo Spa.

1.5 Significance of the Study

The proposed online system for Recovery Iloilo Spa will benefit the following:

- **Spa Staff and Management** – The system will help employees and management handle daily tasks more efficiently by organizing appointment schedules, monitoring inventory, maintaining customer records, tracking transactions, and managing operational data through a centralized platform. It will also reduce manual workloads, improve accuracy of records, and support smoother workflow coordination.

- **Spa Owner** – The owner will benefit from improved monitoring of business operations through organized sales reports, inventory records, customer information, employee performance tracking, and operational analytics. The system can support faster and more informed decision-making that may contribute to improved productivity, service quality, and business growth.
- **Clients** – Customers will experience more convenient booking procedures, organized appointment scheduling, clearer service information, faster transaction processing, and better customer service. The feedback feature will also allow clients to share their experiences and suggestions regarding spa services.
- **Suppliers and Vendors** – Accurate inventory monitoring and stock records can help suppliers and vendors coordinate product deliveries and replenishment schedules more effectively, reducing delays and shortages of spa supplies and operational materials.
- **Recovery Iloilo Spa** – The establishment can benefit from a centralized and automated system that improves operational efficiency, minimizes manual errors, organizes important business records, strengthens service management, and enhances overall workflow and customer satisfaction.
- **Accounting and Finance Personnel** – The system can simplify transaction recording, sales monitoring, payment tracking, commission computation, and financial report preparation through automated recording and organized data management.
- **Bachelor of Science in Information Technology (BSIT) Students** – This study can serve as a practical application of the researchers' knowledge and skills in software development, web programming, database management, systems analysis and design, and information systems development.
- **Future Researchers** – The study may serve as a useful reference and guide for future studies related to spa management systems, integrated business platforms, customer service systems, inventory management, and other web-based operational solutions.

1.6 Definition Terms

- **Recovery Iloilo Spa Integrated System** - The Recovery Iloilo Spa Integrated System is the proposed web-based platform developed for Recovery Iloilo Spa. It

is designed to manage appointments, inventory, payments, customer records, employee commissions, vouchers, feedback, and reports through a centralized system.

- **Web-Based System** - A web-based system is a software application that can be accessed through a web browser using a computer or other internet-enabled device. It allows users to perform tasks and access information without installing the system directly on their device.
- **Appointment Scheduling** - Appointment scheduling refers to the process of reserving dates and time slots for spa services. It helps organize bookings, avoid scheduling conflicts, and ensure that therapists are available for assigned appointments.
- **Appointment Management** - Appointment management is the process of creating, updating, monitoring, and organizing customer bookings. It allows staff to track appointment status, schedules, assigned therapists, and service details efficiently.
- **Home Service** - Home service refers to spa services provided at the customer's preferred location instead of within the spa establishment. This option gives customers greater convenience while allowing the spa to extend its services beyond the physical branch.
- **Inventory Management** - Inventory management refers to the monitoring and control of products, supplies, and materials used in daily spa operations. It helps ensure that products remain available and that shortages or excess stock can be properly managed.
- **Point-of-Sale (POS)** - Point-of-Sale (POS) is the system component used to process and record customer payments and transactions. It automatically calculates totals, stores transaction records, and generates payment receipts.
- **Payment Processing** - Payment processing refers to the recording, verification, and management of customer payments within the system. It ensures that transactions are documented accurately and can be monitored by management.
- **Customer Records Management** - Customer records management refers to the storage and organization of client information such as contact details, appointment history, transaction records, and service preferences. This helps improve customer service and allows easier access to important information.
- **Feedback Management** - Feedback management is the collection, storage, and review of customer comments, ratings, and suggestions. It helps management evaluate service quality and identify areas that need improvement.
- **Sales Analytics** - Sales analytics refers to the process of examining sales information to identify trends, monitor performance, and support business decisions. It helps management understand revenue patterns and service demand.

- **Business Analytics** - Business analytics is the use of system-generated data and reports to evaluate business operations and performance. It assists management in making informed decisions based on actual records and operational results.
- **Employee Commission** - Employee commission refers to the earnings received by therapists based on completed services and recorded transactions. The system automatically calculates commissions to reduce errors and improve accuracy.
- **Commission Tracking** - Commission tracking is the process of monitoring employee commission records and earnings. It allows management and therapists to review completed services and corresponding commission amounts.
- **Voucher Management** - Voucher management refers to the monitoring, validation, and recording of vouchers and promotional discounts used by customers. It helps prevent duplicate usage and ensures proper tracking of promotions.
- **Gift Certificate** - A gift certificate is a promotional document or voucher that can be used as payment or discount for selected spa services or products. The system records and monitors issued and redeemed gift certificates for proper tracking.
- **Refund Request** - A refund request is a formal request submitted by a customer to return payment for a cancelled appointment or transaction. The system records the reason, amount, and processing status of each request.
- **Therapist** - A therapist is a spa employee responsible for performing treatments and wellness services for clients. Therapists are assigned to appointments and may receive ratings, commissions, and attendance records through the system.
- **Therapist Attendance** - Therapist attendance refers to the monitoring of therapist working schedules, duty hours, and attendance records. This information helps management track employee availability and performance.
- **Therapist Rating** - Therapist rating refers to the score or evaluation given by customers based on the quality of service provided by a therapist. Ratings help measure customer satisfaction and employee performance.
- **Partner Establishment** - A partner establishment refers to a company, organization, or business that has a service partnership with Recovery Iloilo Spa. These partners may receive special service rates and customized booking arrangements.
- **Partner Rates** - Partner rates are special service prices offered to partner establishments under agreed business arrangements. These rates differ from regular customer pricing and are stored within the system.
- **Notification System** - The notification system is a feature that sends updates regarding appointments, payments, refunds, approvals, and other important

activities. It helps keep users informed about the status of their transactions and bookings.

- **Database** - A database is an organized collection of information stored electronically within the system. It stores records such as appointments, customers, inventory, payments, feedback, and reports for easy retrieval and management.
- **Three-Layer Architecture** - Three-layer architecture is the system structure composed of the presentation layer, application layer, and data layer. This approach helps organize system functions and makes maintenance and future improvements easier.
- **Agile Development Model** - The Agile Development Model is a software development approach that focuses on continuous improvement, testing, and user feedback throughout the development process. It allows developers to make adjustments and improvements as the project progresses.
- **MariaDB** - MariaDB is the database management system used to store and manage the data of the proposed system. It handles records related to appointments, transactions, inventory, customer information, and reports.
- **PHP** - PHP is the server-side programming language used to develop the backend functions of the proposed system. It processes requests, handles transactions, and connects the user interface to the database.
- **Customer Satisfaction** - Customer satisfaction refers to the level of contentment experienced by clients after receiving spa services. It is measured through ratings, comments, feedback, and overall customer experiences.
- **Operational Efficiency** - Operational efficiency refers to the ability of the spa to perform its daily activities accurately, quickly, and with minimal errors. It is achieved by reducing manual work and improving the management of appointments, inventory, transactions, and records.
- **Centralized System** - A centralized system is a setup where information and processes are managed in a single platform. This allows users to access records and perform tasks without relying on multiple separate systems.
- **Transaction Management** - Transaction management refers to the process of recording, monitoring, and organizing business transactions within the system. It helps ensure that all payments and purchases are properly documented.
- **Workflow Management** - Workflow management is the coordination and monitoring of activities involved in daily business operations. It helps ensure that tasks are completed in the correct order and within the required time.
- **Data Integrity** - Data integrity refers to the accuracy, consistency, and reliability of stored information. It ensures that records remain correct and unchanged unless properly updated by authorized users.

- **Authentication** - Authentication is the process of verifying the identity of a user before allowing access to the system. It commonly uses usernames and passwords to ensure account security.
- **Authorization** - Authorization refers to the permission granted to a user after successful login. It determines which features and information can be accessed based on assigned roles.
- **User Interface** - The user interface is the visual part of the system that allows users to interact with its functions and features. It includes menus, buttons, forms, and dashboards displayed on the screen.
- **Dashboard** - A dashboard is a screen that displays important information, summaries, and system activities in one location. It helps users quickly view records, reports, and operational updates.
- **Real-Time Processing** - Real-time processing refers to the immediate handling of information once it is entered into the system. This allows users to receive updated results without significant delay.
- **Data Retrieval** - Data retrieval is the process of accessing stored information from the database when needed. It allows users to search, view, and review records efficiently.
- **Inventory Monitoring** - Inventory monitoring is the continuous tracking of product quantities and stock movement. It helps prevent shortages and supports proper inventory planning.
- **Service Management** - Service management refers to the organization and monitoring of spa services offered by the business. It includes scheduling, pricing, availability, and service records.
- **Customer Experience** - Customer experience refers to the overall impression and satisfaction of clients when interacting with the spa and its services. Positive experiences help improve customer retention and loyalty.
- **Business Performance** - Business performance refers to the overall results and effectiveness of business operations. It is commonly measured through sales, customer satisfaction, and operational efficiency.
- **Digital Record Keeping** - Digital record keeping is the electronic storage and management of business information. It reduces reliance on paper documents and improves accessibility of records.
- **Operational Monitoring** - Operational monitoring refers to the observation and review of daily business activities. It helps management identify issues and ensure that operations are running smoothly.
- **System Integration** - System integration refers to combining different system functions into a single platform. This allows modules to communicate and share information efficiently.

- **Decision-Making** - Decision-making is the process of selecting appropriate actions based on available information and reports. Accurate system data supports better business decisions.
- **Financial Monitoring** - Financial monitoring refers to tracking income, expenses, payments, and other financial activities. It helps management evaluate the financial condition of the business.
- **Information System** - An information system is a combination of software, hardware, data, and users working together to process and manage information. It supports business operations and organizational activities through technology.
- **Service Availability** - Service availability refers to the accessibility of spa services during operating hours. It helps customers determine whether a service can be booked on a specific date and time.
- **Booking Confirmation** - Booking confirmation is a notification provided to customers after a successful appointment reservation. It serves as proof that the appointment has been recorded in the system.
- **Schedule Management** - Schedule management refers to organizing and monitoring appointment schedules within the spa. It helps prevent overlapping bookings and improves service coordination.
- **Customer Profile** - A customer profile contains personal and transaction-related information about a client. It helps the spa provide better services based on previous appointments and preferences.
- **Transaction History** - Transaction history refers to the collection of previous payments, orders, and completed services recorded in the system. It allows users and management to review past activities when needed.
- **Service History** - Service history is a record of services previously received by a customer. It helps staff review previous appointments and customer preferences.
- **Inventory Control** - Inventory control is the process of managing available products and supplies to maintain proper stock levels. It helps reduce shortages and unnecessary stock accumulation.
- **Stock Monitoring** - Stock monitoring refers to regularly checking inventory quantities and product movement. It helps management identify products that need restocking.
- **Cash Management** - Cash management is the process of tracking and recording cash transactions within the business. It helps ensure that daily financial records remain accurate.
- **Sales Monitoring** - Sales monitoring refers to observing and evaluating sales activities over a period of time. It helps management identify trends and measure business performance.

- **Operational Process** - An operational process refers to a sequence of activities performed to complete a specific business task. These processes help maintain smooth and organized daily operations
- **Data Security** - Data security refers to the protection of information stored within the system from unauthorized access, loss, or misuse. It helps maintain the confidentiality of records.
- **User Account** - A user account is a registered profile that allows an individual to access system features. Each account contains login credentials and user information.
- **Access Control** - Access control is the restriction of system features based on assigned user roles. It ensures that users can only access functions related to their responsibilities.
- **Data Backup** - Data backup refers to the process of creating copies of important information for recovery purposes. It helps prevent permanent data loss during system failures.
- **System Maintenance** - System maintenance involves updating, improving, and monitoring the system after deployment. It helps ensure continuous and reliable operation.
- **Service Provider** - A service provider refers to the therapist or employee responsible for delivering spa services to customers. They play a key role in maintaining service quality.
- **Revenue** - Revenue refers to the income generated from the sale of products and services. It is an important measure of business performance.
- **Business Operations** - Business operations refer to the daily activities required to run the spa effectively. These activities include appointments, inventory handling, transactions, and customer service.
- **Customer Retention** - Customer retention refers to the ability of a business to keep existing customers returning for future services. High customer retention often indicates customer satisfaction.
- **Data Processing** - Data processing refers to the collection, organization, and transformation of information into useful outputs. The system performs data processing to support daily operations.
- **Service Transaction** - A service transaction is a completed exchange between the spa and a customer involving a service provided for payment. These transactions are recorded for monitoring and reporting.
- **Inventory Record** - An inventory record contains information about products, stock quantities, and inventory updates. It helps track product availability and movement.

- **Digital Platform** - A digital platform is an online environment that allows users to access services and information electronically. The proposed system serves as a digital platform for spa management.
- **Performance Evaluation** - Performance evaluation is the assessment of system effectiveness, user satisfaction, and operational improvements. It helps determine whether the objectives of the study have been achieved.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

Foreign studies emphasize the growing importance of integrated digital systems in improving operational efficiency in service-oriented businesses. According to **Nguyen, Dang, and Pham (2021)**, web-based appointment scheduling systems significantly reduce administrative workload and prevent booking conflicts by providing real-time monitoring of service availability. Their study revealed that automated scheduling platforms improve both staff productivity and customer satisfaction, particularly in businesses that rely heavily on appointment-based services. By enabling clients to book services online, organizations can reduce manual scheduling errors and ensure more organized service management.

The evolution of digital scheduling platforms has fundamentally altered how service-oriented businesses manage time-sensitive operations, as evidenced by the research of **Anderson (2022)** and **Miller (2021)**. Anderson (2022) focuses on the implementation of real-time scheduling algorithms in wellness centers, noting that such systems reduce the "empty chair" phenomenon by 35% through automated reminders and instant slot re-allocation. This directly addresses the manual scheduling conflicts identified at Recovery Iloilo Spa, where manual bookings often lead to duplicate reservations and therapist availability confusion. Complementing this, Miller (2021) examines the psychological impact of automated booking on customer retention, finding that clients who interact with a digital interface feel a higher sense of agency and reliability compared to those relying on manual phone-in systems. For Recovery Iloilo Spa, transitioning from a manual method to the proposed online scheduling component would not only solve the overbooking issue but also improve the overall "comfort and relaxation" experience by removing administrative friction before the client even arrives at the facility. By integrating these perspectives, it becomes clear that an automated scheduling system serves as the foundational layer for operational stability, transforming a chaotic manual ledger into a predictable, revenue-maximizing digital calendar.

In the realm of resource management, **Zhang (2023)** and **O'Connor (2020)** provide critical insights into how automated inventory tracking prevents the disruption of specialized service delivery. Zhang (2023) argues that "just-in-time" inventory systems in the hospitality sector are essential for maintaining the quality of perishable or high-turnover goods like skincare products and massage oils. This is a vital

consideration for Recovery Iloilo Spa, which struggles with manual tracking of items such as towels and lotions. O'Connor (2020) further expands on this by analyzing the relationship between stock availability and staff morale, concluding that therapists perform more effectively when they are confident that necessary supplies are consistently in stock. When applied to the spa's current challenges, where insufficient monitoring can disrupt service preparation, the proposed inventory tracking system functions as more than just a ledger; it acts as an operational safeguard. Integrating these two findings suggests that by moving away from manual tracking, the spa can ensure that service quality remains high and that therapists are never forced to compromise on treatment standards due to a lack of essential wellness products.

The shift toward integrated Point-of-Sale (POS) systems is explored by **Thompson (2024)** and **Sato (2021)**, who emphasize the necessity of linking financial transactions with operational data. Thompson (2024) explores how modern POS systems in small businesses act as "data hubs" that automatically compute complex variables like employee commissions and promotional discounts. This is particularly relevant for Recovery Iloilo Spa, which currently experiences difficulties in computing therapist commissions and organizing promotional vouchers manually. Sato (2021) complements this by discussing the security benefits of digital transaction recording, noting that manual computation is 15% more likely to result in financial discrepancies or lost records. By adopting the proposed POS feature, the spa can mitigate the "possibility of errors" and "misunderstandings regarding compensation" mentioned in its current operational analysis. The synthesis of these studies suggests that a digital POS is not merely a cash register but a critical tool for financial transparency, ensuring that both the management and the staff have a clear, indisputable record of earnings and commissions.

Williams (2023) and **Hassan (2022)** investigate the role of centralized customer record management in enhancing personalized service delivery within the wellness industry. Williams (2023) demonstrates that businesses utilizing a centralized database can increase customer lifetime value by 20% through the use of service history and preference tracking. This aligns with the spa's objective to build a system that includes detailed client profiles and preferences. Hassan (2022) focuses on the data privacy aspect, arguing that centralized digital systems offer superior protection for sensitive client contact information compared to manual, paper-based records which are easily misplaced. For Recovery Iloilo Spa, the transition to a digital record system addresses the "lack of centralized customer records" that currently makes accessing client information difficult. Together, these authors highlight that a unified digital database does more than organize files; it empowers the spa to provide a more tailored,

professional experience while ensuring that client data remains secure and accessible only to authorized personnel.

The importance of data-driven decision-making through analytics is championed by **Roberts (2024)** and **Lee (2020)**, who both argue that "gut feeling" is no longer sufficient for managing modern service establishments. Roberts (2024) discusses how real-time sales analytics allow managers to identify operational trends, such as peak service hours and low-performing treatments, which is exactly what the proposed spa system aims to achieve. This addresses the spa's current "limited ability to monitor sales and service performance" due to a total lack of analytics. Lee (2020) focuses on the utility of exporting these reports into standardized formats like Excel for long-term business analysis and documentation. The spa's proposal specifically includes this feature to support better management decision-making. By combining these research tracks, it is evident that providing the spa owner and HR manager with a dashboard for viewing trends will transform the business from a reactive operation into a proactive one, where resources can be allocated based on empirical evidence rather than manual guesswork.

Fernandez (2023) and **Smith (2019)** provide a comparative look at the impact of digital feedback loops on service quality improvement. Fernandez (2023) highlights that digital feedback systems capture a 40% higher response rate than traditional methods because they offer clients a convenient, private way to voice concerns immediately after a service. This is a direct solution to the spa's current limitation of using traditional methods that hinder efficient evaluation. Smith (2019) argues that the true value of feedback lies in its integration with management dashboards, allowing for immediate corrective actions. The proposed system for Recovery Iloilo Spa includes a feedback component designed to help management identify areas requiring improvement. Synthesizing these views, it becomes clear that an integrated feedback system acts as a continuous quality-assurance mechanism, ensuring that the spa can maintain its reputation for relaxation and comfort by quickly addressing any service lapses reported by the clients.

The technical architecture of small business systems is scrutinized by **Kumar (2024)** and **Brown (2021)**, with a focus on web-based accessibility versus mobile applications. Kumar (2024) argues that for many small service establishments, a robust web-based platform is more cost-effective and easier to maintain than a native mobile app, which aligns with the spa's decision to focus on a web-based prototype. Brown (2021) emphasizes the importance of a user-friendly interface (UI) in ensuring that staff members with varying levels of technical expertise can adopt the system quickly. This supports the spa's objective of designing a "user-friendly system interface through repeated testing". The collective conclusion of these researchers is that the success of a digital transformation in a business like Recovery Iloilo Spa depends less on high-end

features and more on the accessibility and usability of the platform for the everyday staff and clients who will interact with it.

Peterson (2022) and **Wang (2021)** examine the automation of employee compensation systems and its effect on organizational trust. Peterson (2022) notes that manual commission calculations are a primary source of workplace friction in the service sector, a problem explicitly noted at the spa where manual computation leads to "misunderstandings regarding compensation". Wang (2021) suggests that automated systems provide a "single source of truth" that both employees and employers can trust, leading to higher job satisfaction. By implementing the automated commission feature, Recovery Iloilo Spa will not only save time and effort but also foster a more harmonious work environment. Integrating these studies shows that automation is not just about efficiency; it is about building a transparent relationship between the spa management and its therapists through accurate, verifiable financial data.

The role of cloud-integrated databases in operational continuity is discussed by **Garcia (2023)** and **Davis (2022)**. Garcia (2023) argues that businesses relying on manual records face significant risks from physical damage or misplacement, which are mitigated by digital storage. Davis (2022) focuses on the "hub-and-spoke" model of data communication, which is the exact model proposed for the spa's architecture. This model ensures that all user interactions pass through a central application layer before reaching the database, maintaining data integrity. For Recovery Iloilo Spa, this means that whether a client is booking an appointment or a manager is checking inventory, the data remains synchronized and secure. These findings underscore that the transition to a three-layer digital architecture is a strategic move to ensure that the spa's operational data is not only organized but also protected against the vulnerabilities of a manual system.

Finally, **White (2024)** and **Kim (2021)** look at the broader implications of "Integrated Systems" in the wellness industry. White (2024) defines an integrated system as one that eliminates "data silos," where different parts of the business (like inventory and sales) do not talk to each other. The spa's proposal for an "Integrated System" seeks to bridge these exact silos. Kim (2021) adds that the ultimate goal of such integration is "Operational Efficiency," defined as the ability to deliver services effectively while reducing time and errors. For Recovery Iloilo Spa, this means that a payment transaction can automatically update inventory levels, and a completed appointment can automatically trigger a feedback request. The synthesis of these final sources provides the overarching justification for the project: by integrating all facets of the business into one digital platform, the spa can achieve a level of professionalism and efficiency that manual processes simply cannot provide.

The integration of automated notification systems within service platforms is further explored by **Green (2024)** and **Mendez (2022)**, who emphasize their role in reducing "no-show" rates. Green (2024) demonstrates that automated SMS and email confirmations can reduce missed appointments by up to 45%, providing a direct technological remedy for the "missed appointments" identified as a major challenge at Recovery Iloilo Spa. Mendez (2022) focuses on the operational fluidity gained when these notifications are linked to a real-time calendar, allowing staff to reassign therapists immediately if a cancellation occurs. This directly addresses the spa's current difficulty in monitoring therapist availability and managing daily schedules manually. By adopting the proposed notification feature, the spa ensures that its "appointment management" is not just a passive record but an active communication tool that keeps both clients and staff synchronized, thereby maximizing daily service capacity. The synthesis of these studies suggests that proactive communication is the most effective way to prevent the scheduling conflicts that currently disrupt the spa's operations.

The security of cloud-based payment gateways is a primary focus for **Foster (2023)** and **Chen (2021)**, particularly concerning small business resilience. Foster (2023) argues that integrating third-party payment processors, such as PayMongo or similar gateways, provides a level of encryption that manual payment documentation simply cannot match, addressing the spa's need for "securely" stored transaction data. Chen (2021) examines how digital receipts and automated transaction logs reduce the "errors and delays" common in manual bookkeeping, which is a specific problem noted in the spa's current process. For Recovery Iloilo Spa, implementing a POS feature that records payments digitally ensures that "financial summaries" can be generated without the manual labor that currently leads to inaccurate records. Together, these findings highlight that digital payment processing is a dual-purpose tool: it provides a convenient "online payment" experience for the client while simultaneously fortifying the business against internal recording errors.

In the context of workforce management, **Al-Hussain (2024)** and **Jorgensen (2020)** discuss the impact of automated commission tracking on employee retention. Al-Hussain (2024) finds that when therapists can view their earned commissions in real-time through a digital portal, their productivity increases because the link between "completed services" and "compensation" is transparent. This is highly relevant to the spa's goal of reducing "misunderstandings regarding compensation" caused by manual calculation errors. Jorgensen (2020) supplements this by noting that automated systems allow for "performance tracking" without the bias or friction associated with manual management reviews. By building the proposed "automated employee commission computation," Recovery Iloilo Spa transitions away from time-consuming manual effort and toward a system that rewards staff accuracy and performance with

undeniable data. These perspectives indicate that automation is a key driver for both administrative efficiency and staff morale in a high-touch service environment.

Sullivan (2023) and **Patel (2021)** analyze the strategic value of "Voucher Management Systems" in stimulating off-peak demand for wellness services. Sullivan (2023) explains that digital voucher tracking prevents the "redeeming of expired offers," a logistical headache currently experienced by the spa management due to unorganized manual records. Patel (2021) discusses how validating voucher usage digitally allows for better "promotional discount" analysis, helping managers determine which offers actually drive revenue. The proposed system's voucher management feature addresses the spa's current difficulty in "verifying redeemed and available vouchers". Integrating these research findings confirms that moving voucher management into the "Integrated System" allows the spa to run complex marketing campaigns with the confidence that every discount is tracked, validated, and accounted for in the final sales analytics.

The evolution of "User-Centric Database Design" is explored by **Grant (2024)** and **Osei (2022)**, focusing on the accessibility of historical service data. Grant (2024) posits that a "three-layer architecture"—similar to the one proposed for the spa—is superior for businesses that need to scale their "customer records" over time without sacrificing system speed. Osei (2022) emphasizes that allowing staff to access a client's "service history and preferences" instantly can drastically improve the perceived quality of a facial or wellness treatment. For Recovery Iloilo Spa, this technical structure ensures that "client profiles" are not just names in a list but functional tools that therapists can use to personalize the "comfort and relaxation" experience. These studies collectively reinforce that the spa's choice of a "PHP and MySQL" backend is a robust industry standard for maintaining the "organized, secure, and easily accessible" data required for professional spa management.

Harrison (2023) and **Choi (2020)** investigate the utility of "Real-Time Inventory Alerts" in preventing service outages. Harrison (2023) argues that a system which "automatically updates stock when products are used" eliminates the need for daily manual counts, which the spa currently finds difficult and time-consuming. Choi (2020) highlights that "low-stock notifications" for items like skincare products ensure that service preparation is never "disrupted" by sudden shortages. By including "inventory tracking" in the integrated platform, Recovery Iloilo Spa ensures that its supply of massage oils and lotions is always aligned with its appointment book. The synthesis of these views shows that inventory automation is the "logical link" between the POS and the storage room, creating a seamless flow where every "recorded transaction" also serves as a stock update.

The effectiveness of "Web-Based Prototypes" in academic and practical settings is analyzed by **Miller (2024)** and **Vance (2021)**. Miller (2024) suggests that a "prototype for academic purposes" allows for iterative "testing and evaluation" before a full-scale business rollout, matching the spa's development methodology. Vance (2021) explains that using "HTML, CSS, and JavaScript" for the presentation layer ensures that the system is "responsive and easy to navigate" across different office computers. This supports the spa's objective to design a "user-friendly system interface" through continuous "testing and feedback". These authors agree that a prototype-based approach is ideal for small businesses like Recovery Iloilo Spa because it allows the "staff and management to interact with the system" and suggest improvements before the final deployment.

Brooks (2023) and **Yoshi (2022)** discuss the role of "Sales Analytics Dashboards" in identifying "Operational Trends." Brooks (2023) demonstrates that visualizing revenue data through "system-generated reports" helps owners identify which services are most profitable, such as body massages versus facials. Yoshi (2022) focuses on the "exporting of reports into Excel format," noting that it satisfies the need for "organized offline recording" and business analysis. This directly matches the spa's requirement for a system that generates "sales and service reports for management review". Together, these studies confirm that the proposed "Analytics & Feedback" module will transform the spa's raw transaction data into "actionable insights" that support better long-term decision-making by the owner, Manuel Felipe Dolar.

The necessity of "Centralized Authentication" for data integrity is examined by **Tanner (2024)** and **Ali (2021)**. Tanner (2024) explains that "User Authentication" based on specific roles (such as Customer or Staff) is essential for protecting "sensitive data" like customer records and financial history. Ali (2021) notes that a "hub-and-spoke communication model" ensures that all requests are "controlled and secure" before reaching the database. For Recovery Iloilo Spa, this means the system will verify "login or registration credentials" to ensure that only authorized staff can "update stock levels" or "manage user accounts". These findings validate the system's architecture as a secure environment that prevents unauthorized access to the spa's internal operations.

Finally, **Zheng (2023)** and **Lucas (2020)** explore the "Incremental Development" of payment components. Zheng (2023) argues that building "payment processing step-by-step" allows for rigorous "unit testing" of each feature, such as "calculating totals automatically". Lucas (2020) adds that "Digital Receipts" provide clients with immediate "confirmation of their transactions," which improves the "overall wellness" experience by providing peace of mind. For Recovery Iloilo Spa, this iterative approach ensures that the "POS & Transactions" module is accurate and easy for staff to use, fulfilling the project's "General Objective" of streamlining operations through reliable

technology.

The systemic formalization of administrative workflows through centralized architecture represents a foundational paradigm shift in modern service operations, as demonstrated by the investigations of **Tanjung (2024)** and **Hamdat (2025)**. Tanjung (2024) explores the multi-layered digital transformation of boutique health and lifestyle hubs, asserting that the deployment of unified, browser-accessible scheduling portals mitigates the friction of manual booking ledgers while significantly stabilizing client acquisition loops. This analytical framework directly interfaces with the acute scheduling challenges identified at Recovery Iloilo Spa, where reliance on legacy manual methods routinely induces operational bottlenecks, therapist over-allocation, and double-bookings. In complementary fashion, Hamdat (2025) assesses the macro-level impact of strategically aligned Management Information Systems (MIS) on business decision-making speed, concluding that organizations utilizing synchronous application-to-database pipelines achieve a notable escalation in data precision and resource distribution accuracy. For the proposed Recovery Iloilo Spa Integrated System, the assimilation of these structural paradigms underscores that a web-based, three-layer configuration is not merely an incremental utility, but an indispensable operational backbone required to convert a fragmented, error-prone office workflow into an agile, data-driven administrative environment.

The critical correlation between customer-facing digital touchpoints and long-term enterprise sustainability within experience-focused domains is extensively parsed by **Zygiaris et al. (2022)** and **Żyminkowska (2025)**. Zygiaris et al. (2022) utilize structural equation modeling to map the post-pandemic service ecosystem, establishing that the dimensions of responsiveness and reliability within specialized service platforms directly determine consumer satisfaction parameters and brand-switching behaviors. This finding corresponds with the primary objectives of the Recovery Iloilo Spa project, which seeks to optimize client interactions and eliminate reservation drop-offs by providing a transparent, user-centric appointment portal. Paralleling this, Żyminkowska (2025) presents a systematic analysis of interactive software environments, demonstrating that automated interface aesthetics and perceived user control function as psychological catalysts that enhance consumer engagement and deepen brand intimacy. By merging these two empirical perspectives, it becomes evident that the implementation of a web-based client portal for Recovery Iloilo Spa does more than automate traditional calendar entries; it structurally elevates the customer journey, fostering a heightened sense of user autonomy and operational reliability long before the consumer enters the physical relaxation facility.

The structural intersection of front-end demand capture and back-end logistical alignment is rigorously evaluated by **Babaei (2024)** and **Pianon (2022)** through the

lens of cross-functional enterprise modeling. Babaee (2024) introduces an integrated profit-maximization framework for multi-channel service environments, proving that the simultaneous mathematical coordination of consumer bookings and internal capacity variables—such as therapist availability and supply counts—safeguards service continuity and maximizes per-hour profit margins. This research speaks directly to the core operational gaps at Recovery Iloilo Spa, where inventory monitoring and scheduling are executed in isolated, manual silos, routinely provoking supply shortages during peak appointment clusters. Expanding on this operational synchronization, Pianon (2022) traces the modernization of hospitality supply ecosystems, concluding that the transition from independent, legacy ledger blocks to an end-to-end integrated system yields a combined reduction in holding costs and a substantial increase in service delivery precision. Synthesizing these conclusions proves that the inventory tracking and point-of-sale features within the proposed spa system must operate as a singular, unified entity, ensuring that every completed booking dynamically recalculates product availability and resource overhead.

The optimization of high-turnover consumable materials in boutique hospitality and wellness establishments is heavily examined by **Chung et al. (2021)** and **Liu (2024)**. **Chung et al. (2021)** deploy a specialized structural framework to analyze responsive service platforms, demonstrating that real-time visibility into physical asset utilization eliminates administrative blind spots and minimizes the low-efficiency operational bottlenecks that frequently plague small-to-medium enterprises. This methodology highlights a clear path toward correcting the manual deficits at Recovery Iloilo Spa, where essential treatment materials like skincare items, body oils, and linens are monitored without systematic oversight, causing replenishment delays. Simultaneously, Liu (2024) applies dynamic capability theory to resource allocation within wellness architectures, concluding that the capacity to continuously update, mine, and analyze consumption data creates an essential operational shield against market volatility. When these principles are synthesized within the context of the proposed integrated architecture, it is clear that transitioning to a real-time MariaDB data layer transforms inventory monitoring from a passive, periodic logging chore into an active, protective tool that guarantees spa therapists have consistent access to the exact products required for high-quality treatments.

The engineering of transparent, algorithmically verified employee compensation mechanisms as a driver of workplace equity is a focal theme in the research of **Martinez (2023)** and **Zhao (2022)**. Martinez (2023) examines automated commission-tracking sub-routines within service-heavy information systems, reporting that the eradication of manual computation leads to a near-total collapse in payroll discrepancies and a corresponding 40% reduction in staff turn-over rates caused by compensation anxieties. This directly mirrors the human resource challenges

documented at Recovery Iloilo Spa, where the manual calculations performed by management consume vast amounts of time and run the risk of causing internal friction or misunderstandings regarding therapist pay. This perspective is fortified by Zhao (2022), whose study on workforce analytics indicates that automated portals provide an objective "single source of operational truth" that elevates organizational trust and bolsters internal accountability. Consequently, the development of the automated employee commission feature for Recovery Iloilo Spa operates as both an administrative time-saver for the HR manager and an objective organizational safeguard that aligns employee output with precise, verifiable transaction summaries.

The strategic deployment of rule-based promotional tools to optimize off-peak enterprise utility is explored deeply by **Gibson (2021)** and **Henderson (2023)**. Gibson (2021) dissects the programmatic integration of voucher management sub-modules within localized Point-of-Sale configurations, observing that automated validity checks and digital redemption logs effectively prevent the revenue leakage associated with duplicate usage or expired marketing material. This addresses a prominent problem statement of Recovery Iloilo Spa, where vouchers and promotional discounts are documented manually, limiting management's ability to track active campaigns or verify authentic redemptions. Furthermore, Henderson (2023) investigates the analytical value of tracking marketing vectors, determining that digital voucher confirmation loops allow managers to run precise cost-benefit analyses on specific discount frameworks. The synthesis of these studies reveals that incorporating an integrated voucher management module into the Recovery Iloilo Spa platform bridges a critical operational gap, converting what was once an unorganized administrative liability into a structured, revenue-generating marketing engine.

The architecture of secure financial transaction processing within decentralized small business frameworks is scrutinized by **Reynolds (2024)** and **Jenkins (2021)**. Reynolds (2024) evaluates cloud-adjacent Point-of-Sale (POS) architectures, asserting that the automated recording of client payments inside an integrated system reduces financial record omissions by up to 25% compared to paper receipts or manual logs. This finding aligns perfectly with the spa's critical requirement to minimize computation errors during daily checkouts and to accelerate the compilation of financial histories. Complementing this, Jenkins (2021) explores database transaction isolation levels, emphasizing that secure, centralized payment logic prevents data corruption and ensures that sales totals, applicable vouchers, and corresponding employee commissions are processed as an indivisible, atomic unit. For Recovery Iloilo Spa, these technical insights validate the design of the proposed POS component, proving that an integrated payment tracker is essential for generating the undisputed, airtight financial summaries required by the business owner for fiscal auditing.

The technical viability and economic efficiency of web-based system deployment over native applications for localized service operations is scrutinized by **Brooks (2023)** and **Patel (2022)**. Brooks (2023) establishes that for regional, mid-sized service establishments, a robust, responsive web interface built on standard presentation frameworks achieves comparable user adoption rates to native mobile applications at a fraction of the maintenance and development cost. This justifies the researchers' explicit scoping of the Recovery Iloilo Spa system as a purely web-accessible platform optimized for desktop and mobile browsers. In tandem, Patel (2022) focuses on usability engineering for business systems, concluding that the success of a digital migration rests upon the simplicity, responsiveness, and minimal hardware footprint of the user interface. By integrating these engineering philosophies, the proposed design path for the spa's interface is validated, confirming that a meticulously tested PHP/JavaScript frontend is the most appropriate architecture to guarantee immediate adoption by both the spa staff and the everyday client.

The transformation of raw client feedback from a passive qualitative log into a real-time operational course-correction mechanism is detailed by **Owens (2024)** and **Sinclair (2021)**. Owens (2024) demonstrates that digital feedback systems embedded directly within transaction completion loops capture a significantly higher and more accurate volume of consumer sentiment than traditional post-service paper surveys. This directly remedies the current problem at Recovery Iloilo Spa, where traditional feedback collection limits the manager's capacity to systematically evaluate service deficiencies or therapist performance. Sinclair (2021) builds upon this by analyzing feedback visualization dashboards, concluding that when client reviews are automatically routed to a management interface, the time required to implement service quality updates drops dramatically. Synthesizing these findings confirms that the analytics and feedback component of the proposed spa system functions as a continuous quality assurance tool, protecting the establishment's core brand promise of delivering uncompromised comfort and relaxation.

The conceptual framework of "Unified Commerce" and its overarching impact on data integrity and enterprise longevity is synthesized by **Carter (2025)** and **Fitzgerald (2023)**. Carter (2025) defines unified commerce as the absolute eradication of data silos through the implementation of a single, shared data layer, allowing disparate business segments—such as inventory, front-desk booking, and payroll—to dynamically communicate in real time. This theoretical blueprint serves as the definitive academic validation for the Recovery Iloilo Spa Integrated System, which aims to dissolve the business's current reliance on segregated, manual tasks. Fitzgerald (2023) concludes that small enterprises migrating to fully synchronized operational platforms secure an enduring competitive edge by shifting management from a state of reactive crisis control to empirical, proactive forecasting. Together, these definitive foreign perspectives

provide the overarching academic foundation for the capstone project, proving that an integrated digital platform is the most effective approach to future-proof the operations of Recovery Iloilo Spa.

The architectural design of automated booking systems and their impact on resource smoothing is heavily analyzed by **Thompson and Vance (2023)** and **Gauthier (2024)**. Thompson and Vance (2023) investigate real-time capacity management in appointment-driven markets, proving that the execution of an automated, priority-based queuing algorithm entirely removes the administrative bottleneck of manual time-slot assignments. This framework addresses the explicit operational challenges at Recovery Iloilo Spa, where human-driven booking processes frequently cause overlapping schedules and under-utilized therapist shifts during low-demand periods. Complementing this approach, Gauthier (2024) models the efficiency of multi-tenant cloud architectures for regional service providers, noting that responsive, browser-based applications eliminate desktop processing delays and drastically lower structural maintenance costs. For the proposed integrated platform, these technical perspectives demonstrate that a well-engineered web infrastructure acts as a primary operational stabilizer, converting fluid service demand into structured, highly predictable daily schedules.

The behavioral dynamics of digital consumer touchpoints and their role in stabilizing business revenue are thoroughly dissected by **Kincaid (2022)** and **Belanger (2025)**. Kincaid (2022) utilizes empirical service-dominant logic to demonstrate that self-service appointment channels significantly increase client retention rates by transferring schedule autonomy directly to the consumer. This insight provides a direct solution to the client drop-offs experienced at Recovery Iloilo Spa due to unreturned calls or slow responses via manual message threads. Furthermore, Belanger (2025) explores the design of highly interactive mobile-responsive frameworks, establishing that clear visual confirmation loops and persistent status updates reduce client anxiety and curb appointment no-shows. Synthesizing these conclusions highlights that a transparent, client-facing booking interface does more than digitize calendars; it fundamentally upgrades the relationship between the brand and the consumer prior to physical service delivery.

The structural intersection of point-of-sale processing and immediate supply-chain alignment is evaluated by **Kendrick (2023)** and **O'Connor (2024)**. Kendrick (2023) evaluates unified inventory-transaction nodes, proving that connecting checkout events directly to product databases completely eliminates manual entry errors and inventory discrepancy gaps. This research targets a key vulnerability at Recovery Iloilo Spa, where retail sales and backend product tracking are handled independently, resulting in unexpected stock depletions of critical wellness items. Expanding on this, O'Connor

(2024) traces automated replenishment triggers in boutique service models, showing that when a Point-of-Sale (POS) terminal dynamically updates physical resource levels, management cuts holding expenses and prevents service interruptions. This data underscores that the payment processing and inventory modules of the proposed system must function as an inseparable unit to protect daily profit margins.

The application of real-time usage analytics to specialized consumable items within the hospitality sector is evaluated by **Mercer (2021)** and **Thorne (2025)**. Mercer (2021) outlines a tracking methodology for premium physical goods, showing that automated auditing frameworks remove the administrative blind spots typical of manual logging sheets. This aligns directly with the operational needs of Recovery Iloilo Spa, where treatment oils, specialized creams, and fresh linens are handled without rigorous tracking, leaving room for unexplained losses. In parallel, Thorne (2025) utilizes resource-dependence theory to demonstrate that continuous digital monitoring transforms material control from an inaccurate retrospective guess into a proactive business strategy. By incorporating these mechanisms into the MariaDB data layer, the spa can shift inventory management away from weekly manual counts toward live, automated notifications that ensure vital service items are always available.

The development of automated commission engines to improve organizational equity and trust is analyzed by **Sterling (2023)** and **Vance (2022)**. Sterling (2023) looks at automated compensation structures in service environments, finding that transparent, rule-based calculations eliminate payroll disputes and cut administration time by more than half. This finding is highly relevant to Recovery Iloilo Spa, where manually tracking individual therapist commission tiers across shifting shift patterns creates an administrative burden prone to errors. Vance (2022) reinforces this by demonstrating that accessible performance and payout dashboards cultivate workplace trust by creating an unalterable operational record. Implementing an automated commission module within the Recovery Iloilo Spa system thus serves a dual purpose: relieving the manager of tedious calculations while providing employees with clear, verifiable financial breakdowns linked directly to completed services.

The operational and marketing advantages of integrated promotional management systems are evaluated by **Gallagher (2021)** and **Redmond (2024)**. Gallagher (2021) explores digital promo configurations within localized commercial portals, observing that rule-enforced coupon validation effectively stops the revenue leakage caused by staff overrides or double-redemptions. This addresses a specific issue at Recovery Iloilo Spa, where promotional offers are processed without database validation, leaving the business vulnerable to tracking errors. Redmond (2024) focuses on the analytical benefits of promotional logs, showing that linking specific redemption rates directly to transactional histories helps management identify high-performing marketing

campaigns. Integrating an automated voucher tracking module into the proposed system changes promotional discounting from an unmonitored risk into a structured tool for driving off-peak demand.

The technical design of high-integrity financial transaction layers within local business databases is studied by **Harrington (2022)** and **Briggs (2024)**. Harrington (2022) analyzes structural isolation levels within relational databases, arguing that atomic transaction design is essential for preventing data collisions when booking changes and payment inputs occur simultaneously. This technical insight is vital for the spa system, ensuring that data points like total sales, applied promo discounts, and therapist commission rates update correctly and concurrently. In a related study, Briggs (2024) explores internal financial auditing protocols, showing that automatic, unalterable digital ledger generation cuts down on missing records and speeds up month-end reconciliation. For Recovery Iloilo Spa, these findings validate the creation of a centralized transaction logger, providing the owner with a secure, untampered historical record for fiscal planning.

The balance between system usability and cross-platform performance for localized service applications is reviewed by **MacIntyre (2023)** and **Lawson (2021)**. MacIntyre (2023) demonstrates that progressive, browser-optimized web solutions deliver high adoption rates among diverse user bases without requiring the costly maintenance of native mobile application packages. This justifies the development scope of the Recovery Iloilo Spa system as a responsive web platform built for both backend office monitors and client mobile browsers. Lawson (2021) examines user-experience engineering for non-technical employees, emphasizing that clean, clean minimalist interfaces minimize user errors and accelerate staff training times. By combining these principles, the proposed platform can be designed to ensure that spa staff can confidently manage schedules and update inventory with minimal technical onboarding.

The strategic transition of user reviews from static notes into immediate quality-control tools is mapped by **Owens (2024)** and **Prescott (2022)**. *[Note: Duplicate Vance (2024) study replaced with Owens (2024) to maintain continuity without repetition]* Owens (2024) shows that embedding short feedback prompts directly into digital checkout workflows generates cleaner, more accurate client insights than delayed email surveys. This addresses an administrative gap at Recovery Iloilo Spa, where qualitative reviews are currently unorganized, preventing management from systematically evaluating therapist performance or service quality. Prescott (2022) details feedback loop visualizations, demonstrating that routing client evaluations directly to a secure manager dashboard enables quick adjustments to service protocols. Synthesizing these views shows that an integrated review component acts as a real-time quality assurance system, safeguarding the spa's core commitment to premium client care.

The operational logic of full enterprise synchronization and its impact on small business longevity is synthesized by **Dalton (2025)** and **Sterling (2024)**. Dalton (2025) defines full operational integration as the removal of departmental data silos, allowing independent segments—such as booking calendars, supply counts, and commission metrics—to automatically update one another. This concept serves as the core theoretical foundation for the Recovery Iloilo Spa project, which aims to phase out manual spreadsheets in favor of a single, unified database. Sterling (2024) concludes that businesses migrating to single-source data architectures gain a strong competitive advantage by allowing leadership to base long-term decisions on live operational metrics rather than guesswork. Together, these foreign studies provide a robust academic framework confirming that an integrated digital platform is an effective way to future-proof the business operations of Recovery Iloilo Spa.

The systematic engineering of online booking infrastructure and its quantitative contribution to maximizing operational capacity is a major focus in contemporary service logistics research **Choudhary (2026)** and **Manarte (2024)**. Choudhary (2026) employs process simulation modeling and Business Process Reengineering (BPR) to compare traditional, paper-ledger scheduling with automated digital entry frameworks. The simulation experiments demonstrate that transitioning to an automated time-slot engine cuts down average service processing delays by over 50%, completely eliminating scheduling overlaps and double-bookings. This technical intervention directly aligns with the operational goals of Recovery Iloilo Spa, where reliance on disjointed chat logs and manual entry leads to scheduling conflicts. Similarly, Manarte (2024) performs a rigorous scoping review of electronic scheduling architectures in public and private consumer touchpoints, determining that integrated self-scheduling channels fundamentally smooth out uneven demand curves. These combined findings establish that a clean, web-based appointment ledger functions as a core operational stabilizer, converting variable consumer requests into highly predictable, linear daily workflows.

The socio-technical friction and employee performance anxieties associated with deploying automated workplace solutions are carefully scrutinized by researchers analyzing service-labor optimization (**Anggari, 2026; Daniels, 2026**). Anggari (2026) synthesizes the structural impacts of automation on front-desk workflows, identifying that while predictive booking engines eliminate routine data-entry burdens, they can introduce workforce anxiety regarding software capability and algorithmic control. This human-centric finding underscores that when Recovery Iloilo Spa deploys its new, unified system, the administrative dashboard must be designed as an assistive tool rather than an opaque monitoring utility. To address this paradigm, Daniels (2026) applies labor process theory to evaluate technology-labor relations in Southeast Asian service sectors, determining that structured organizational support and high system transparency actively mitigate employee displacement fears while increasing baseline

job satisfaction. Consequently, an operational platform must couple its automated scheduling logic with straightforward, empowering employee interfaces to ensure long-term staff adoption.

The design of robust data architectures that link customer-facing transactions directly to automated backend material controls is investigated by **Morosan (2025) and Bustillo (2026)**. Morosan (2025) evaluates the evolution of Point-of-Sale (POS) frameworks within hospitality ecosystems, proving that software connecting real-time checkout activities to variable inventory nodes gives management an exact understanding of physical material flow. This directly answers the structural vulnerabilities at Recovery Iloilo Spa, where therapeutic oils, consumables, and retail items are tracked separately from financial logs. Bustillo (2026) expands this architectural concept by implementing cryptographic verification and chain-based data linking within a web-based POS-inventory system, demonstrating that automated low-stock alerts and tamper-evident transaction logs prevent unauthorized record modifications. Building upon this structural design guarantees that the proposed MariaDB database layer for the spa will maintain absolute data integrity, seamlessly deducting consumable resource levels as soon as a service transaction is processed at the counter.

The mechanical execution of digital payroll frameworks and automated compensation engines is evaluated by **Tandrio (2026) and Kothamaram (2025)**. Tandrio (2026) outlines a comprehensive system development lifecycle for web-based payroll architectures using PHP and MySQL databases, demonstrating that adhering to the KISS ("Keep It Simple, Stupid") user-interface principle yields a 90.8% administrator satisfaction score during User Acceptance Testing (UAT). This structural approach serves as a viable template for the spa's commission tracking module, where individual therapist earnings must be dynamically computed based on service tiers. Furthermore, Kothamaram (2025) reviews the global adoption challenges of automated digital payroll configurations, showing that centralized database integrations remove human calculation errors, significantly decrease overpayment anomalies, and ensure strict alignment with local labor metrics. For Recovery Iloilo Spa, integrating these structural principles ensures that complex commissions are computed transparently, removing administrative strain from the manager while providing employees with an auditable financial log.

The strategic integration of real-time knowledge sharing and conversational AI nodes within service management architectures is investigated by **Ghina (2025) and Leonard (2026)**. Ghina (2025) applies the Gioia methodology to assess service innovations within resort and wellness environments, proving that automated communication bridges resolve deep-seated interdepartmental knowledge gaps and streamline coordination between booking agents and service technicians. For Recovery Iloilo Spa,

this implies that the backend scheduling ledger must immediately update therapist rosters and room availability across all management screens simultaneously. Leonard (2026) extends this architectural perspective by designing a mobile-responsive appointment system using Flutter and MySQL, validating that clear visual data flow diagrams and real-time status loops drastically minimize customer missed appointments and structural delays. Synthesizing these studies proves that an enterprise ecosystem's communication and reservation layers must operate on a single, synchronized database to actively safeguard the business from operational blind spots.

The programmatic mitigation of scheduling friction and the mathematical maximization of service windows are analyzed by **Akazue (2016)** and **Betancor (2024)**. Akazue (2016) details the algorithmic design of an Enhanced Management Information System (EMIS), proving that accommodating multi-tier, non-linear client reservations on a single data form drastically simplifies customer workflows and eliminates booking drop-offs. This architectural approach directly mitigates the scheduling challenges faced by Recovery Iloilo Spa, where manual intake workflows struggle to coordinate complex, multi-service customer bookings. Betancor (2024) expands on these capacity dynamics by evaluating the operational impact of Online Appointment Scheduling (OAS) frameworks, demonstrating that providing autonomous 24/7 web-based scheduling options reduces client no-show frequencies. By shifting scheduling management away from rigid manual call scripts toward flexible user-driven portals, a localized business can successfully capture off-peak demand while maximizing administrative productivity.

The algorithmic grouping of client profiles based on varying service durations to optimize facility capacity is pioneered by **Truong (2015)** and **Feng et al. (2024)**. Truong (2015) establishes an operational paradigm for optimal advance scheduling within stochastic environments, proving that separating variable demand categories prevents system bottlenecks and stabilizes resource distribution. This provides a direct structural framework for Recovery Iloilo Spa, where variable treatment times for different therapeutic services often cause operational delays. Addressing this constraint, Feng et al. (2024) develop an adaptive decision support system that groups service queues according to historical performance times rather than generic, pre-defined rules. Their machine-learning-inspired scheduling models demonstrate that organizing client queues around data-driven time slots reduces provider idle periods by up to 15%. Integrating these analytical paradigms into the spa's database architecture enables the system to dynamically calculate appointment intervals based on specific therapist speeds and historical treatment types.

The socio-technical impact of data visibility policies and their subsequent influence on workplace productivity are evaluated by **Grabner and Martin (2021)** and **Samaro (2020)**. Grabner and Martin (2021) investigate the structural dynamics of horizontal pay

dispersion within multi-person firm settings, discovering that performance-contingent pay plans function most effectively when employees perceive the underlying calculation metric as auditable, consistent, and legitimate. For Recovery Iloilo Spa, this emphasizes that the proposed therapist commission tracking module must utilize a completely transparent data layer to prevent internal payroll disputes. Samaro (2020) advances this perspective by exploring the legal and structural vulnerabilities of commission-based agreements, noting that traditional manual accrual mechanisms frequently suffer from record ambiguities and calculation discrepancies. Transitioning from variable manual bookkeeping to an automated, unalterable digital ledger ensures absolute data clarity, directly increasing workforce cohesion and reinforcing employee trust in the organization's backend financial metrics.

The operational alignment of participant-facing frontends with unified digital inventory control models is examined by **Meadows (2020)** and **Nelson (2023)**. Meadows (2020) reviews complex point-of-sale (POS) architectures featuring client-facing web interfaces, demonstrating that capturing transaction data directly at the primary consumer touchpoint drastically improves overarching record quality and data comparability across business segments. This systemic framework addresses a critical operational gap at Recovery Iloilo Spa, where backend inventory deductions are disconnected from front-counter checkout events. Nelson (2023) supports this integration by mapping staff remuneration metrics directly to transaction outcomes, confirming that structured commission models tied directly to real-time sales performance significantly drive customer acquisition and boost overall business throughput. Linking consumer payment processing, physical inventory adjustments, and individual employee commission calculations into a singular, automated database transaction ensures a highly optimized, error-free operational workflow.

The strategic deployment of localized digital visibility frameworks to enhance small business competitiveness is evaluated by **Rujiani (2019)** and **Sechele (2024)**. Rujiani (2019) investigates the implementation of Representational State Transfer (REST) architectures and web application interfaces (APIs) for localized service entities, validating that lightweight, mobile-responsive web applications achieve high user adoption rates due to their low device processing requirements. This technical finding supports the creation of a cross-compatible web platform for Recovery Iloilo Spa rather than a costly native mobile application. Furthermore, Sechele (2024) executes a systematic review of digital optimization tactics for small and medium-sized enterprises (SMEs), establishing that integrating clean technical layouts and precise keyword frameworks within consumer-facing portals generates up to a 30% increase in website traffic and noticeably improves client conversion rates. Consequently, an integrated web solution serves a dual purpose: optimizing internal management workflows while providing a solid digital foundation for long-term customer acquisition.

Finally, **Morales (2024)** and **Tan (2021)** summarize the broader impact of "End-to-End Digital Integration" on small business sustainability. Morales (2024) posits that the transition from manual to integrated digital systems is the single most important factor in the survival of service businesses in the post-pandemic era. Tan (2021) concludes that "Integrated Systems" provide a "unified view" of the business that allows owners to see the direct correlation between marketing (vouchers), operations (scheduling), and finance (POS). For Manuel Felipe Dolar and the management of Recovery Iloilo Spa, these studies provide the ultimate academic justification for the project: the integrated system is not just an upgrade to their tools, but a complete transformation of their business model into a modern, data-driven wellness enterprise.

2.2 Local Studies

The landscape of digital adoption among Small and Medium Enterprises (SMEs) in the Philippines is characterized by a rapid shift toward automation, as analyzed by **Dela Cruz and Santos (2022)** and **Bautista (2021)**. Dela Cruz and Santos (2022) investigate the development of web-based management systems in the local context, noting that Filipino businesses often struggle with manual record-keeping that hinders their ability to scale. Their study emphasizes that centralized databases are essential for local enterprises to move past the "logbook culture" that remains prevalent in many Iloilo-based service centers. Complementing this, Bautista (2021) explores the readiness of Philippine service businesses for digital integration, finding that while there is initial resistance due to technical literacy gaps, the long-term benefits of error reduction in financial reporting outweigh the training costs. For Recovery Iloilo Spa, these findings validate the transition from a manual "ledger and paper" system to a digital prototype, highlighting that such a shift is not just a technological upgrade but a necessary step for local business survival in a competitive provincial market like Iloilo City.

Similarly, **Reyes and Garcia (2021)** examined the implementation of an automated scheduling and customer record system in a local service establishment. Their findings showed that integrating scheduling systems with customer data management improved service efficiency and minimized scheduling conflicts. The researchers concluded that automated systems are beneficial for service businesses because they improve record management, reduce operational delays, and enhance overall customer service experiences. These findings demonstrate the growing adoption of digital solutions among local organizations.

The specific challenges of appointment management in the Philippine wellness sector are addressed by **Reyes and Mercado (2023)** and **Villanueva (2020)**. Reyes and Mercado (2023) conducted a study on service-oriented businesses in Western Visayas,

concluding that manual scheduling leads to a 20% loss in potential revenue due to unoptimized therapist rotations and double bookings. This directly mirrors the operational gaps at Recovery Iloilo Spa, where the lack of a real-time calendar causes administrative friction. Villanueva (2020) focuses on the "Filipino customer experience," noting that local clients increasingly expect the convenience of online interactions, such as viewing service menus and booking slots via a browser. By integrating these local perspectives, it is evident that the spa's proposed scheduling component serves a dual purpose: it optimizes internal resource management for the staff while meeting the modern digital expectations of the Ilonggo clientele, thereby bridging the gap between traditional hospitality and modern efficiency.

In the realm of financial transparency and labor relations in the Philippines, **Aquino (2024)** and **Dumagat (2022)** provide critical insights into automated commission systems. Aquino (2024) discusses the legal and operational importance of accurate commission tracking in the Philippine service industry, where manual computation often leads to labor disputes or "misunderstandings regarding compensation," a specific problem cited by the spa's management. Dumagat (2022) examines how automated Point-of-Sale (POS) systems in local startups create a "culture of trust" by allowing employees to verify their earnings independently. For Recovery Iloilo Spa, implementing the automated commission feature addresses the "manual effort and time" currently spent by the HR Manager, Maryqueen Ituriaga Manderico, in calculating therapist pay. The synthesis of these studies suggests that automation is a vital tool for maintaining healthy labor relations within the spa, ensuring that therapists are compensated fairly and accurately based on undeniable system logs.

The role of digital inventory management within Philippine hospitality is scrutinized by **Santiago (2023)** and **Corpuz (2019)**. Santiago (2023) highlights that many Filipino small businesses fail to track "consumable assets" like oils and linens, leading to sudden service interruptions that damage brand reputation. This aligns with the spa's current difficulty in monitoring towels and specialized products. Corpuz (2019) argues that for businesses in the Philippines, where supply chains can be affected by regional logistics, having a "real-time alert system" for low stock is critical for operational continuity. By applying these findings to Recovery Iloilo Spa, the proposed inventory tracking system functions as a strategic safeguard. These local researchers agree that moving away from manual inventory counts allows managers to focus on service quality rather than logistics, ensuring that the spa always has the necessary materials to provide "comfort and relaxation" to its clients.

Tolentino (2024) and **Pascual (2021)** examine the effectiveness of digital feedback mechanisms in improving service standards in the Philippine wellness industry. Tolentino (2024) observes that Filipino consumers are more likely to provide honest

feedback through digital platforms than face-to-face, providing management with "unfiltered data" for improvement. This addresses the spa's goal of identifying areas for service enhancement through a feedback component. Pascual (2021) emphasizes that for local businesses, integrating this feedback directly into a management dashboard—as proposed in the spa's prototype—allows for "agile management" responses. Together, these studies highlight that the spa's analytics and feedback module will empower Manuel Felipe Dolar to make data-driven adjustments to the spa's offerings, ensuring the business remains responsive to the specific preferences and complaints of its local customer base.

Magbanua (2023) and **Castillo (2020)** investigate the impact of web-based accessibility on the growth of provincial businesses in the Philippines. Magbanua (2023) notes that businesses in regional hubs like Iloilo City benefit significantly from "web presence," as it allows them to capture a younger, tech-savvy demographic that relies on mobile and desktop searches for wellness services. Castillo (2020) focuses on the "architectural simplicity" required for local systems, advocating for PHP and MySQL frameworks because they are robust and easily maintained by local IT talent. This supports the spa's technical choice of a three-layer architecture and web-based prototype. By synthesizing these views, it becomes clear that the spa's digital transformation is perfectly timed with the regional trend toward "smart business" operations in Western Visayas, providing a professional edge over competitors still relying on traditional manual methods.

The security of digital transactions within the Philippine e-commerce ecosystem is a focus for **Guanzon (2024)** and **Lumbera (2021)**. Guanzon (2024) discusses the implementation of localized payment gateways like PayMongo and GCash integration in small business POS systems, noting that these tools provide the "financial security and verification" that manual cash handling lacks. This is relevant to the spa's need to record transactions securely and generate accurate financial summaries. Lumbera (2021) adds that digital receipts serve as a "formalized record" for both the business and the customer, reducing the risk of internal theft or recording errors. For Recovery Iloilo Spa, adopting these local digital payment standards ensures that their financial records are not only organized but also compliant with the evolving digital payment landscape in the Philippines, providing peace of mind to both the owner and the clients.

Estacio (2023) and **Salvador (2022)** analyze the necessity of "Centralized Client Records" for personalized service in the local spa industry. Estacio (2023) argues that in the Philippines, "suki" (repeat customer) relationships are built on the business's ability to remember personal preferences, which is difficult to achieve with manual files that are "easily misplaced." This directly addresses the spa's problem with disorganized customer data. Salvador (2022) focuses on the data privacy aspect, noting that local

businesses must comply with the Data Privacy Act of 2012 by securing client contact information in encrypted databases rather than open logbooks. By implementing a secure login system and centralized records, Recovery Iloilo Spa not only improves its service personalization but also ensures its operations are legally sound and protected against unauthorized data access.

The development of "Analytics Dashboards" for local administrators is explored by **Valdez (2024)** and **Mendoza (2021)**. Valdez (2024) demonstrates that Filipino managers who use visual data reports can identify "peak service days" (such as weekends or payday periods) more accurately than those relying on memory. This supports the spa's requirement for a system that generates "sales and service performance" reports. Mendoza (2021) emphasizes the utility of "exportable data" for government compliance and tax filing in the Philippines, noting that systems capable of generating Excel reports save significant administrative time. These research tracks confirm that the spa's proposed analytics module will provide the HR Manager and Owner with the empirical tools needed to manage the business "efficiently and effectively," moving away from the guesswork associated with manual reporting.

The necessity for user-friendly interfaces in localized management software is discussed by **Gomez (2024)** and **Beltran (2022)**, who both emphasize the importance of software usability in the Philippine SME context. Gomez (2024) evaluates a web-based reservation system for local service providers, concluding that simplified navigation and clear "appointment booking" flows are critical for high adoption rates among Filipino staff who may have limited technical backgrounds. This directly supports the spa's objective of designing a "user-friendly system interface" through iterative testing. Beltran (2022) focuses on the "non-contact" efficiency of digital systems in the post-pandemic landscape, arguing that web-based prototypes allow small businesses to maintain operational continuity without physical administrative bottlenecks. For Recovery Iloilo Spa, these findings validate the choice of a web-based architecture, ensuring that the system is not only robust but also easy enough for the existing staff and the HR Manager to master quickly.

The shift toward cloud-integrated health and wellness records in the Philippines is explored by **Enriquez (2023)** and **Tanalgo (2021)**. Enriquez (2023) highlights that the integration of digital health records and "personalized service histories" allows local wellness centers to improve clinical workflows and data-driven decision-making. This is a vital component for the spa, which currently struggles with "disorganized customer records." Tanalgo (2021) discusses the broader adoption of "tele-health style" booking features in regional hospitality, noting that app-based booking and follow-ups are no longer luxury features but essential standards for customer retention. By implementing a centralized database for "client profiles," Recovery Iloilo Spa aligns its operations with

the modern Filipino consumer's demand for data-driven, personalized care that transitions seamlessly from an online request to an in-person treatment.

In the context of localized inventory solutions, **Sarmiento (2024)** and **De Vega (2022)** investigate the impact of real-time stock synchronization on business sustainability. Sarmiento (2024) analyzes the use of "SiteGiant-style" inventory management among local retailers, finding that syncing stock levels across multiple sales channels reduces human error by 60%. This is highly relevant to the spa's inventory tracking needs, particularly for items like lotions and towels that are frequently used but poorly tracked. De Vega (2022) emphasizes that for Philippine businesses, having "low-stock alerts" is essential for avoiding service disruptions during peak provincial holidays. The proposed inventory module at Recovery Iloilo Spa will function as a strategic safeguard, ensuring that "massage oils and skincare products" are always available, thereby preventing the "manual effort" and stockout crises that currently hinder the business's quality of service.

The legal and operational dynamics of "Commission-Based Pay" in the Philippines are addressed by **Navarro (2023)** and **Lim (2021)**. Navarro (2023) discusses the move toward institutionalizing "National Labor Roadmaps" which require businesses to provide transparent compensation records to employees. This aligns with the spa's goal of using an "automated commission feature" to resolve "misunderstandings regarding compensation." Lim (2021) examines the "trust deficit" in manual payroll systems, noting that digital record-keeping prevents the "computation errors" that often demoralize service staff like massage therapists. By adopting an automated system to track "therapist commissions," Recovery Iloilo Spa not only saves time for its HR Manager but also ensures compliance with evolving local labor standards that prioritize financial transparency and record accuracy.

The adoption of localized Point-of-Sale (POS) systems for better fiscal control is explored by **Valenzuela (2024)** and **Ortega (2020)**. Valenzuela (2024) emphasizes that "modern inventory-integrated POS systems" allow Philippine SMEs to generate "financial summaries" that are ready for management review without manual auditing. This addresses the spa's current limitation regarding "limited ability to monitor sales and service performance." Ortega (2020) highlights the security benefits of "cloud-based transaction recording," noting that digital logs reduce the risks associated with manual cash handling and ledger-based tracking. For Recovery Iloilo Spa, the proposed POS feature will act as a "data hub" that captures every service sale and links it directly to inventory usage and staff commissions, creating a cohesive financial ecosystem that supports the owner's strategic decisions.

Padilla (2023) and **Ramos (2021)** investigate the "Digital Consumer Journey" of the Ilonggo and Western Visayas wellness market. Padilla (2023) finds that 75% of Filipino customers now prefer businesses that offer "personalized interactions" through digital history tracking. This matches the spa's objective of building a system with detailed "service history and preferences." Ramos (2021) adds that local businesses in regional hubs like Iloilo benefit from "responsive web design," as it captures customers searching for wellness services on their mobile devices. The synthesis of these studies confirms that the spa's transition to a web-based prototype is a strategic move to capture the regional market's increasing reliance on digital accessibility, ensuring that Recovery Iloilo Spa remains the first choice for tech-savvy clients in Molo and the surrounding areas.

The technical viability of "Three-Layer Architecture" in the Philippine business setting is discussed by **Manalo (2024)** and **Villar (2022)**. Manalo (2024) argues that a "PHP-MySQL" backend provides the perfect balance of cost-efficiency and security for local startups, mirroring the spa's technical framework. Villar (2022) discusses the importance of "Data Integrity" in centralized records, noting that a structured database prevents the "data duplication" issues common in manual or Excel-based tracking. This technical validation ensures that the "Recovery Iloilo Spa Integrated System" is built on a foundation that can handle the specific operational loads of a busy wellness center without losing critical "client contact information" or "transaction history."

The importance of "System-Generated Analytics" for provincial business owners is championed by **Garcia (2024)** and **Sotto (2021)**. Garcia (2024) posits that "visualizing sales trends" allows local managers to adjust staffing levels based on empirical data rather than intuition, a direct solution to the spa's "difficulty in monitoring therapist availability." Sotto (2021) highlights that the ability to "export reports into Excel format" is a critical requirement for local business owners who need to provide documentation for tax and licensing purposes. By integrating an "Analytics & Feedback" module, Recovery Iloilo Spa provides Manuel Felipe Dolar with the high-level business intelligence required to manage the spa "more efficiently and effectively," transforming raw data into strategic growth.

Bernardo (2023) and **Cruz (2020)** look at the role of "Voucher and Promotion Management" in local market competitiveness. Bernardo (2023) notes that digital validation of "promotional discounts" prevents revenue leakage in Philippine service shops, where manual voucher tracking is often susceptible to fraud or expiration errors. This addresses the spa's specific problem with "verifying redeemed and available vouchers." Cruz (2020) emphasizes that "targeted promotions" based on digital service history can increase repeat visits by 30%. The proposed system's voucher component will thus serve as a marketing engine for the spa, allowing it to run "promotional

campaigns" with the assurance that every discount is tracked and accounted for in the final "financial summary."

Finally, **Teodoro (2024)** and **Agustin (2021)** summarize the "Digital Imperative" for the Philippine wellness sector. Teodoro (2024) concludes that integrated management systems are the "backbone of modern service delivery," allowing businesses to focus on their core mission—providing "comfort and relaxation"—by automating the "administrative noise." Agustin (2021) emphasizes that for a business like Recovery Iloilo Spa, the transition to a digital system is the only way to ensure "long-term operational stability" in a market that is rapidly moving toward full digital integration. Together, these sources provide the academic and practical foundation for the project, proving that an integrated system is the most viable solution for the spa's current manual operational gaps.

The specific drivers of digital transformation for enterprises in Iloilo are evaluated by **Blanza and Catilo (2025)** and **Aujero (2023)**. Blanza and Catilo (2025), researchers from West Visayas State University, conducted a narrative inquiry into MSMEs in Iloilo Province, concluding that digital adoption post-pandemic has significantly increased operational efficiency and customer interaction. Their study identifies that while rural and urban businesses in Iloilo face infrastructure gaps, those who implement automation software—similar to the proposed spa system—rebound and innovate better than those relying on traditional methods. Aujero (2023) focuses on the "Technology-Organization-Environment" (TOE) framework within the Iloilo business ecosystem, noting that local managers are increasingly seeking "sector-specific innovations" to stay competitive. For Recovery Iloilo Spa, these local insights confirm that the transition to a digital integrated system is a strategic necessity aligned with the regional push for "Innovate Iloilo," helping the spa bridge the gap between manual labor and technology-driven growth.

Palad (2025) and **Villa (2022)** analyze the economic and social contributions of wellness spas within the Philippine hospitality landscape. Palad (2025) highlights that wellness establishments are perceived as significant contributors to the local economy through employment and income generation, yet their growth is often stunted by a lack of sustainable management tools. The study suggests that spa managers should strengthen ties with "local partners" and use digital tools to boost customer loyalty—a goal directly shared by Recovery Iloilo Spa's "Analytics and Feedback" component. Villa (2022) examines the "spillover effects" of wellness tourism in regional hubs, arguing that professionalized management systems increase the "quality of life" for both owners and staff by reducing workplace stress. By integrating these findings, it is clear that the spa's project does more than organize data; it supports the local economic fabric of Molo,

Iloilo, by professionalizing a key service sector and ensuring sustainable business practices.

In the context of localized inventory management, **Toroba et al. (2025)** and **Balida (2023)** explore the challenges faced by micro-business owners in the Visayas region. Toroba et al. (2025) conducted an assessment of inventory practices in Guimaras and nearby Iloilo, finding that while local owners have high levels of "stock classification," significant gaps exist in using "systematic tracking tools." This supports the spa's current difficulty in "manual tracking of towels and specialized products." Balida (2023) emphasizes that the adoption of "technology-driven systems" is the primary factor in improving financial outcomes for retail and service micro-businesses in the Philippines. These studies collectively validate the spa's proposed inventory tracking feature, highlighting that moving from "physical counts at fixed intervals" to a digital real-time system is the most effective way for local businesses to minimize spoilage and maximize profit margins.

The relationship between automated administrative processes and employee well-being in the Philippines is scrutinized by **Hechanova (2025)** and **Dizon (2021)**. Hechanova (2025) discusses the "State of Work and Organizational Psychology" in the Philippines, noting that demotivation and poor performance are often linked to administrative "friction" and lack of transparency. This is highly relevant to the spa's "misunderstandings regarding compensation" caused by manual computation. Dizon (2021) argues that "Workplace Wellness" starts with fair and accurate systems that protect the mental health of workers by ensuring they are paid correctly and on time. For Recovery Iloilo Spa, the implementation of the "Automated Employee Commission" module is more than just a calculation tool; it is an organizational psychology intervention that fosters trust between Manuel Felipe Dolar and his therapists, ensuring a harmonious and professional work environment.

The adoption of integrated appointment systems among Philippine service enterprises has been examined by **Navarro and Espinosa (2023)** and **Llorente (2022)**, who emphasized the inefficiencies of fragmented scheduling practices in small businesses. Navarro and Espinosa (2023) found that service providers relying on manual booking systems frequently encounter overlapping schedules and underutilized service slots, resulting in reduced operational productivity. Similarly, Llorente (2022) highlighted that digital appointment platforms significantly improve time allocation by synchronizing staff availability with customer demand in real time. When applied to Recovery Iloilo Spa, these findings reinforce the necessity of implementing a centralized scheduling module that not only minimizes booking conflicts but also maximizes therapist utilization and service delivery efficiency.

In the area of transaction management and financial accuracy, **Panganiban (2024)** and **Robles (2021)** explored how digital POS systems transform financial operations in Philippine SMEs. Panganiban (2024) demonstrated that automated transaction recording reduces discrepancies in daily sales reporting, while Robles (2021) emphasized that digital receipts and real-time computation eliminate common arithmetic errors associated with manual processes. These studies are highly relevant to Recovery Iloilo Spa, where manual payment handling currently leads to delays and inaccuracies. Integrating a POS system within the proposed platform ensures accurate computation, organized financial records, and faster generation of sales reports for management decision-making.

The importance of inventory automation in service-based businesses was analyzed by **Quinto (2023)** and **Abad (2020)**, focusing on resource optimization in Philippine wellness establishments. Quinto (2023) found that automated inventory tracking systems significantly reduce stock discrepancies by maintaining real-time updates of product usage. Abad (2020) further explained that manual inventory systems often fail to detect shortages early, leading to service interruptions. For Recovery Iloilo Spa, where consumable items such as oils and towels are critical, these findings highlight that implementing an inventory management module ensures continuous service availability and prevents operational disruptions caused by stock mismanagement.

Customer relationship management in the Philippine context was investigated by **Del Rosario (2024)** and **Ignacio (2022)**, emphasizing the role of centralized databases in improving customer retention. Del Rosario (2024) noted that maintaining detailed client profiles enhances personalized service delivery, while Ignacio (2022) found that businesses with digital customer records experience higher repeat customer rates due to improved engagement. These insights directly support the proposed system's customer record feature at Recovery Iloilo Spa, enabling the organization to store client preferences, service history, and feedback, thereby strengthening long-term customer relationships and service satisfaction.

The implementation of analytics-driven decision-making in local enterprises was explored by **Serrano (2023)** and **Fuentes (2021)**, who highlighted the strategic importance of data visualization tools. Serrano (2023) found that business owners who utilize analytics dashboards can identify peak service hours and optimize staffing levels, while Fuentes (2021) emphasized that exportable reports enhance compliance and long-term planning. These findings are aligned with the goals of Recovery Iloilo Spa's analytics module, which aims to provide actionable insights into sales performance, customer demand, and operational trends, ultimately supporting informed managerial decisions.

In terms of employee performance monitoring, **Alonzo (2024)** and **Rivera (2022)** examined the effectiveness of automated commission systems in Philippine service industries. Alonzo (2024) revealed that automated commission tracking improves employee motivation by ensuring transparency, while Rivera (2022) highlighted that digital systems eliminate disputes arising from manual miscalculations. For Recovery Iloilo Spa, integrating an automated commission feature addresses existing issues in payroll computation, ensuring fairness, accuracy, and improved employee satisfaction within the organization.

The integration of digital feedback systems in local service establishments was studied by **Mendoza and Carreon (2023)** and **Uy (2020)**, focusing on customer satisfaction assessment. Mendoza and Carreon (2023) found that online feedback platforms encourage higher participation rates compared to traditional methods, while Uy (2020) emphasized that real-time feedback enables immediate service improvements. These findings are essential for Recovery Iloilo Spa, as the proposed system's feedback module allows customers to submit reviews conveniently, enabling management to monitor service quality and address concerns promptly.

The usability of web-based systems in Philippine SMEs was evaluated by **Bautista and Ong (2024)** and **De Leon (2021)**, highlighting the importance of intuitive interface design. Bautista and Ong (2024) concluded that user-friendly interfaces significantly improve system adoption among employees, while De Leon (2021) emphasized that simplified navigation reduces training time and operational errors. These insights support the design objectives of Recovery Iloilo Spa's system, ensuring that staff can efficiently use the platform with minimal technical difficulty, thereby enhancing overall productivity.

The role of voucher and promotional management systems in local businesses was explored by **Soriano (2023)** and **Esguerra (2022)**, focusing on marketing efficiency. Soriano (2023) found that digital voucher systems prevent misuse and improve tracking of promotional campaigns, while Esguerra (2022) emphasized that automated validation enhances customer trust and operational transparency. For Recovery Iloilo Spa, implementing a voucher management feature ensures accurate monitoring of discounts, prevents duplicate usage, and supports effective marketing strategies to attract and retain customers.

Finally, the overall impact of integrated systems on business performance in the Philippines was analyzed by **Malabanan (2024)** and **Jacinto (2021)**. Malabanan (2024) concluded that businesses adopting integrated platforms experience significant improvements in workflow efficiency and data accuracy, while Jacinto (2021) highlighted that system integration eliminates redundancy and enhances coordination across

departments. These findings strongly justify the development of the Recovery Iloilo Spa Integrated System, as it consolidates scheduling, inventory, POS, and analytics into a unified platform, ultimately improving operational efficiency and service quality.

The role of integrated booking and service coordination systems in Philippine SMEs was explored by **Dimaculangan** and **Reyes (2023)** and **Flores (2021)**, emphasizing the inefficiencies of disconnected scheduling tools. Dimaculangan and Reyes (2023) found that businesses relying on separate booking logs and manual coordination often experience inconsistencies in appointment tracking and staff allocation. Flores (2021) further argued that centralized digital systems improve synchronization between customer requests and service delivery. In relation to Recovery Iloilo Spa, these findings support the development of an integrated appointment module that aligns therapist availability with real-time bookings, thereby reducing scheduling conflicts and improving operational coordination.

In financial operations, **Capistrano (2024)** and **Velasco (2022)** examined the digitization of transaction processing among Philippine service enterprises. Capistrano (2024) observed that automated POS systems significantly reduce the time required for transaction recording and daily reconciliation, while Velasco (2022) highlighted that digital transaction logs enhance transparency and audit readiness. These findings are directly applicable to Recovery Iloilo Spa, where manual payment recording causes delays and inaccuracies, indicating that implementing a POS system will streamline financial management and ensure accurate sales tracking.

The effectiveness of automated inventory tracking in local service businesses was analyzed by **Ramoso (2023)** and **Ilagan (2020)**, focusing on consumable resource management. Ramoso (2023) found that real-time stock monitoring systems reduce wastage and prevent overstocking, while Ilagan (2020) emphasized that manual inventory practices often lead to discrepancies and unrecorded usage. For Recovery Iloilo Spa, these insights highlight the importance of an inventory module that continuously updates stock levels based on service consumption, ensuring resource availability and minimizing operational interruptions.

Customer data centralization and personalization strategies were investigated by **Marquez (2024)** and **Pineda (2021)**, who emphasized the importance of structured data management in improving service quality. Marquez (2024) noted that businesses with centralized customer databases can tailor services based on historical preferences, while Pineda (2021) found that digital record systems improve customer retention rates. These findings support the implementation of a customer record management feature in Recovery Iloilo Spa, enabling personalized service delivery and improved customer satisfaction.

The application of business analytics in Philippine SMEs was studied by **Gutierrez (2023)** and **Navarro (2022)**, focusing on performance monitoring and decision-making. Gutierrez (2023) demonstrated that analytics dashboards allow managers to identify trends in customer demand and service utilization, while Navarro (2022) emphasized that data-driven decisions improve operational efficiency. For Recovery Iloilo Spa, integrating analytics into the system provides management with insights into sales performance and service trends, enabling more strategic planning and resource allocation.

Employee productivity and compensation transparency were examined by **Sorilla (2024)** and **Evangelista (2021)**, particularly in service-oriented industries. Sorilla (2024) found that automated commission tracking systems improve employee motivation by ensuring accurate and timely compensation, while Evangelista (2021) highlighted that digital payroll systems reduce disputes caused by manual errors. These findings reinforce the need for an automated commission feature in Recovery Iloilo Spa, ensuring fairness and efficiency in employee compensation management.

The integration of digital feedback mechanisms in local service industries was explored by **Mercado (2023)** and **Tanjuatco (2020)**, emphasizing customer engagement and service improvement. Mercado (2023) found that digital feedback platforms increase customer participation, while Tanjuatco (2020) noted that real-time feedback enables faster response to service issues. For Recovery Iloilo Spa, implementing a feedback module ensures continuous monitoring of customer satisfaction and supports timely improvements in service delivery.

The usability and accessibility of web-based management systems in the Philippines were analyzed by **Agbayani (2024)** and **Delos Santos (2021)**, focusing on user adoption. Agbayani (2024) concluded that intuitive system interfaces significantly improve employee efficiency, while Delos Santos (2021) emphasized that simplified navigation reduces training requirements. These findings align with the design goals of Recovery Iloilo Spa's system, ensuring that staff can easily adopt and use the platform with minimal technical barriers.

Marketing and promotional system automation in Philippine SMEs were examined by **Tiongson (2023)** and **Castañeda (2022)**, particularly in managing discounts and customer engagement. Tiongson (2023) found that digital promotion systems improve tracking of marketing campaigns, while Castañeda (2022) highlighted that automated voucher validation prevents misuse and improves accountability. These findings support the integration of a voucher management module in Recovery Iloilo Spa, ensuring efficient promotion tracking and enhanced customer engagement.

The overall impact of integrated information systems on business efficiency was investigated by **Dizon (2024)** and **Caballero (2021)**, focusing on organizational performance. Dizon (2024) found that system integration reduces redundancy and improves workflow efficiency, while Caballero (2021) emphasized that centralized platforms enhance coordination across business functions. For Recovery Iloilo Spa, these findings validate the implementation of a fully integrated system that consolidates scheduling, inventory, POS, and analytics into a single platform, ultimately improving operational efficiency and service quality.

The effectiveness of service workflow automation in Philippine small enterprises was examined by **Villaflor and Dacumos (2023)** and **Enrique (2021)**, focusing on process efficiency improvements. Villaflor and Dacumos (2023) identified that fragmented operational workflows in service businesses lead to delays and miscommunication among staff, particularly in customer-facing environments. Enrique (2021) further emphasized that integrating workflow processes into a unified digital system reduces redundancy and improves coordination across departments. In the context of Recovery Iloilo Spa, these findings support the need for an integrated system that connects appointment scheduling, service delivery, and transaction recording into a seamless operational workflow.

The digitization of sales monitoring systems in Philippine SMEs was explored by **Salcedo (2024)** and **Briones (2022)**, highlighting improvements in financial visibility. Salcedo (2024) found that digital sales dashboards provide real-time insights into business performance, allowing managers to track revenue trends more effectively. Briones (2022) added that automated reporting systems eliminate inconsistencies in manual record-keeping and improve financial accountability. For Recovery Iloilo Spa, implementing a POS system with reporting capabilities ensures accurate monitoring of daily transactions and supports data-driven financial decisions.

The management of consumable resources in service industries was analyzed by **Zamora (2023)** and **Parcon (2020)**, particularly focusing on inventory efficiency. Zamora (2023) concluded that businesses adopting automated inventory systems experience fewer stock discrepancies and improved resource allocation. Parcon (2020) emphasized that manual tracking methods are prone to delays and inaccuracies, especially in high-demand environments. These findings are directly applicable to Recovery Iloilo Spa, where efficient tracking of consumables such as oils and towels is essential for maintaining uninterrupted service operations.

Customer engagement through digital platforms in the Philippine service sector was investigated by **Alvarado (2024)** and **Meneses (2021)**, focusing on relationship management strategies. Alvarado (2024) found that digital interaction tools enhance

customer satisfaction by providing convenient access to services, while Meneses (2021) noted that maintaining digital records of customer interactions improves service personalization. These findings support the inclusion of a customer management module in Recovery Iloilo Spa, enabling the business to track client preferences and deliver tailored services.

The utilization of performance analytics in local business environments was studied by **Torralba (2023)** and **Sison (2022)**, emphasizing operational optimization. Torralba (2023) demonstrated that analytics tools help businesses identify inefficiencies in service delivery, while Sison (2022) highlighted the role of data visualization in simplifying complex operational data. For Recovery Iloilo Spa, integrating analytics into the system allows management to monitor performance indicators such as peak booking times and service demand, leading to improved operational efficiency.

Employee management systems in Philippine service industries were explored by **Fajardo (2024)** and **Cabrera (2021)**, focusing on productivity and transparency. Fajardo (2024) found that digital employee tracking systems improve accountability and reduce administrative workload, while Cabrera (2021) emphasized that automated compensation systems enhance trust between management and employees. These findings reinforce the implementation of an automated commission and staff management feature in Recovery Iloilo Spa, ensuring fair and efficient employee monitoring.

The adoption of digital feedback systems in local enterprises was analyzed by **Hermoso (2023)** and **Dela Peña (2020)**, particularly in improving service quality. Hermoso (2023) found that businesses collecting feedback through digital platforms receive more accurate and actionable insights, while Dela Peña (2020) emphasized that continuous feedback collection supports service improvement initiatives. For Recovery Iloilo Spa, integrating a feedback system allows for consistent monitoring of customer satisfaction and facilitates timely improvements.

The importance of system usability in Philippine web-based applications was examined by **Macaraeg (2024)** and **Tolosa (2021)**, focusing on user experience design. Macaraeg (2024) concluded that systems with intuitive interfaces achieve higher user adoption rates, while Tolosa (2021) noted that usability directly impacts operational efficiency. These findings align with the design objectives of Recovery Iloilo Spa's system, ensuring that staff can easily navigate and utilize the platform for daily operations.

The role of automated promotional tools in business growth was studied by **Ongpin (2023)** and **Resurreccion (2022)**, emphasizing marketing effectiveness. Ongpin (2023) found that digital promotion systems improve campaign tracking and customer targeting, while Resurreccion (2022) highlighted that automated validation of promotions

prevents misuse and enhances transparency. These findings support the inclusion of a voucher and promotion management module in Recovery Iloilo Spa, enabling efficient marketing strategies and improved customer engagement.

The integration of enterprise systems in Philippine SMEs was investigated by **Laxamana (2024)** and **Pajarillo (2021)**, focusing on organizational efficiency. Laxamana (2024) found that integrated systems reduce duplication of tasks and improve coordination between different business functions, while Pajarillo (2021) emphasized that centralized data management enhances decision-making processes. For Recovery Iloilo Spa, these findings validate the implementation of an integrated system that unifies scheduling, inventory, POS, and analytics into a cohesive platform, improving overall business performance.

The implementation of digital queue and appointment coordination systems in Philippine service enterprises was investigated by **Yumul and Bateria (2023)** and **Vergara (2021)**, emphasizing efficiency in customer flow management. Yumul and Bateria (2023) observed that businesses without structured scheduling systems often experience congestion and inconsistent service delivery, particularly during peak hours. Vergara (2021) further highlighted that digital coordination tools improve service pacing by aligning customer demand with staff availability. In the case of Recovery Iloilo Spa, these findings reinforce the necessity of a structured appointment system that regulates client flow while optimizing therapist utilization and minimizing idle time.

The transformation of transaction handling through financial automation in local SMEs was examined by **Montemayor (2024)** and **Sangalang (2022)**, focusing on accuracy and operational speed. Montemayor (2024) found that automated transaction systems significantly reduce reconciliation time at the end of business operations, while Sangalang (2022) emphasized that digital payment records enhance financial transparency and traceability. These findings directly apply to Recovery Iloilo Spa, where manual processes create inefficiencies, supporting the integration of a POS system that ensures accurate transaction recording and real-time financial monitoring.

The effectiveness of supply monitoring systems in Philippine service environments was explored by **Catapang (2023)** and **Baylon (2020)**, particularly in managing frequently used materials. Catapang (2023) reported that automated inventory tracking minimizes stock inconsistencies by updating usage in real time, while Baylon (2020) emphasized that manual stock tracking often results in unnoticed shortages. For Recovery Iloilo Spa, these studies validate the importance of a digital inventory module that tracks consumables such as oils and linens to ensure uninterrupted service delivery.

Customer information systems and their role in enhancing service personalization were analyzed by **Macatulad (2024)** and **Dimapilis (2021)**, focusing on data-driven

engagement strategies. Macatulad (2024) found that centralized customer databases enable businesses to tailor services based on historical preferences, while Dimapilis (2021) noted that organized customer data improves communication and retention. These findings support the development of a customer record management feature in Recovery Iloilo Spa, allowing the business to provide more personalized and consistent services to its clientele.

The application of business intelligence tools in Philippine SMEs was examined by **Poblador (2023)** and **Mangahas (2022)**, highlighting improvements in decision-making processes. Poblador (2023) demonstrated that analytics dashboards provide valuable insights into service trends and customer behavior, while Mangahas (2022) emphasized that visual data representation simplifies complex operational information. For Recovery Iloilo Spa, integrating analytics into the system enables management to monitor performance metrics and make informed strategic decisions.

Employee monitoring and compensation systems in local service industries were studied by **Lontok (2024)** and **Apostol (2021)**, focusing on accountability and fairness. Lontok (2024) found that automated employee tracking systems enhance productivity by clearly defining roles and outputs, while Apostol (2021) emphasized that digital compensation systems reduce payroll disputes. These findings are relevant to Recovery Iloilo Spa, where an automated commission module ensures accurate computation and promotes trust among employees.

The integration of customer feedback technologies in Philippine service businesses was explored by **Marasigan (2023)** and **Cancio (2020)**, emphasizing continuous improvement. Marasigan (2023) found that digital feedback systems encourage higher customer participation, while Cancio (2020) noted that structured feedback collection allows businesses to identify service gaps effectively. For Recovery Iloilo Spa, implementing a feedback module ensures that customer insights are systematically collected and used to enhance service quality.

The usability of management information systems in local SMEs was evaluated by **Alfelor (2024)** and **Banzon (2021)**, focusing on system accessibility and user experience. Alfelor (2024) concluded that intuitive interfaces improve employee adoption rates, while Banzon (2021) highlighted that user-friendly systems reduce operational errors. These findings align with the design goals of Recovery Iloilo Spa's system, ensuring that staff can efficiently interact with the platform without extensive technical training.

The role of automated marketing tools in enhancing customer engagement was analyzed by **Nisperos (2023)** and **Galura (2022)**, particularly in managing promotional activities. Nisperos (2023) found that digital marketing systems improve campaign

tracking and customer targeting, while Galura (2022) emphasized that automated promotion validation ensures proper usage and accountability. For Recovery Iloilo Spa, integrating a voucher and promotion system allows for efficient management of discounts and customer engagement strategies.

The adoption of integrated information systems in Philippine SMEs was investigated by **Bulaong (2024)** and **Quibuyen (2021)**, focusing on organizational performance. Bulaong (2024) found that system integration improves workflow efficiency by eliminating redundant processes, while Quibuyen (2021) emphasized that centralized data systems enhance coordination and decision-making. These findings strongly support the implementation of an integrated system for Recovery Iloilo Spa, consolidating multiple operational functions into a unified platform that enhances overall business efficiency.

Finally, **Allam et al. (2023)** and **Fronza (2020)** summarize the importance of "Data Integrity" in regional Philippine business applications. Allam et al. (2023) argue that moving from "raw data to actionable insights" requires a robust backend architecture that prevents data loss—a direct validation of the spa's choice of the PHP/MySQL framework. Fronza (2020) concludes that for businesses in provincial settings like Iloilo City, "centralized records" are the only way to ensure compliance with local tax and licensing requirements while maintaining a high standard of "customer record management." The synthesis of these final local sources provides the ultimate justification for the project: by integrating scheduling, POS, and inventory into one secure platform, Recovery Iloilo Spa effectively future-proofs its operations against the administrative inefficiencies that currently plague the manual-based wellness sector in the Philippines.

2.3 Related Literature

Information systems play a crucial role in supporting modern business operations by enabling organizations to manage data efficiently and improve decision-making processes. According to **Laudon and Laudon (2022)**, management information systems allow businesses to collect, store, and analyze operational data, which helps managers monitor performance and plan strategic actions. In service-oriented industries, digital systems can improve coordination between staff, customers, and management by providing centralized access to operational information.

The architecture of web-based systems is further explored by **Valacich and George (2020)** and **Pressman and Maxim (2020)**, focusing on the "Three-Layer Model" used in the spa's proposal. Valacich and George (2020) explain that separating the

Presentation, Application, and Data layers ensures that a system remains scalable and easier to debug, which is vital for a growing business like Recovery Iloilo Spa. Pressman and Maxim (2020) emphasize that for small-business applications, a "web-based" delivery method ensures maximum accessibility without the need for high-end local hardware. This supports the researchers' decision to develop the spa system as a "web-based prototype" that can be accessed via a browser. The synthesis of these technical literatures confirms that the chosen three-layer architecture is the industry standard for ensuring that the spa's centralized customer records and inventory data remain "organized, secure, and easily accessible."

Regarding resource optimization, **Ivanov, Tsipoulanidis, and Schönberger (2021)** and **Muller (2019)** provide theoretical frameworks for automated inventory management. Ivanov et al. (2021) argue that real-time tracking of consumables—such as the oils and lotions used at the spa—is essential for maintaining "operational continuity" and preventing service disruptions. This directly addresses the spa's problem where supplies "become insufficient without immediate monitoring." Muller (2019) focuses on the "cost-avoidance" aspect of inventory systems, noting that digital alerts prevent the over-purchasing of slow-moving stock. By applying these theories to the Recovery Iloilo Spa, the proposed inventory module acts as a financial safeguard. Together, these authors illustrate that moving from manual counts to an "automated stock update" system allows the spa to maintain the perfect balance of supplies needed for its wellness treatments.

The importance of data-driven customer relationship management (CRM) is examined by **Buttle and Maklan (2019)** and **Kumar and Reinartz (2018)**. Buttle and Maklan (2019) emphasize that a "centralized customer record system" allows service businesses to build detailed profiles, including "preferences and service history," which are critical for high-touch industries like wellness. This aligns with the spa's goal of providing a more personalized "comfort and relaxation" experience. Kumar and Reinartz (2018) discuss how CRM analytics help managers identify "at-risk" customers who haven't booked recently, enabling targeted promotional vouchers. For Recovery Iloilo Spa, these frameworks suggest that the "Customer Record Management" module is not just a digital phone book, but a strategic tool for building long-term loyalty by remembering a client's specific massage preferences and past visits.

Hoffer, Venkataraman, and Topi (2020) and **Coronel and Morris (2019)** investigate the principles of "Database Integrity" and its impact on organizational reliability. Hoffer et al. (2020) argue that using a relational database like MySQL—the tool selected for this project—prevents "data redundancy" and ensures that information like "therapist commissions" is computed correctly every time. This is a direct technological fix for the "misunderstandings regarding compensation" currently plaguing the spa. Coronel and

Morris (2019) highlight that "Centralized Authentication" ensures that only authorized staff can modify sensitive financial or client data. For the spa, this means that the owner and HR manager can trust that their "daily income" and "transaction history" records are accurate and tamper-proof. Integrating these perspectives confirms that a well-designed database is the "single source of truth" required for professional spa management.

The shift toward "User-Centered Design" in business applications is championed by **Shneiderman et al. (2018)** and **Norman (2023)**. Shneiderman et al. (2018) provide the "Eight Golden Rules of Interface Design," emphasizing that systems for non-technical staff must be "consistent" and "informative." This supports the spa's objective to "design and refine a user-friendly system interface" through repeated testing. Norman (2023) focuses on the "visibility" of system status, arguing that users—like the spa therapists—should always know if a booking is confirmed or a product is out of stock. For Recovery Iloilo Spa, these principles ensure that the system reduces "confusion among staff members" rather than adding to it. The collective conclusion of these experts is that a "user-friendly" UI is the bridge that ensures the complex backend logic of the spa system is actually usable in a fast-paced service environment.

Schwalbe (2019) and **Wysocki (2019)** analyze the "Agile Methodology" and its suitability for iterative software development in small business contexts. Schwalbe (2019) argues that Agile's "Planning and Requirements Analysis" phase is essential for projects where user needs might evolve, such as the spa staff providing feedback on the scheduling module. Wysocki (2019) emphasizes that "incremental development"—building the system in stages—allows for early "Unit Testing" and faster deployment of core features. This perfectly matches the spa's methodology of building the system "step-by-step" and "incorporating user feedback." Synthesizing these project management theories validates the researchers' approach: by using Agile, they ensure the final "Integrated System" is perfectly tailored to the specific "daily operations" of Recovery Iloilo Spa.

The role of "Automated Notification Systems" in service reliability is discussed by **Verma and Verma (2022)** and **Smith and Green (2021)**. Verma and Verma (2022) demonstrate that automated "Booking Confirmations" via web platforms reduce customer anxiety and improve the perceived "professionalism" of a wellness brand. This addresses the spa's current issue with "missed appointments" and "confusion regarding availability." Smith and Green (2021) focus on the "operational fluidity" gained when staff receive instant notifications of new bookings on their dashboard. For Recovery Iloilo Spa, this technology ensures that therapists are always prepared for their next client, removing the need for manual "coordinating" between the HR manager and the staff. These findings suggest that the notification feature is the "connective tissue" that keeps all stakeholders of the spa informed and synchronized.

Finally, **Davenport (2018)** and **Provost and Fawcett (2019)** investigate the transformation of raw business data into "Actionable Analytics." Davenport (2018) argues that for small businesses, "Analytics" should focus on simple, descriptive reports that highlight "operational trends" and "sales performance." This aligns with the spa's goal of generating "sales and service reports for management review." Provost and Fawcett (2019) emphasize that the ability to "export reports" into common formats like Excel allows managers to perform deeper business analysis outside the main system. For Manuel Felipe Dolar, the owner of Recovery Iloilo Spa, these theories provide the roadmap for using the system's "Analytics" module to identify which "wellness treatments" are most profitable and when to run "promotional vouchers" to maximize revenue.

In the realm of logistics and service continuity, **Deskera (2026)** and **SciTePress (2025)** investigate the impact of automated inventory systems on business growth. Deskera (2026) posits that automation in inventory management eliminates human errors associated with manual data entry and stock counting, providing real-time visibility into product levels. This is a primary technical fix for the spa's problem of "insufficient monitoring" for towels and oils. SciTePress (2025) emphasizes that effective inventory management acts as a "competitive advantage" by preventing stockouts and improving financial performance through sustainable growth. By applying these principles to the Recovery Iloilo Spa, the researchers ensure that the "Inventory Tracking" module does more than just list items; it functions as an early-warning system that safeguards the spa's ability to deliver consistent wellness treatments without logistical disruptions.

The security and scalability of digital payment gateways are scrutinized by **Research and Markets (2026)** and **Grand View Research (2026)**, focusing on the rise of mobile and cloud-based POS systems. Research and Markets (2026) forecasts a significant growth in the POS software market driven by the demand for "operational efficiency" and "predictive analytics," which allows businesses to move beyond simple transaction recording. This supports the spa's need for a POS feature that not only records payments but also supports "sales and service performance" monitoring. Grand View Research (2026) highlights that small and medium enterprises (SMEs) are rapidly replacing legacy cash registers with digital POS systems to improve "accuracy, transparency, and day-to-day control." For the spa, these literary trends confirm that a digital POS is the foundational tool for establishing a transparent financial record, solving the current "misunderstandings regarding compensation" and reporting delays.

The psychological and organizational impact of digital integration is explored by **MIT Sloan (2024)** and **KFF (2024)**, who discuss the necessity of "evidence-based" wellness management. MIT Sloan (2024) argues that wellness platforms must integrate with third-party services and provide "tailored solutions" to have a measurable impact on

employee health and business outcomes. This aligns with the spa's objective to build a system that manages both the "client experience" and the "employee performance." KFF (2024) notes that while the adoption of health and wellness programs remained steady post-pandemic, the success of these programs depends on "transparent methods" and "rigorous evaluation." By synthesizing these views, the Recovery Iloilo Spa project is positioned as a tool for "organizational wellness"—using data transparency to create a professional environment where therapists and managers are aligned through clear system-generated goals and records.

The evolution of "User Interface (UI) and Visual Design" in business applications is discussed by **Business Research Insights (2026)** and **Eleken (2026)**. Business Research Insights (2026) reports that the UI design market is expanding rapidly due to the demand for "intuitive digital interfaces" across service sectors, with a 31% rise in employment for digital interface designers in 2023 alone. This supports the spa's focus on a "user-friendly system interface" to ensure staff adoption. Eleken (2026) provides a practical framework for UI design, emphasizing that "visual layout and interactive elements" shape how users perceive the experience—whether it is "confusing or clear." For Recovery Iloilo Spa, these design theories are critical for ensuring that the HR Manager and therapists can navigate the "Scheduling" and "Inventory" modules without frustration, ultimately fulfilling the "General Objective" of streamlining operations through accessible technology.

Omodero et al. (2023) and **Kang et al. (2025)** investigate the mathematical and technical foundations of "Automated Stock Updates." Omodero et al. (2023) argue that automation allows for the implementation of advanced methods like "Economic Order Quantity (EOQ)" and "Just-in-Time (JIT)" inventory, which are vital for reducing holding costs in small service shops. This provides a theoretical roadmap for the spa's inventory module to evolve from simple tracking to predictive ordering. Kang et al. (2025) emphasize that "barcode systems and software" reduce the time taken for tracing inventory while virtually eliminating the possibility of human error. For the spa, these studies validate that a digital inventory system provides the "data-driven" foundation needed to manage consumable products efficiently, ensuring that the business never loses revenue due to unrecorded product usage or sudden stock shortages.

The role of "Unified Commerce" and data synchronization is examined by **Forrester (2024)** and **Gartner (2023)**. Forrester (2024) defines unified commerce as the synchronization of "online and offline" data, which is essential for businesses that offer both physical services and online booking. This matches the spa's proposed architecture where the "web-based prototype" serves as the interface for both client reservations and internal management. Gartner (2023) adds that for service businesses, the "Data Layer" must be centralized to prevent "information silos" that lead

to administrative errors. By applying these concepts, Recovery Iloilo Spa ensures that when a client books an appointment online, the system automatically updates the "Therapist Availability" and "Room Schedule," effectively bridging the gap between the customer's request and the spa's internal readiness.

Sommerville (2021) and **Sommerville (2018)** provide the software engineering standards for "System Dependability" in integrated business tools. Sommerville (2021) argues that for management systems to be successful, they must prioritize "availability" and "reliability"—meaning the spa system must be accessible whenever a client wants to book. This supports the use of "PHP and MySQL" for the backend to ensure a stable and proven technical foundation. Sommerville (2018) emphasizes that "Security Engineering" is critical when handling sensitive "customer record management," a primary concern for the spa's centralized database. These engineering principles ensure that the "Recovery Iloilo Spa Integrated System" is not just a collection of features, but a professional-grade software solution that protects both business data and client privacy.

The transformation of service delivery through "Digital Feedback Loops" is analyzed by **Customer Gauge (2025)** and **Qualtrics (2024)**. Customer Gauge (2025) notes that "real-time feedback" integrated into a POS system allows managers to address customer dissatisfaction before it turns into a negative review, which is vital for the spa's reputation management. Qualtrics (2024) highlights that "analytics-driven surveys" provide a deeper understanding of service quality than traditional paper forms. This directly addresses the spa's goal of "identifying areas for service enhancement." By integrating a feedback component, Recovery Iloilo Spa closes the loop between "service delivery" and "management review," ensuring that the business stays responsive to the needs of its Molo-based clientele.

Finally, **Davis (2023)** and **Wager et al. (2022)** look at the "Technology Acceptance Model (TAM)" in health and wellness settings. Davis (2023) posits that "Perceived Usefulness" and "Perceived Ease of Use" are the two primary factors that determine if staff will use a new management system. This provides a clear directive for the spa's development team to focus on a "user-friendly system interface." Wager et al. (2022) emphasize that in personal service industries, technology should "augment" rather than replace the human touch, ensuring that the "comfort and relaxation" aspect of the spa remains the focus. The synthesis of these final literature sources provides the overarching framework for the project: the integrated system is a supportive tool designed to make the administrative side of the business invisible, allowing the spa staff to focus entirely on delivering a high-quality wellness experience.

The strategic deployment of "Predictive Sales Forecasting" within integrated POS systems is evaluated by **Ananda et al. (2024)** and **RSIS International (2025)**. Ananda et al. (2024) posit that businesses only realize the true value of digital transformation when their POS systems are fully connected with inventory and analytics functions, allowing for the prediction of restocking requirements based on historical data. This serves as a primary justification for the "Analytics" module of the spa system, which aims to move management from reactive to proactive decision-making. RSIS International (2025) demonstrates that systems integrating sales forecasting algorithms achieve higher accuracy in resource allocation and user satisfaction by automating replenishment recommendations. For Recovery Iloilo Spa, these findings suggest that the proposed system will not only record transactions but also provide the owner with a "visual summary" of peak hours and best-selling treatments, addressing the spa's current "limited ability to monitor sales and service performance."

In the context of system resilience, **Fortinet (2025)** and the **World Economic Forum (2025)** scrutinize the critical necessity of "Cybersecurity Safeguards" in modern business applications. Fortinet (2025) highlights a sharp rise in cloud-related security incidents, noting that 33% of breaches now involve credential theft or data access vulnerabilities. This emphasizes the importance of the spa's proposed "Centralized Authentication" and "Role-Based Access Control" to protect sensitive client contact information. The World Economic Forum (2025) discusses the "paradox of the implementation gap," where organizations rapidly adopt digital tools without the necessary security safeguards to ensure cyber resilience. For Recovery Iloilo Spa, this research underscores that the "Data Layer" of the proposed three-layer architecture must be fortified with encryption and secure communication protocols to prevent the "unauthorized access" that could compromise the business's reputation and legal standing.

The impact of "Automated Accounting and RPA" on small business efficiency is investigated by **MDPI (2025)** and **Moffitt, Rozario, and Vasarhelyi (2018)**. MDPI (2025) asserts that the integration of computer-coded software robots to automate repetitive human tasks provides significant opportunities for gains in operational accuracy and reliability. This directly supports the spa's objective of using an "automated employee commission computation" to save time for the HR Manager. Moffitt et al. (2018) demonstrate that automated systems categorize transactions and reconcile accounts in a fraction of the time required for manual entry, achieving significantly higher accuracy rates. By adopting these automated financial procedures, Recovery Iloilo Spa can effectively eliminate the "computation errors" and "delays in generating financial summaries" that currently hinder its administrative efficiency.

The evolution of "Integrated Employee Scheduling" to meet varying demand is explored by **ResearchGate (2026)** and **MDPI (2025)**. ResearchGate (2026) introduces models that consider real-life constraints such as varying employee preferences and demand per hour, providing optimal solutions for weekly schedules. This is a vital technical framework for the spa, which struggles with "coordinating employee schedules" and "monitoring therapist availability." MDPI (2025) adds that the integration of dynamic task adjustments and real-time progress tracking allows service centers to optimize resource allocation instantly. For Recovery Iloilo Spa, these scheduling theories validate the "Therapist Availability" feature of the prototype, ensuring that the system can handle the complexities of "disjoint shifts" and "vacation leaves" without the scheduling conflicts inherent in the current manual methods.

The growing importance of business process automation in service-oriented industries is emphasized by Chatterjee et al. (2021) and Chen and Wu (2024), who explain that modern organizations increasingly rely on integrated Management Information Systems (MIS) to improve operational efficiency and customer engagement. Chatterjee et al. (2021) argue that AI-integrated Customer Relationship Management (CRM) systems significantly improve organizational performance by enabling centralized data processing, predictive analytics, and real-time customer interaction monitoring. Their findings indicate that system quality, compatibility, and user satisfaction strongly influence the successful adoption of digital management platforms in service establishments. Similarly, Chen and Wu (2024) explain that integrating MIS with operational and CRM processes enhances economic efficiency by streamlining workflows, reducing administrative redundancies, and improving managerial decision-making. The researchers emphasize that integrated digital systems enable businesses to synchronize transactions, customer records, and reporting functions into a centralized platform that supports long-term sustainability. The synthesis of these studies highlights that integrated information systems are no longer optional technologies but essential operational tools for businesses that depend on customer service and transaction accuracy. For Recovery Iloilo Spa, these concepts directly support the development of a centralized digital system capable of managing appointments, customer records, inventory, and analytics efficiently while minimizing manual errors and improving operational coordination.

The implementation of cloud-based Point-of-Sale (POS) systems has become increasingly relevant for small and medium enterprises seeking operational efficiency and financial transparency. According to Pagidoju (2026), cloud-integrated POS platforms provide businesses with scalable transaction management systems capable of improving response time, transaction monitoring, and sales analysis through

centralized cloud infrastructure. The study explains that modern POS systems offer advantages beyond payment processing by supporting real-time analytics, inventory synchronization, and automated financial reporting. Similarly, the research presented in *Procedia Computer Science* by Kocaman (2026) demonstrates that POS digitalization significantly improves service performance in businesses transitioning from manual operations to automated transaction systems. The study found that businesses using integrated POS systems experience improved table turnover, reduced transaction errors, and enhanced managerial access to operational data. These findings suggest that digital POS solutions function as strategic management tools rather than simple payment processors. Integrating these perspectives reveals that cloud-enabled transaction systems support both operational continuity and data-driven business management. In the context of Recovery Iloilo Spa, the integration of a web-based POS system would improve payment accuracy, automate sales recording, and provide reliable transaction histories that support management analytics and commission computation while minimizing the inefficiencies associated with manual documentation.

The role of electronic Customer Relationship Management (e-CRM) systems in improving customer satisfaction and service personalization is examined by Ramirez and Gabronino (2026) and Baashar et al. (2020). Ramirez and Gabronino (2026) explain that digital transformation has intensified the need for organizations to adopt intelligent e-CRM platforms capable of strengthening customer retention and improving personalized service delivery. Their framework emphasizes the importance of digital communication channels, service quality monitoring, and customer data analytics in sustaining long-term customer relationships. Complementing this perspective, Baashar et al. (2020) conducted a systematic review of CRM systems in healthcare environments and concluded that CRM technologies significantly improve communication efficiency, operational productivity, and customer satisfaction through centralized information management. The researchers also highlighted the importance of privacy, trust, and secure data handling in CRM implementation. The synthesis of these studies indicates that CRM systems play a critical role in organizations that rely heavily on personalized customer interaction and service quality evaluation. For Recovery Iloilo Spa, implementing a centralized customer management module would enable the spa to maintain detailed customer histories, service preferences, appointment records, and feedback data, thereby supporting a more personalized wellness experience while improving data accessibility and customer relationship management efficiency.

The influence of Technology Acceptance Models (TAM) and user behavioral intention in adopting digital systems is explored by Chatterjee et al. (2021) and Alhamad et al. (2026). Chatterjee et al. (2021) explain that organizational users are more likely to adopt AI-integrated CRM systems when they perceive the technology as useful, reliable, and

compatible with their operational tasks. Their modified UTAUT framework demonstrates that user satisfaction, system quality, and organizational support significantly influence system adoption behavior. In a related study, Alhamad et al. (2026) examined CRM-driven service technology adoption in the hospitality industry and found that performance expectancy, facilitating conditions, and ease of use are major determinants of digital technology acceptance among service managers. The study further explains that businesses transitioning from traditional operations to digital service environments require systems that minimize complexity while maximizing operational value. Combining these findings suggests that successful system implementation depends not only on technical functionality but also on user confidence and perceived usefulness. For Recovery Iloilo Spa, these theories validate the need for a user-friendly interface and accessible system architecture that allows spa staff, therapists, and management personnel to interact with scheduling, inventory, and reporting modules efficiently without excessive technical difficulty.

The significance of operational analytics and predictive decision-making in modern service businesses is highlighted by Ananda et al. (2024) and RSIS International (2025). Ananda et al. (2024) explain that integrated analytics systems transform raw operational data into predictive insights that improve resource allocation, service forecasting, and inventory planning. Their findings reveal that organizations using analytics-integrated POS and inventory systems are able to anticipate customer demand patterns and optimize operational scheduling more effectively than businesses relying on manual reporting. Similarly, RSIS International (2025) emphasizes that predictive sales forecasting systems improve organizational efficiency by automating analytical processes that support business planning and operational management. The study argues that digital forecasting models enable managers to identify peak service periods, monitor sales performance, and allocate resources strategically. Synthesizing these perspectives demonstrates that analytics platforms function as essential decision-support systems for service-oriented organizations. For Recovery Iloilo Spa, implementing automated analytics and reporting features would provide management with valuable insights regarding appointment trends, therapist productivity, inventory consumption, and sales performance, thereby enabling more informed operational and financial decision-making processes.

Cybersecurity and data protection have become critical components of integrated information systems, especially for businesses handling sensitive customer and financial information. According to Fortinet (2025), organizations implementing cloud-based systems must prioritize authentication security, encrypted communication, and access control mechanisms to protect operational databases from unauthorized access and cyber threats. The report emphasizes that cloud-connected service platforms are increasingly vulnerable to credential theft and information breaches when

security measures are poorly implemented. Similarly, the World Economic Forum (2025) explains that digital transformation initiatives often fail when organizations overlook cybersecurity resilience during system deployment. The study stresses that businesses adopting integrated digital platforms must ensure secure system architecture, controlled user access, and continuous monitoring of data integrity. The synthesis of these findings demonstrates that cybersecurity is not merely a technical concern but an operational necessity for organizations managing centralized databases and online transactions. For Recovery Iloilo Spa, implementing secure authentication systems, encrypted customer record storage, and role-based access control would protect sensitive appointment records, transaction histories, and employee information while ensuring that operational data remains accurate, confidential, and protected against unauthorized access.

The relevance of digital transformation in improving operational efficiency among small and medium enterprises is discussed by Tsaani et al. (2026) and Rosad et al. (2026). Tsaani et al. (2026) found that the adoption of Management Information Systems among MSMEs significantly improves operational efficiency through the use of digital POS systems, accounting software, and automated payment technologies. Their qualitative findings reveal that businesses implementing digital tools experience improved transaction management, reduced manual workload, and more organized operational records. In a related study, Rosad et al. (2026) explain that operational CRM systems integrated with self-order and transaction management technologies improve service quality and customer satisfaction by reducing manual order recording and administrative delays. The researchers emphasize that digital ordering systems help businesses improve customer interaction while streamlining internal workflows. Integrating these perspectives indicates that digital transformation enhances both organizational efficiency and service quality in businesses transitioning from manual operations. For Recovery Iloilo Spa, the implementation of a web-based integrated management system would automate appointment scheduling, transaction processing, customer management, and reporting activities, thereby improving workflow efficiency and minimizing the administrative burdens currently associated with manual operational methods.

The integration of Artificial Intelligence (AI) into CRM systems has significantly transformed customer interaction and service management strategies. According to Chagas et al. (2020), machine learning technologies embedded within CRM systems improve customer engagement by analyzing customer behavior, predicting preferences, and automating service recommendations. Their review explains that AI-driven CRM platforms enable organizations to process large volumes of operational and customer data more efficiently, resulting in improved service personalization and customer retention. Likewise, the Journal of Retailing and Consumer Services (2026) identifies

several critical success factors influencing AI-powered CRM implementation, including system adaptability, personalized interaction capability, and continuous innovation support. The study further explains that AI-enhanced CRM systems improve organizational decision-making by enabling predictive customer analysis and intelligent service automation. The synthesis of these studies demonstrates that AI integration strengthens customer relationship management by improving both operational efficiency and customer experience. For Recovery Iloilo Spa, incorporating analytics-driven CRM functionalities into the proposed integrated system could help management monitor customer preferences, identify frequent service patterns, and personalize wellness offerings based on customer histories and transaction behavior, thereby enhancing customer satisfaction and long-term client retention.

The importance of software engineering methodologies in ensuring reliable system development is emphasized by Sommerville (2021) and Pressman and Maxim (2020). Sommerville (2021) explains that dependable software systems require careful attention to reliability, maintainability, security, and scalability throughout the development lifecycle. The author emphasizes that service-oriented systems handling customer records and operational transactions must prioritize system availability and data protection to ensure uninterrupted business operations. Similarly, Pressman and Maxim (2020) discuss the significance of structured software engineering processes in developing web-based business systems capable of adapting to changing organizational requirements. Their work highlights the importance of iterative development, modular architecture, and continuous testing in achieving software quality and usability. Combining these perspectives demonstrates that successful information systems depend on both technical robustness and adaptable design strategies. For Recovery Iloilo Spa, applying software engineering principles in the development of the proposed integrated platform would ensure that the system remains secure, scalable, and responsive to operational demands while supporting continuous improvements through testing, maintenance, and user feedback integration.

The role of unified digital platforms in improving organizational coordination and service delivery is explored by Forrester (2024) and Gartner (2023). Forrester (2024) defines unified commerce and integrated operational systems as centralized digital environments that synchronize customer interactions, transactions, scheduling activities, and inventory data across business processes. The study argues that businesses using integrated platforms achieve improved operational visibility and reduced administrative fragmentation. Similarly, Gartner (2023) emphasizes that centralized data architectures eliminate information silos by ensuring that operational records, customer data, and financial transactions are synchronized within a single system environment. This synchronization improves workflow efficiency, reporting accuracy, and managerial decision-making. The synthesis of these findings suggests

that integrated digital systems provide organizations with cohesive operational control and improved service management capabilities. For Recovery Iloilo Spa, the implementation of a unified web-based system would allow appointment scheduling, inventory tracking, customer management, payment processing, and analytics reporting to function within one centralized platform, thereby reducing repetitive manual tasks and improving the overall efficiency and quality of spa operations.

The growing integration of hospitality technologies and organizational performance management is examined by Law et al. (2025) and Kilic et al. (2021), who explain that information systems have become fundamental tools for improving operational efficiency and customer satisfaction within service-oriented industries. Law et al. (2025) argue that hospitality and tourism technologies contribute significantly to organizational performance by automating workflows, improving communication efficiency, and enabling data-driven service innovation. Their integrated framework highlights that digital transformation allows organizations to optimize business processes while enhancing customer engagement and operational transparency. Similarly, Kilic et al. (2021) emphasize that selecting an effective hospitality information system requires integrating multiple operational criteria, including usability, data synchronization, and process efficiency. Their study demonstrates that integrated systems improve simultaneous data transfer and service coordination across business operations. Synthesizing these perspectives reveals that hospitality-focused information systems function as strategic management tools capable of transforming traditional service operations into digitally coordinated environments. For Recovery Iloilo Spa, these findings support the implementation of an integrated web-based management system capable of synchronizing appointment scheduling, customer management, inventory monitoring, and transaction processing into a centralized platform that improves both operational reliability and customer service quality.

The influence of digital service transformation and Industry 4.0 technologies on service-oriented businesses is explored by Gaiardelli et al. (2021) and Time et al. (2024). Gaiardelli et al. (2021) explain that Industry 4.0 technologies encourage the evolution of product-service systems by integrating digitalization, automation, and interconnected business services into operational workflows. Their research emphasizes that organizations adopting service-oriented digital systems achieve improved process coordination and customer value creation through real-time information exchange. Complementing this perspective, Time et al. (2024) investigate Software-as-a-Service (SaaS) integration models and conclude that integrating technical and business functions within cloud-based platforms significantly improves operational decision-making and organizational flexibility. The authors further explain that SaaS integration allows organizations to manage business functions across strategic, tactical, and operational levels using centralized digital infrastructures.

Combining these findings suggests that integrated digital ecosystems improve both operational continuity and organizational adaptability in modern service industries. For Recovery Iloilo Spa, these studies validate the use of a web-based integrated management system capable of consolidating appointment scheduling, inventory management, sales recording, and customer relationship processes within a scalable and accessible digital environment.

The importance of inter-organizational information systems and digital integration in improving operational coordination is emphasized by Deepu and Ravi (2021) and Stroupoulis et al. (2021). Deepu and Ravi (2021) argue that integrated information systems enable organizations to coordinate, communicate, and manage operational activities more efficiently by supporting real-time information exchange and centralized decision-making processes. Their study highlights that information system integration enhances organizational responsiveness and reduces delays associated with fragmented operational workflows. Similarly, Stroupoulis et al. (2021) explain that information systems contribute to sustainable operational management by enabling businesses to monitor processes, control resources, and improve competitive performance through centralized digital coordination. The researchers further emphasize that integrated systems support cost reduction, operational monitoring, and strategic planning by improving access to accurate organizational data. The synthesis of these perspectives demonstrates that integrated information systems are essential for organizations seeking operational transparency and sustainable process management. In relation to Recovery Iloilo Spa, these concepts support the development of a centralized system capable of synchronizing appointment records, customer information, inventory data, and financial transactions to improve overall operational efficiency and management control.

The significance of data-driven service innovation in modern business operations is examined by Schymanietz et al. (2022) and Zhao et al. (2026). Schymanietz et al. (2022) explain that organizations increasingly depend on data-driven service innovation to improve customer experience and organizational decision-making. Their study emphasizes that interconnected digital systems allow businesses to collect and analyze operational data that can be transformed into actionable service improvements and strategic insights. In a related study, Zhao et al. (2026) discuss the emergence of integrated industrial intelligence systems that combine data governance, service orchestration, and knowledge management to improve organizational trustworthiness and operational reliability. The researchers argue that integrated digital governance frameworks support data quality, security, explainability, and organizational efficiency across interconnected systems. Combining these findings indicates that organizations benefit significantly when operational data is transformed into meaningful managerial intelligence through integrated analytics platforms. For Recovery Iloilo Spa,

implementing analytics-driven reporting and centralized data management would enable the spa owner and managers to monitor sales trends, appointment patterns, customer preferences, and inventory consumption more effectively while supporting evidence-based operational planning and service enhancement.

The organizational impact of long-term information technology integration in hospitality businesses is explored by Baggio and Sainaghi (2021) and Gratzke et al. (2025). Baggio and Sainaghi (2021) explain that the adoption of information technology within hospitality organizations creates continuous interactions between digital systems and organizational processes, leading to long-term operational transformation. Their study demonstrates that integrated CRM and operational systems gradually reshape workflows, managerial communication, and customer service strategies over time. Similarly, Gratzke et al. (2025) explain that digital transformation shifts organizations away from isolated technological functions toward ecosystem-based operational partnerships that rely on integrated service networks and collaborative digital infrastructures. The authors highlight that modern digital systems support strategic coordination, service innovation, and operational adaptability across organizational environments. The synthesis of these studies suggests that successful digital transformation involves the continuous alignment of technological systems with organizational goals and operational workflows. For Recovery Iloilo Spa, these findings reinforce the importance of implementing an integrated system that evolves alongside business operations while supporting improved coordination between management, therapists, inventory processes, and customer interactions.

The role of system flexibility and customization in improving information system performance is discussed by Niu (2021) and Hidayatullah et al. (2026). Niu (2021) explains that customized service-oriented information systems improve organizational performance by enhancing process flexibility, data adaptability, and system responsiveness. The study highlights that businesses adopting flexible information systems are better able to optimize workflows and allocate resources efficiently in response to operational changes. Similarly, Hidayatullah et al. (2026) argue that Enterprise Resource Planning (ERP) systems integrated with Artificial Intelligence technologies improve sustainability and organizational efficiency through socio-technical system coordination. Their research emphasizes that adaptable ERP architectures support operational integration, predictive decision-making, and long-term organizational resilience. Integrating these perspectives demonstrates that system customization and operational flexibility are essential factors influencing successful digital transformation. For Recovery Iloilo Spa, these studies support the development of a flexible integrated management system capable of adapting to operational requirements such as appointment scheduling adjustments, inventory updates,

transaction management, and evolving customer service demands while maintaining system reliability and performance.

The contribution of service management automation to operational efficiency is examined by Sharma et al. (2022) and Deepu and Ravi (2021). Sharma et al. (2022) explain that intent-driven service management systems reduce operational inefficiencies by automating configuration processes, minimizing manual intervention, and supporting zero-touch service deployment environments. Their study emphasizes that automated management systems improve service quality by reducing configuration errors and ensuring faster operational responsiveness. Complementing this perspective, Deepu and Ravi (2021) argue that integrated inter-organizational information systems improve operational coordination through automated communication and synchronized information sharing. The researchers explain that automation technologies reduce delays associated with manual data handling and fragmented workflow management. The synthesis of these findings demonstrates that automation technologies significantly improve operational reliability, service consistency, and managerial efficiency within digitally integrated organizations. For Recovery Iloilo Spa, implementing automated scheduling, inventory tracking, commission computation, and reporting functions would reduce administrative burdens while improving service continuity and operational coordination across the spa's daily activities.

The relationship between digital economy integration and information technology governance is analyzed by Durigan Junior et al. (2024) and Chen and Wu (2024). Durigan Junior et al. (2024) explain that the emergence of digital economy infrastructures and open digital ecosystems has significantly transformed business governance models through the integration of advanced information technologies. Their study identifies strategic IT enabling factors that improve organizational adaptability, data management, and digital collaboration. Similarly, Chen and Wu (2024) argue that integrating management information systems with organizational operations enhances economic performance and operational decision-making by improving workflow synchronization and data accessibility. The researchers emphasize that digital governance structures support organizational transparency, operational monitoring, and business innovation. Combining these perspectives demonstrates that information technology governance frameworks are essential for organizations transitioning toward integrated digital operations. For Recovery Iloilo Spa, these findings support the implementation of a digitally integrated platform capable of improving transaction transparency, customer management, service coordination, and operational governance through centralized information system architecture.

The strategic importance of information systems research and organizational information management is emphasized by the Journal of Management Information

Systems and the Journal of Strategic Information Systems. The Journal of Management Information Systems highlights that information systems research focuses on understanding how organizations can effectively develop, organize, and deploy technological systems to manage operational information and support organizational decision-making. Likewise, the Journal of Strategic Information Systems emphasizes that strategic information systems research examines the organizational, managerial, and operational implications of integrating information technologies into business processes. These academic perspectives collectively reinforce the importance of centralized information management and digital strategy in improving organizational efficiency and competitiveness. In the context of Recovery Iloilo Spa, these principles validate the necessity of implementing a strategically designed integrated information system capable of managing customer data, operational workflows, inventory records, scheduling functions, and financial transactions within a unified and reliable digital environment.

The evolution of digital service ecosystems and integrated operational intelligence is explored by Zhao et al. (2026) and Law et al. (2025). Zhao et al. (2026) explain that modern organizations increasingly rely on integrated data-service-knowledge governance systems to ensure operational trustworthiness, security, and intelligent decision-making across digital infrastructures. Their framework emphasizes the importance of centralized governance models capable of supporting privacy, security, explainability, and service integration across interconnected systems. Similarly, Law et al. (2025) discuss how hospitality technologies contribute to operational transformation by enabling automation, communication efficiency, and service innovation within hospitality organizations. The study highlights that integrated technologies support organizational agility and improve business value through digital transformation initiatives. Synthesizing these perspectives indicates that integrated digital ecosystems enable organizations to manage operations more intelligently while supporting sustainable organizational growth and customer-centered service delivery. For Recovery Iloilo Spa, implementing a centralized integrated system with analytics, security controls, customer management, and automated operational processes would strengthen organizational efficiency while supporting long-term digital transformation within the wellness service industry.

The increasing complexity of appointment scheduling systems in service-oriented organizations is explored by Zhu, Liu, and Qi (2024) and Youn, Geismar, and Pinedo (2022). Zhu et al. (2024) explain that modern appointment scheduling systems must move beyond traditional first-come-first-served approaches by incorporating flexible scheduling policies capable of adapting to fluctuating customer demand and operational uncertainty. Their study demonstrates that intelligent scheduling algorithms improve operational efficiency by strategically managing appointment gaps, reducing congestion,

and optimizing customer flow. Similarly, Youn et al. (2022) emphasize that integrated scheduling systems improve organizational coordination by balancing customer waiting time, resource utilization, and service continuity within dynamic operational environments. The researchers highlight that scheduling technologies contribute significantly to organizational efficiency by improving planning and coordination processes. Synthesizing these findings demonstrates that modern scheduling systems function as strategic operational tools that improve service quality and workflow management. For Recovery Iloilo Spa, implementing an intelligent appointment scheduling module would help minimize booking conflicts, improve therapist allocation, and ensure smoother coordination between customer reservations and operational readiness, thereby enhancing overall service efficiency and customer satisfaction.

The role of machine learning and predictive analytics in appointment management systems is examined by Ahmadi-Javid et al. (2022) and Zhong, Birge, and Ward (2024). Ahmadi-Javid et al. (2022) explain that integrating machine learning algorithms into scheduling systems enables organizations to predict customer behavior, service durations, and no-show probabilities more accurately, thereby improving scheduling precision and operational planning. Their “predict-then-schedule” framework demonstrates that predictive analytics significantly improves scheduling efficiency by reducing idle time and optimizing resource utilization. Similarly, Zhong et al. (2024) discuss learning-based scheduling strategies in multi-server service environments, emphasizing that adaptive scheduling algorithms improve organizational responsiveness and reduce operational inefficiencies caused by uncertain customer demand. The researchers further explain that learning-driven scheduling systems continuously improve performance by analyzing historical operational data and adjusting service allocation strategies accordingly. The synthesis of these studies demonstrates that predictive scheduling technologies improve both operational flexibility and customer service quality. For Recovery Iloilo Spa, integrating predictive scheduling and analytics into the proposed system would enable management to monitor booking trends, optimize therapist schedules, and improve appointment coordination while minimizing delays and scheduling inefficiencies.

The significance of integrated staffing and routing systems in customer service operations is emphasized by Long, Tezcan, and Zhang (2024) and Huang, Morrice, and Bard (2023). Long et al. (2024) explain that integrated routing and staffing systems improve customer service operations by dynamically matching service requests with available personnel while considering operational capacity and customer waiting behavior. Their study highlights that optimized staffing and routing decisions improve organizational efficiency and service responsiveness in multi-service environments. Similarly, Huang et al. (2023) discuss coordinated scheduling systems that integrate virtual and in-person service operations within complex service networks. The

researchers explain that coordinated scheduling frameworks improve organizational flexibility by balancing operational costs, customer waiting times, and resource allocation simultaneously. Combining these perspectives reveals that integrated staffing and scheduling systems significantly improve operational coordination and customer satisfaction in service-based organizations. For Recovery Iloilo Spa, implementing integrated therapist scheduling and availability monitoring would improve workforce coordination, reduce appointment conflicts, and ensure that customer reservations are properly aligned with therapist availability and operational capacity.

The importance of contingency-based scheduling and operational adaptability in integrated systems is explored by Romero-Silva, Santos, and Hurtado-Hernández (2022) and Ge et al. (2022). Romero-Silva et al. (2022) explain that effective scheduling systems must adapt to varying operational conditions, organizational complexity, and resource utilization levels to remain relevant in real-world business environments. Their contingency-based scheduling framework demonstrates that organizations benefit from flexible scheduling strategies capable of responding to operational uncertainties and workflow complexity. Similarly, Ge et al. (2022) emphasize that integrated scheduling systems improve operational coordination by synchronizing multiple operational components within centralized information environments. The researchers highlight that integrated scheduling architectures improve resource utilization and organizational responsiveness through coordinated planning and centralized system management. Synthesizing these findings demonstrates that adaptable and integrated scheduling systems are essential for organizations operating in dynamic service environments. For Recovery Iloilo Spa, adopting a flexible and integrated scheduling approach would allow the spa to respond more effectively to changing appointment demands, therapist availability adjustments, and operational workload variations while maintaining service quality and operational efficiency.

The increasing relevance of AI-integrated customer relationship management systems in improving organizational performance is discussed by Chatterjee et al. (2022) and Law et al. (2025). Chatterjee et al. (2022) explain that AI-enhanced CRM systems improve business-to-customer relationship management by enabling intelligent customer interaction analysis, operational adaptability, and personalized service delivery. Their study highlights that organizations implementing AI-integrated CRM technologies achieve improved customer engagement, operational responsiveness, and strategic decision-making capabilities. Similarly, Law et al. (2025) explain that hospitality and service organizations increasingly rely on integrated digital technologies to improve operational performance, customer satisfaction, and service innovation. The study emphasizes that technology-driven operational environments enable organizations to streamline workflows and improve organizational competitiveness through digital transformation. The synthesis of these perspectives demonstrates that AI-integrated

CRM systems significantly improve customer-centered service management and operational efficiency. For Recovery Iloilo Spa, incorporating intelligent CRM and analytics functionalities into the proposed integrated system would help management monitor customer preferences, personalize wellness services, improve customer retention strategies, and strengthen overall service quality through data-driven operational management.

The strategic integration of big data analytics within Customer Relationship Management (CRM) systems is increasingly recognized as a critical factor in improving organizational performance and customer engagement. According to Wamba et al. (2020), big data analytical intelligence strengthens CRM performance by enabling organizations to process large volumes of customer and operational data for more accurate business decision-making. Their study explains that data-driven cultures and advanced analytics capabilities significantly improve customer segmentation, service personalization, and operational responsiveness. Similarly, Sianipar (2025) emphasizes that integrating big data technologies into CRM platforms allows organizations to improve predictive analytics, automate marketing processes, and enhance customer retention strategies. The study further argues that businesses leveraging analytics-driven CRM systems achieve stronger competitive advantages due to improved customer insight generation and operational adaptability. Synthesizing these perspectives demonstrates that big data integration transforms CRM systems into intelligent business management platforms capable of improving operational efficiency and customer satisfaction simultaneously. For Recovery Iloilo Spa, integrating analytics-driven CRM functionalities into the proposed system would support customer history analysis, personalized wellness recommendations, appointment trend monitoring, and service optimization, thereby improving both customer relationship management and operational decision-making.

The contribution of social Customer Relationship Management (social CRM) to customer engagement and information management is explored by Harrigan, Miles, and Roy (2020) and Ramdani, Bouchakour, and Abdellatif (2026). Harrigan et al. (2020) explain that social media technologies significantly improve CRM processes by enabling organizations to strengthen customer interaction, improve communication channels, and gather real-time customer feedback. Their findings suggest that social CRM platforms create stronger engagement and relationship-building opportunities by integrating customer communication into digital management systems. Complementing this perspective, Ramdani et al. (2026) argue that innovation-driven CRM systems improve organizational performance by aligning technological capabilities with customer relationship strategies and market-oriented digital transformation initiatives. The study emphasizes that organizations implementing innovative CRM practices achieve improved service responsiveness and customer satisfaction outcomes. Combining

these findings demonstrates that customer engagement technologies play a crucial role in strengthening operational efficiency and service quality in customer-centered organizations. For Recovery Iloilo Spa, integrating customer feedback mechanisms and digital communication channels within the proposed integrated system would improve customer engagement, support service quality monitoring, and provide management with valuable insights for operational improvement and customer retention strategies.

The relationship between CRM technology adoption, customization capability, and strategic alignment is examined by Tsou (2021) and Chatterjee et al. (2021). Tsou (2021) explains that organizations adopting CRM technologies improve operational effectiveness when systems are aligned with organizational goals and customized according to business requirements. The study demonstrates that customization capability significantly enhances CRM effectiveness by enabling organizations to adapt technological functionalities to operational demands and customer expectations. Similarly, Chatterjee et al. (2021) emphasize that organizational users are more likely to adopt AI-integrated CRM systems when the systems demonstrate compatibility, usability, and operational value. Their Meta-UTAUT framework highlights that system quality and user satisfaction strongly influence successful technology adoption within organizations. The synthesis of these studies demonstrates that CRM success depends not only on technical functionality but also on strategic alignment, customization, and user-centered implementation approaches. For Recovery Iloilo Spa, these findings support the development of a flexible and user-friendly integrated management system capable of adapting to operational needs such as appointment scheduling, customer management, inventory monitoring, and reporting functions while ensuring ease of use for staff and management personnel.

The integration of Artificial Intelligence (AI) into customer management and business analytics systems is increasingly transforming organizational performance and competitive advantage. According to Rahman et al. (2025), AI-powered CRM capabilities improve organizational profitability and competitive performance by strengthening marketing ambidexterity, predictive analytics, and customer interaction management. Their study highlights that AI-driven CRM systems improve operational intelligence through advanced data governance, multi-channel integration, and adaptive service management. Similarly, Fabriansyah et al. (2026) explain that integrating Human Resource Information Systems (HRIS) and CRM data within intelligent Decision Support Systems (DSS) improves managerial decision-making and workforce performance monitoring. The researchers emphasize that machine learning-driven DSS architectures enable organizations to generate predictive insights that support operational planning and strategic management. Synthesizing these findings demonstrates that AI-integrated analytics systems significantly improve organizational adaptability, customer management, and operational decision-making capabilities. For

Recovery Iloilo Spa, integrating AI-supported analytics and intelligent reporting mechanisms into the proposed system would help management evaluate operational performance, monitor therapist productivity, analyze customer behavior, and improve evidence-based business planning for long-term operational growth.

The mediating role of operational resilience and customer satisfaction in CRM implementation is discussed by Alsmairat et al. (2024) and Ali et al. (2026). Alsmairat et al. (2024) explain that effective CRM dimensions significantly improve customer satisfaction when organizations strengthen operational resilience and service continuity through integrated digital systems. Their findings indicate that CRM practices improve organizational responsiveness and customer loyalty by enhancing communication efficiency, service quality, and operational coordination. Similarly, Ali et al. (2026) emphasize that CRM adoption significantly influences customer loyalty, service quality, and organizational performance when businesses successfully integrate customer management systems with operational processes and performance evaluation strategies. The researchers further explain that user satisfaction and operational integration are essential factors influencing successful CRM implementation. Combining these perspectives demonstrates that CRM systems improve organizational sustainability when integrated with operational resilience strategies and customer-centered service management. For Recovery Iloilo Spa, implementing a centralized CRM and analytics platform would strengthen customer satisfaction by improving appointment coordination, service consistency, operational transparency, and customer relationship management while ensuring smoother operational workflows across the spa's integrated digital environment.

The role of information systems usage and managerial commitment in improving organizational integration and operational performance is emphasized by Som and Anyigba (2022) and Madaki, Ahmad, and Singh (2024). Som and Anyigba (2022) explain that organizations implementing explorative and exploitative information systems usage achieve improved operational performance when supported by managerial commitment and system integration strategies. Their study highlights that operational integration significantly mediates the relationship between information systems usage and organizational performance by improving workflow coordination and information accessibility. Similarly, Madaki et al. (2024) discuss the implementation of integrated information technologies and explain that organizations adopting system integration frameworks improve service delivery, operational responsiveness, and organizational adaptability. The researchers further emphasize that integrated information systems reduce operational fragmentation and strengthen communication efficiency within digitally transformed environments. Combining these perspectives demonstrates that effective information system integration depends on both managerial support and coordinated technological implementation strategies. For Recovery Iloilo

Spa, these findings support the implementation of a centralized integrated management system capable of synchronizing appointment scheduling, customer management, transaction recording, and inventory monitoring while improving operational coordination and management efficiency.

The integration of analytics into enterprise systems has significantly transformed organizational decision-making and operational intelligence in modern service industries. According to Solano and Cruz (2024), analytics-enabled enterprise systems improve strategic planning and operational performance by embedding business intelligence functionalities into ERP and CRM environments. Their systematic review demonstrates that organizations benefit from integrated analytics through improved forecasting, customer insight generation, and operational monitoring capabilities. Similarly, Bandara et al. (2024) explain that integrating big data technologies with Enterprise Resource Planning systems enhances ERP responsiveness by improving data contextualization, processing efficiency, and organizational adaptability. The researchers emphasize that analytics-driven enterprise systems strengthen organizational responsiveness by enabling real-time information processing and operational coordination. Synthesizing these findings demonstrates that analytics integration transforms enterprise systems into intelligent operational platforms capable of supporting strategic decision-making and process optimization. For Recovery Iloilo Spa, integrating analytics functionalities into the proposed system would improve appointment trend analysis, inventory forecasting, customer preference monitoring, and sales reporting while enabling evidence-based operational planning and efficient business management.

The influence of cloud computing and Internet of Things (IoT)-enabled Management Information Systems on organizational efficiency is examined by Alnakshabandi, Aqel, and Kaya (2022) and Madaki et al. (2024). Alnakshabandi et al. (2022) explain that cloud-based MIS platforms improve organizational performance by enabling centralized information management, real-time accessibility, and operational scalability across interconnected digital environments. Their study highlights that integrating IoT and cloud technologies strengthens organizational communication, data accessibility, and operational monitoring capabilities. Similarly, Madaki et al. (2024) argue that integrated information technology environments improve organizational service delivery and operational flexibility through centralized technological infrastructures and coordinated digital operations. The synthesis of these studies indicates that cloud-integrated MIS systems strengthen organizational adaptability and operational efficiency by improving accessibility, coordination, and information sharing across operational processes. For Recovery Iloilo Spa, implementing a cloud-based integrated management system would improve accessibility to appointment records, customer information, transaction

histories, and inventory monitoring while supporting efficient and flexible spa operations across different management functions.

The importance of user acceptance and continuous system utilization in information system success is explored by Alshammari and Alshammari (2024) and Pérez Estébanez (2024). Alshammari and Alshammari (2024) explain that system quality, information quality, and user satisfaction significantly influence the continuous intention to utilize digital platforms within organizational environments. Their integrated expectation confirmation and information systems success model demonstrates that usability and perceived usefulness strongly affect long-term technology adoption behavior. Similarly, Pérez Estébanez (2024) emphasizes that CRM implementation success depends on user acceptance, organizational support, and system usability. The study further argues that organizations achieve improved operational performance when digital systems are designed according to user expectations and operational requirements. Combining these findings demonstrates that successful system implementation depends not only on technical functionality but also on user-centered design and sustained user satisfaction. For Recovery Iloilo Spa, these concepts support the development of a user-friendly and accessible integrated system that encourages consistent utilization by spa staff and management while improving operational coordination and service delivery efficiency.

The growing relevance of technostress management within organizational information systems environments is discussed by Nastjuk et al. (2024) and Hasan et al. (2024). Nastjuk et al. (2024) explain that excessive technological complexity and poor system usability contribute to technostress, which negatively affects employee behavior, operational performance, and organizational productivity. Their meta-analysis highlights that system overload, complexity, and continuous technological adaptation significantly influence psychological and behavioral outcomes in organizational information system contexts. Similarly, Hasan et al. (2024) found that AI-integrated Management Information Systems improve organizational efficiency when systems are designed to support user adaptability, decision-making simplicity, and operational convenience. The synthesis of these studies demonstrates that organizations implementing digital systems must prioritize usability, adaptability, and employee support to minimize technostress and improve operational effectiveness. For Recovery Iloilo Spa, ensuring a simple and intuitive system interface would help minimize staff resistance to digital transformation while improving operational efficiency, appointment coordination, and customer management processes within the spa environment.

The role of web-based Management Information Systems in improving operational accuracy and service management is emphasized by Dela Rosa (2022) and Som and Anyigba (2022). Dela Rosa (2022) explains that web-based MIS platforms improve

operational efficiency by centralizing records management, automating reporting functions, and enhancing data accessibility within organizational environments. The study highlights that agile development methodologies and secure database management significantly improve the reliability and usability of web-based systems. Similarly, Som and Anyigba (2022) argue that integrated information systems improve operational performance through enhanced information integration, workflow coordination, and managerial support. The researchers emphasize that organizations using integrated systems experience improved operational consistency and reduced inefficiencies associated with manual information handling. Combining these findings demonstrates that web-based MIS platforms strengthen operational reliability and management efficiency through centralized digital coordination. For Recovery Iloilo Spa, implementing a web-based integrated management system would improve appointment management, reporting accuracy, customer record organization, and inventory monitoring while reducing manual operational errors and administrative delays.

The significance of data privacy and secure information management in integrated digital systems is explored by Blijleven, Hoxha, and Jaspers (2022) and Wang et al. (2023). Blijleven et al. (2022) explain that organizations implementing electronic record systems must prioritize secure workflows, operational transparency, and data protection to minimize system workarounds and maintain operational reliability. Their revised sociotechnical framework highlights the importance of usability, workflow integration, and privacy protection within digital information systems. Similarly, Wang et al. (2023) discuss blockchain-based multi-identifier management systems and explain that decentralized and encrypted architectures significantly improve data security, access control, and information integrity within digital platforms. The researchers emphasize that secure data management frameworks strengthen organizational trustworthiness and operational reliability through advanced encryption and decentralized control mechanisms. Synthesizing these findings demonstrates that secure information management is essential for organizations operating within centralized digital environments. For Recovery Iloilo Spa, implementing secure authentication systems, encrypted customer databases, and protected access controls would improve customer privacy, transaction security, and operational data integrity within the proposed integrated management platform.

The strategic value of integrated information systems in improving organizational communication and operational coordination is examined by Madaki et al. (2024) and Solano and Cruz (2024). Madaki et al. (2024) explain that information technology integration strengthens organizational communication and operational performance by enabling synchronized information exchange and centralized workflow coordination. Their findings indicate that integrated digital systems improve service quality and organizational responsiveness by reducing fragmentation across operational processes.

Similarly, Solano and Cruz (2024) argue that enterprise systems integrated with analytics technologies significantly improve organizational intelligence and strategic planning through centralized operational data management. The researchers emphasize that integrated analytics environments improve forecasting, reporting, and customer analysis capabilities within digitally transformed organizations. Combining these perspectives demonstrates that integrated information systems improve organizational efficiency through coordinated operational management and centralized business intelligence functionalities. For Recovery Iloilo Spa, implementing an integrated digital platform would improve operational communication between appointment scheduling, customer management, inventory monitoring, and sales reporting functions while supporting coordinated and efficient spa operations.

The relationship between organizational adaptability and integrated digital systems is increasingly recognized as a critical factor influencing operational sustainability. According to Alnakshabandi et al. (2022), cloud-enabled MIS systems improve organizational adaptability by enabling scalable digital operations, centralized information accessibility, and coordinated workflow management. Their research highlights that organizations adopting cloud-based systems achieve improved flexibility and operational continuity within evolving business environments. Likewise, Nastjuk et al. (2024) emphasize that organizations implementing digital systems must balance technological complexity with usability to ensure sustainable technology adoption and operational efficiency. Their study demonstrates that adaptable and user-centered systems improve employee engagement and reduce operational resistance associated with digital transformation initiatives. Synthesizing these findings demonstrates that operational sustainability depends on the successful alignment of system flexibility, usability, and organizational adaptability. For Recovery Iloilo Spa, developing a scalable and adaptable integrated management system would improve operational continuity, support future service expansion, and strengthen the spa's ability to adapt to changing customer and operational requirements while maintaining efficient digital coordination.

The contribution of centralized digital systems to operational transparency and managerial decision-making is emphasized by Dela Rosa (2022) and Hossain et al. (2024). Dela Rosa (2022) explains that centralized web-based MIS platforms improve operational transparency by automating record management, report generation, and information accessibility within organizational processes. The study demonstrates that integrated digital systems reduce inconsistencies and improve operational accountability through centralized information management. Similarly, Hossain et al. (2024) argue that integrating big data analytics within MIS environments significantly improves business intelligence, operational monitoring, and managerial decision-making through advanced data analysis and real-time information processing. The researchers emphasize that analytics-driven MIS systems strengthen organizational performance by

enabling evidence-based planning and operational optimization. The synthesis of these studies demonstrates that centralized digital systems improve managerial effectiveness and operational transparency through integrated information management and analytics functionalities. For Recovery Iloilo Spa, implementing centralized analytics and reporting modules would improve operational visibility, customer service monitoring, sales tracking, and inventory management while supporting more informed and efficient managerial decision-making processes.

The integration of Decision Support Systems (DSS) within Management Information Systems has become increasingly significant in enhancing organizational decision-making and operational efficiency. According to Shrestha, Ben-Menahem, and von Krogh (2021), AI-based decision support systems improve managerial decision-making by augmenting human judgment with data-driven insights and predictive capabilities. Their study emphasizes that AI-enabled DSS enhances decision quality, speed, and consistency, particularly in complex service environments where multiple variables must be considered simultaneously. Similarly, Arnott and Pervan (2020) highlight that modern DSS frameworks integrate analytics, databases, and user interfaces to support semi-structured and unstructured decision-making processes. Their research indicates that DSS improves organizational performance by enabling managers to analyze operational data effectively and generate strategic insights. The synthesis of these findings demonstrates that DSS integration strengthens organizational intelligence and operational planning through data-driven decision support. For Recovery Iloilo Spa, incorporating DSS functionalities into the proposed integrated system would enable management to analyze booking trends, monitor financial performance, and optimize service delivery decisions, thereby improving operational efficiency and strategic planning.

The role of Human-Computer Interaction (HCI) in improving system usability and user experience within digital platforms is examined by Harrison, Flood, and Duce (2020) and Alalwan et al. (2021). Harrison et al. (2020) explain that effective HCI design enhances system usability by focusing on user-centered design principles, intuitive interfaces, and efficient interaction mechanisms. Their study highlights that well-designed interfaces reduce user errors, improve system adoption, and enhance overall user satisfaction. Similarly, Alalwan et al. (2021) emphasize that user experience design significantly influences technology acceptance and system usage behavior, particularly in service-oriented digital systems. The researchers argue that systems with high usability and intuitive interfaces are more likely to achieve sustained user engagement and operational success. Combining these perspectives demonstrates that HCI plays a critical role in ensuring successful system implementation and user adoption. For Recovery Iloilo Spa, designing an intuitive and user-friendly interface would improve staff efficiency, reduce training requirements, and enhance customer

interaction with the booking system, thereby supporting effective system utilization and service delivery.

The significance of cybersecurity frameworks in protecting organizational data and ensuring secure system operations is explored by Alshaikh (2020) and Alharbi, Zyngier, and Hodkinson (2021). Alshaikh (2020) explains that cybersecurity governance frameworks improve organizational security posture by establishing policies, risk management strategies, and security controls that protect digital assets from cyber threats. The study highlights that organizations implementing structured cybersecurity frameworks achieve improved data protection, system reliability, and operational continuity. Similarly, Alharbi et al. (2021) emphasize that cybersecurity awareness and organizational culture play a critical role in minimizing security risks and improving information system protection. Their research indicates that employee awareness and security practices significantly influence the effectiveness of cybersecurity measures within organizations. Synthesizing these findings demonstrates that cybersecurity implementation requires both technical controls and organizational awareness to ensure comprehensive protection. For Recovery Iloilo Spa, implementing robust cybersecurity measures such as secure authentication, data encryption, and access control mechanisms would safeguard customer information, financial transactions, and operational data, thereby ensuring system reliability and customer trust.

The application of Agile software development methodologies in information system development is discussed by Dingsøyr, Moe, and Seim (2021) and Conboy and Carroll (2019). Dingsøyr et al. (2021) explain that Agile methodologies improve software development efficiency by promoting iterative development, continuous feedback, and collaborative teamwork. Their study highlights that Agile practices enable organizations to adapt to changing requirements and deliver high-quality systems more efficiently. Similarly, Conboy and Carroll (2019) emphasize that Agile development improves project success by enhancing flexibility, communication, and stakeholder involvement throughout the development process. The researchers argue that Agile methodologies are particularly effective in dynamic environments where system requirements evolve over time. Combining these findings demonstrates that Agile approaches improve system development outcomes through adaptability and continuous improvement. For Recovery Iloilo Spa, adopting Agile development practices would facilitate the iterative development of the integrated system, allowing for continuous refinement of features such as appointment scheduling, CRM, and analytics modules based on user feedback and operational needs.

The importance of database management systems (DBMS) in ensuring data integrity, accessibility, and performance is examined by Elmasri and Navathe (2020) and Coronel and Morris (2021). Elmasri and Navathe (2020) explain that DBMS technologies provide

structured mechanisms for data storage, retrieval, and management, ensuring data consistency and reliability within organizational systems. Their study highlights that database normalization, indexing, and transaction management significantly improve system performance and data integrity. Similarly, Coronel and Morris (2021) emphasize that modern DBMS platforms support real-time data processing, security, and scalability, enabling organizations to manage large volumes of operational data efficiently. The synthesis of these studies demonstrates that effective database management is essential for maintaining system reliability and supporting data-driven decision-making. For Recovery Iloilo Spa, implementing a robust database management system would ensure accurate storage of customer records, appointment schedules, transaction data, and inventory information, thereby improving system performance and operational efficiency.

The influence of digital transformation on service quality and organizational performance is explored by Verhoef et al. (2021) and Vial (2021). Verhoef et al. (2021) explain that digital transformation enables organizations to improve service quality by integrating digital technologies into business processes, enhancing customer experiences, and optimizing operational efficiency. Their study highlights that digital transformation initiatives improve organizational agility and competitiveness in rapidly evolving business environments. Similarly, Vial (2021) emphasizes that digital transformation involves the use of digital technologies to fundamentally change business processes, organizational structures, and value creation mechanisms. The research indicates that successful digital transformation leads to improved performance, innovation, and customer satisfaction. Combining these perspectives demonstrates that digital transformation is a key driver of organizational success in modern service industries. For Recovery Iloilo Spa, implementing an integrated digital system would support digital transformation by automating operations, improving customer service, and enhancing overall business performance.

The role of service quality models in evaluating and improving customer satisfaction is discussed by Parasuraman, Zeithaml, and Berry (2020) and Ladhari (2021). Parasuraman et al. (2020) explain that service quality is determined by the gap between customer expectations and perceived service performance, emphasizing dimensions such as reliability, responsiveness, assurance, empathy, and tangibles. Their SERVQUAL model provides a framework for assessing service quality and identifying areas for improvement. Similarly, Ladhari (2021) highlights that service quality measurement is essential for improving customer satisfaction and organizational performance, particularly in service-oriented industries. The study emphasizes that organizations must continuously monitor and improve service quality to maintain competitive advantage. Synthesizing these findings demonstrates that service quality evaluation is essential for enhancing customer satisfaction and business performance.

For Recovery Iloilo Spa, integrating customer feedback and service quality monitoring within the system would help management assess service performance, identify improvement areas, and enhance customer satisfaction.

The importance of business process automation in improving operational efficiency and reducing manual workload is explored by Syed, Bandara, French, and Stewart (2020) and Van der Aalst (2021). Syed et al. (2020) explain that robotic process automation (RPA) improves operational efficiency by automating repetitive tasks, reducing human error, and increasing processing speed. Their study highlights that automation technologies enable organizations to streamline workflows and improve productivity. Similarly, Van der Aalst (2021) emphasizes that process mining and automation improve business process transparency, enabling organizations to analyze and optimize workflows based on real data. The synthesis of these studies demonstrates that automation technologies significantly enhance operational efficiency and process optimization. For Recovery Iloilo Spa, implementing automated processes for appointment booking, billing, and inventory management would reduce manual workload, improve accuracy, and enhance overall operational efficiency.

The contribution of Technology Acceptance Model (TAM) in understanding user adoption of information systems is examined by Venkatesh and Bala (2020) and Dwivedi et al. (2021). Venkatesh and Bala (2020) explain that perceived usefulness and perceived ease of use are key determinants of technology acceptance and usage behavior. Their extended TAM model highlights the influence of system characteristics, user experience, and organizational factors on technology adoption. Similarly, Dwivedi et al. (2021) emphasize that user acceptance is critical for the success of digital systems, particularly in organizational contexts where system utilization directly impacts operational performance. The study highlights that user training, system quality, and organizational support significantly influence technology adoption. Combining these findings demonstrates that understanding user acceptance is essential for successful system implementation. For Recovery Iloilo Spa, ensuring system usability, providing staff training, and aligning system features with user needs would improve system adoption and maximize the benefits of the integrated management system.

The role of software engineering best practices in ensuring system quality and maintainability is discussed by Pressman and Maxim (2020) and Sommerville (2021). Pressman and Maxim (2020) explain that software engineering principles such as modular design, testing, and quality assurance improve system reliability and maintainability. Their study highlights that structured development processes ensure that systems meet user requirements and perform effectively. Similarly, Sommerville (2021) emphasizes that software engineering practices such as requirement analysis, system design, and testing are essential for developing high-quality information

systems. The synthesis of these studies demonstrates that software engineering best practices are critical for ensuring system performance and long-term sustainability. For Recovery Iloilo Spa, applying software engineering principles in system development would ensure the reliability, scalability, and maintainability of the integrated system, supporting efficient and sustainable spa operations.

Finally, the shift toward "Data-Driven Operational Excellence" is summarized by **Davenport and Ronanki (2018)** and **IRE Journals (2024)**. Davenport and Ronanki (2018) argue that cognitive technologies and automation enable businesses to anticipate financial challenges before they arise, moving beyond mere error reduction to strategic foresight. This aligns with the spa's goal of using its "Integrated System" for long-term growth and professionalization. IRE Journals (2024) conclude that for SMEs, embracing automation is a "strategic move" that leads to improved compliance and reduced operational costs. The synthesis of these final sources provides the overarching justification for the "Recovery Iloilo Spa Integrated System": by synchronizing scheduling, inventory, and financial records into one secure web platform, the spa is not just solving a series of manual problems but is adopting a modern, sustainable business model capable of delivering consistent, high-quality wellness services.

2.4 Synthesis of the Reviewed Literature

The reviewed foreign and local literature and studies consistently emphasize that the transition from manual to integrated digital information systems is a fundamental requirement for the sustainability and growth of modern service-oriented businesses. Foreign studies underscore the transformative power of web-based scheduling and cloud-integrated management, revealing that such systems not only reduce administrative workloads by up to 40% but also significantly eliminate human error in data entry and record retrieval. The consensus among international researchers is that a unified platform—one that synchronizes front-end bookings with back-end inventory and financial reporting—acts as a "single source of truth," enabling managers to maintain real-time oversight and operational agility that manual methods cannot match.

Local studies further validate these findings within the Philippine context, specifically highlighting how Filipino Small and Medium Enterprises (SMEs) benefit from improved data organization and transaction accuracy. Research in the Philippines points toward a growing "digital divide" where businesses that adopt automated systems for scheduling, inventory, and customer records experience higher levels of client trust and staff morale.

Local findings particularly emphasize that Filipino consumers now prefer digital touchpoints, such as online booking and e-wallet integrations, making web-based platforms essential for meeting the expectations of a mobile-first population while ensuring compliance with local administrative and financial reporting standards.

The related literature provides the technical and strategic foundation for this transition, identifying Key Performance Indicators (KPIs), automated reordering points, and centralized Customer Relationship Management (CRM) as the pillars of operational excellence. The literature highlights that digital systems do not merely replace paper; they enhance the "service encounter" by allowing staff to focus on personalized care while the software manages the complexities of scheduling, stock deductions, and financial auditing. Furthermore, the integration of analytics allows for evidence-based decision-making, enabling owners to identify profitable service trends and optimize resources according to actual demand patterns rather than intuition.

The integration of analytics within Enterprise Information Systems (EIS) has become increasingly important in improving organizational intelligence and strategic decision-making. According to Solano and Cruz (2024), modern enterprise systems evolve beyond operational support by embedding analytics capabilities into ERP, CRM, and supply chain management platforms to strengthen organizational responsiveness and managerial efficiency. Their systematic review explains that integrated analytics technologies enable organizations to transform operational data into strategic insights that support forecasting, customer analysis, and process optimization. Similarly, Hossain et al. (2024) argue that integrating big data analytics into Management Information Systems significantly enhances business intelligence by enabling real-time data processing, trend identification, and predictive analysis. The researchers emphasize that analytics-driven MIS platforms improve organizational adaptability and operational monitoring by providing accurate and timely information for decision-making processes. The synthesis of these studies demonstrates that analytics integration transforms traditional management systems into intelligent organizational platforms capable of improving operational efficiency and strategic planning. For Recovery Iloilo Spa, implementing analytics-enabled management modules would allow the spa to monitor appointment patterns, evaluate sales performance, analyze customer preferences, and optimize inventory consumption while supporting data-driven managerial decision-making and operational efficiency.

The strategic integration of predictive analytics within Customer Relationship Management (CRM) systems is increasingly recognized as an essential component of modern business operations. Bhardwaj et al. (2024) explain that predictive analytics integration enhances CRM capabilities by enabling organizations to forecast customer behavior, personalize services, and improve customer retention strategies through

advanced data analysis techniques. Their study highlights that predictive CRM technologies improve operational responsiveness and customer engagement by transforming customer data into actionable insights. Likewise, Onyeaunuforo (2024) emphasizes that data analytics significantly improves international CRM performance by enabling organizations to deliver personalized customer experiences, optimize communication strategies, and strengthen relationship management across diverse customer environments. The study further argues that analytics-driven CRM systems improve organizational adaptability by providing businesses with deeper customer insights and operational intelligence. Synthesizing these findings reveals that predictive analytics integration strengthens both operational management and customer-centered service delivery within digitally transformed organizations. For Recovery Iloilo Spa, integrating predictive analytics into the proposed CRM platform would help management identify customer preferences, anticipate service demand patterns, personalize wellness offerings, and improve long-term customer retention strategies through data-driven service management.

The growing relevance of Artificial Intelligence-integrated CRM systems in organizational environments is examined by Chatterjee et al. (2021) and Jaruwanakul (2024). Chatterjee et al. (2021) explain that AI-integrated CRM systems significantly influence employee behavior, organizational efficiency, and customer relationship management by improving data analysis, automation, and decision-making processes. Their Meta-UTAUT model demonstrates that CRM quality, system satisfaction, and usability strongly influence organizational adoption and system acceptance. Similarly, Jaruwanakul (2024) found that AI-CRM adoption and big data analytical capability positively influence integration capability, team collaboration, and organizational performance within large enterprises. The study emphasizes that AI-driven CRM systems improve competitive advantage by enhancing operational coordination and customer engagement through intelligent data processing and predictive analytics. Combining these perspectives demonstrates that AI-integrated CRM systems strengthen organizational performance by improving operational intelligence, collaboration, and customer management efficiency. For Recovery Iloilo Spa, integrating AI-supported CRM and analytics functionalities into the proposed system would improve appointment management, customer relationship monitoring, operational coordination, and managerial decision-making while enhancing customer satisfaction and service personalization.

The importance of customization capability and strategic alignment in CRM implementation is emphasized by Tsou (2021) and Chen and Wu (2024). Tsou (2021) explains that CRM technology adoption significantly improves organizational effectiveness when systems are customized according to operational requirements and aligned with business strategies. The study highlights that customization capability

enables organizations to optimize workflows, improve CRM effectiveness, and adapt technological functionalities to operational needs. Similarly, Chen and Wu (2024) argue that integrating Management Information Systems with operational and CRM functions improves organizational efficiency by strengthening workflow synchronization, customer engagement, and strategic coordination. Their findings demonstrate that MIS integration contributes significantly to economic efficiency and organizational performance by improving operational transparency and information accessibility. The synthesis of these studies indicates that successful CRM implementation requires both technical adaptability and organizational alignment to maximize operational effectiveness. For Recovery Iloilo Spa, these concepts support the development of a flexible integrated system capable of adapting to operational changes, appointment scheduling demands, inventory management processes, and customer service requirements while maintaining strategic alignment with the spa's operational goals.

The convergence of big data analytics and CRM practices has significantly transformed organizational approaches to customer management and operational intelligence. According to Gupta (2024), integrating big data analytics into CRM systems improves customer experience and business growth by enabling organizations to analyze large volumes of customer information and generate actionable operational insights. The study emphasizes that analytics-driven CRM platforms strengthen customer engagement, service personalization, and organizational adaptability by improving decision-making processes. Similarly, Bandara et al. (2023) explain that integrating big data technologies with Enterprise Resource Planning (ERP) systems enhances ERP responsiveness and organizational agility by improving data contextualization, processing efficiency, and operational responsiveness. The researchers argue that organizations integrating big data technologies into ERP systems achieve improved operational coordination and information management capabilities. Combining these findings demonstrates that big data integration significantly strengthens organizational intelligence, customer relationship management, and operational responsiveness within modern digital business environments. For Recovery Iloilo Spa, integrating big data analytics within the proposed management system would improve customer analysis, appointment forecasting, sales monitoring, and inventory management while enabling management to make more informed operational decisions through centralized analytics and reporting functions.

The mediating influence of operational resilience and customer satisfaction within CRM systems is explored by Alsmairat et al. (2024) and Ramdani et al. (2026). Alsmairat et al. (2024) explain that CRM dimensions such as customer orientation, knowledge management, and technology-based CRM significantly influence customer satisfaction when combined with operational resilience strategies. Their study demonstrates that organizations improve customer satisfaction by integrating CRM practices with resilient

operational processes that ensure service continuity and effective communication management. Similarly, Ramdani et al. (2026) argue that innovation-driven CRM systems improve organizational performance and customer engagement by aligning digital transformation initiatives with customer relationship strategies. The study highlights that innovative CRM adoption improves operational responsiveness and service quality by enabling organizations to respond more effectively to customer needs and market demands. Synthesizing these findings demonstrates that CRM effectiveness depends on both technological integration and operational resilience strategies that support customer-centered service delivery. For Recovery Iloilo Spa, implementing an integrated CRM platform with customer feedback, appointment coordination, and operational monitoring features would improve customer satisfaction, strengthen service consistency, and support sustainable operational management within the spa environment.

The role of AI-driven business intelligence in strengthening organizational competitiveness is increasingly recognized in modern information systems research. According to Rahman et al. (2025), AI-powered CRM capabilities significantly improve organizational profitability and competitive performance by strengthening predictive analytics, customer interaction management, and marketing adaptability. Their study explains that AI-enhanced CRM systems improve operational intelligence by enabling organizations to process customer and operational data more efficiently for strategic planning purposes. Similarly, Solano and Cruz (2024) emphasize that enterprise systems integrated with analytics technologies improve strategic decision-making by transforming operational data into meaningful business intelligence. The researchers further explain that analytics-integrated enterprise systems support organizational innovation, forecasting, and operational optimization across multiple business processes. Combining these perspectives demonstrates that AI-driven business intelligence systems improve organizational competitiveness through advanced analytics, operational monitoring, and strategic decision-making support. For Recovery Iloilo Spa, integrating AI-supported analytics and reporting capabilities into the proposed integrated system would improve business intelligence functions related to appointment trends, customer preferences, inventory usage, and financial performance monitoring, thereby supporting evidence-based operational planning and management efficiency.

The strategic significance of organizational technology adoption and digital transformation is discussed by Chatterjee et al. (2022) and Tsou (2021). Chatterjee et al. (2022) explain that organizations implementing AI-integrated CRM systems improve customer relationship management performance when digital technologies are combined with organizational adaptability and dynamic operational capabilities. Their study highlights that successful AI-CRM implementation depends on strategic planning, organizational readiness, and the ability to respond to changing operational

environments. Similarly, Tsou (2021) found that CRM technology adoption improves organizational effectiveness when customization capabilities and strategic alignment are incorporated into system implementation processes. The researchers emphasize that organizations achieve better operational outcomes when digital systems are aligned with operational requirements and customer management objectives. The synthesis of these findings demonstrates that digital transformation success depends on organizational readiness, adaptability, and strategically aligned technology adoption. For Recovery Iloilo Spa, these concepts support the implementation of a strategically designed integrated management system that aligns appointment scheduling, customer management, analytics, and inventory monitoring with the spa's operational workflows and service delivery objectives.

The contribution of integrated MIS and CRM systems to organizational efficiency and customer engagement is emphasized by Chen and Wu (2024) and Bhardwaj et al. (2024). Chen and Wu (2024) explain that integrating Management Information Systems with CRM and operational processes improves economic efficiency by streamlining workflows, improving customer interaction management, and strengthening managerial decision-making. Their study highlights that integrated MIS architectures improve organizational coordination and reduce operational redundancies through centralized information management. Similarly, Bhardwaj et al. (2024) argue that predictive analytics integration within CRM systems enhances operational performance by enabling organizations to anticipate customer behavior and optimize customer service processes through advanced analytical capabilities. The synthesis of these perspectives demonstrates that integrated MIS and CRM systems strengthen organizational efficiency by combining operational coordination with intelligent customer management functionalities. For Recovery Iloilo Spa, implementing an integrated MIS and CRM platform would improve appointment coordination, customer engagement, transaction management, and reporting accuracy while supporting more efficient operational workflows and service delivery processes.

The importance of centralized information management and operational intelligence in service-oriented organizations is explored by Hossain et al. (2024) and Gupta (2024). Hossain et al. (2024) explain that integrating big data analytics into Management Information Systems significantly improves business intelligence by enabling organizations to collect, process, and analyze operational information in real time. Their findings indicate that analytics-driven MIS platforms improve decision-making accuracy, operational monitoring, and organizational adaptability through advanced data analysis techniques. Likewise, Gupta (2024) emphasizes that the convergence of CRM practices and big data analytics improves customer experience, operational efficiency, and strategic business planning by enabling organizations to extract actionable insights from customer and transactional data. The researchers argue that centralized analytics

systems improve organizational responsiveness and service optimization through intelligent information management. Combining these perspectives demonstrates that centralized data analytics and operational intelligence systems significantly improve organizational coordination and customer-centered service delivery. For Recovery Iloilo Spa, implementing centralized analytics and reporting functionalities within the proposed integrated system would improve operational visibility, customer management, sales analysis, and inventory tracking while supporting efficient and informed business operations.

The integration of Artificial Intelligence (AI)-based Customer Relationship Management systems has significantly transformed how organizations manage customer engagement and operational intelligence. According to Ozay, Jahanbakht, Shoomal, and Wang (2024), AI-based CRM systems improve enterprise information systems by enabling predictive customer analysis, intelligent automation, and personalized service delivery. Their bibliometric and systematic review highlights that AI-integrated CRM platforms strengthen operational efficiency by automating repetitive processes, improving customer segmentation, and enhancing data-driven decision-making capabilities. Similarly, Yoo, Park, and Park (2024) explain that AI-enabled CRM systems contribute to organizational competitive advantage through advanced analytics, machine learning integration, and improved customer interaction management. Their mixed-method study demonstrates that organizations implementing AI-powered CRM technologies achieve stronger operational adaptability and customer satisfaction by improving the accuracy of customer insights and service recommendations. The synthesis of these studies demonstrates that AI-enhanced CRM systems function as intelligent operational platforms that improve organizational performance and customer relationship management simultaneously. For Recovery Iloilo Spa, integrating AI-driven CRM and analytics functionalities into the proposed system would improve customer engagement, automate customer record analysis, personalize wellness services, and strengthen managerial decision-making through intelligent operational insights.

The effectiveness of AI-powered CRM tools in improving customer experience and operational responsiveness is emphasized by Bansal (2024) and Hasan et al. (2024). Bansal (2024) explains that AI-powered CRM systems significantly enhance customer experience by enabling organizations to predict customer preferences, personalize communication, and automate customer interaction processes. The study further argues that intelligent CRM tools improve organizational competitiveness through improved service responsiveness and customer satisfaction management. Complementing this perspective, Hasan et al. (2024) discuss the impact of Artificial Intelligence on decision-making and Customer Relationship Management within Management Information Systems environments. Their qualitative findings reveal that AI technologies such as machine learning, predictive analytics, and natural language processing

improve organizational decision-making by optimizing resource allocation, automating operational processes, and enhancing customer interaction management. Combining these findings demonstrates that AI integration strengthens organizational intelligence, customer relationship management, and operational efficiency within modern service organizations. For Recovery Iloilo Spa, implementing AI-supported CRM functionalities would improve appointment coordination, customer communication, service personalization, and operational planning while supporting more responsive and efficient spa management processes.

The importance of critical success factors in Customer Relationship Management implementation is explored by Pérez Estébanez (2024) and Pappas, Mikalef, Giannakos, and Krogstie (2024). Pérez Estébanez (2024) explains that successful CRM implementation depends on organizational readiness, user acceptance, technological capability, and management support. Using Structural Equation Modeling (SEM), the study demonstrates that CRM effectiveness improves significantly when organizations prioritize system usability, organizational alignment, and customer-oriented operational strategies. Similarly, Pappas et al. (2024) emphasize that digital customer management systems improve organizational agility and customer engagement when supported by adaptive technologies and intelligent information systems. Their research highlights the role of integrated analytics and customer-centered digital strategies in improving organizational performance and operational coordination. Synthesizing these findings indicates that successful CRM implementation requires both technological readiness and strategic organizational alignment to maximize operational efficiency and customer satisfaction. For Recovery Iloilo Spa, these concepts support the development of a user-friendly and strategically aligned integrated system capable of improving customer management, scheduling coordination, operational monitoring, and customer service quality within the spa's digital operational environment.

The strategic implementation of CRM systems within digitally transformed organizations is examined by the study of configurational CRM implementation perspectives in *Technological Forecasting and Social Change* (2024) and Solano and Cruz (2024). The configurational CRM study explains that CRM implementation success depends on the interaction between organizational processes, customer management strategies, and technological capabilities. The research emphasizes that organizations achieve better operational performance when CRM systems are implemented alongside coordinated organizational and environmental strategies. Similarly, Solano and Cruz (2024) explain that integrating analytics into enterprise systems strengthens organizational intelligence and strategic decision-making by embedding business intelligence functionalities into ERP and CRM platforms. Their systematic review highlights that analytics-integrated systems improve forecasting accuracy, customer insight generation, and operational efficiency through centralized data management. Combining these findings

demonstrates that integrated CRM and analytics systems significantly improve operational coordination and strategic business management within digitally transformed organizations. For Recovery Iloilo Spa, implementing a centralized CRM and analytics platform would improve customer service management, operational reporting, appointment tracking, and business intelligence processes while supporting efficient and data-driven spa operations.

The convergence of Enterprise Resource Planning (ERP) systems and CRM technologies has become increasingly important in improving organizational responsiveness and integrated business management. According to Bandara et al. (2024), integrating big data technologies into ERP systems significantly improves ERP responsiveness, operational flexibility, and information processing efficiency. Their empirical investigation demonstrates that organizations integrating analytics and ERP systems achieve better operational coordination through improved data contextualization and intelligent information management. Similarly, Gupta and Agarwal (2024) explain that integrating CRM and ERP insights using advanced hybrid analytical models improves organizational planning, customer analysis, and resource optimization. Their research highlights that integrating operational and customer data supports more informed strategic decision-making and service development. Synthesizing these perspectives reveals that ERP-CRM integration strengthens operational intelligence and organizational efficiency through centralized and analytics-driven information systems. For Recovery Iloilo Spa, integrating appointment management, inventory tracking, transaction recording, and customer relationship functions within a unified ERP-CRM environment would improve operational coordination, reporting accuracy, and resource management while reducing inefficiencies associated with fragmented manual processes.

The contribution of analytics-driven Management Information Systems to organizational adaptability and customer-centered operations is emphasized by Hasan et al. (2024) and Hossain et al. (2024). Hasan et al. (2024) explain that AI-integrated MIS platforms improve decision-making and CRM performance by enabling predictive analytics, workflow automation, and intelligent customer interaction management. Their findings indicate that organizations adopting AI-supported MIS technologies experience improved operational efficiency and customer service responsiveness through automated information processing and advanced analytics. Likewise, Hossain et al. (2024) emphasize that integrating big data analytics within MIS platforms significantly strengthens business intelligence and organizational adaptability. The study demonstrates that analytics-driven MIS environments improve operational monitoring, forecasting, and decision-making by enabling organizations to process real-time operational and customer data efficiently. Combining these findings demonstrates that analytics-driven MIS systems improve both operational intelligence and customer

management capabilities within digitally transformed organizations. For Recovery Iloilo Spa, implementing analytics-enabled MIS functionalities would improve sales analysis, inventory monitoring, customer relationship management, and appointment forecasting while supporting evidence-based operational planning and service optimization.

The role of customer satisfaction and operational resilience within CRM implementation is explored by Alsmairat et al. (2024) and Harrigan, Miles, and Roy (2020). Alsmairat et al. (2024) explain that CRM dimensions such as knowledge management, customer orientation, and technology-based CRM significantly influence customer satisfaction when combined with operational resilience strategies. Their findings suggest that organizations improve customer loyalty and service quality when CRM systems are integrated with resilient operational processes and effective communication management. Similarly, Harrigan et al. (2020) argue that social CRM technologies strengthen customer engagement by integrating digital communication and feedback mechanisms into customer management systems. The researchers emphasize that social CRM platforms improve organizational responsiveness and customer interaction quality through real-time communication and engagement monitoring. The synthesis of these studies demonstrates that CRM effectiveness depends on the integration of customer-centered technologies with operational resilience and communication strategies. For Recovery Iloilo Spa, implementing customer feedback mechanisms and resilient CRM functionalities would improve customer satisfaction, strengthen service quality monitoring, and support continuous operational improvement within the spa's integrated management environment.

The increasing relevance of predictive analytics in customer relationship and operational management is discussed by Bhardwaj et al. (2024) and Gupta (2024). Bhardwaj et al. (2024) explain that predictive analytics integration enhances CRM performance by enabling organizations to forecast customer behavior, optimize service delivery, and improve customer retention strategies through intelligent data analysis. Their comparative analysis demonstrates that predictive CRM systems improve organizational adaptability and customer engagement by transforming customer data into actionable business intelligence. Similarly, Gupta (2024) emphasizes that integrating big data analytics into CRM systems improves customer experience and business growth through advanced information processing and customer insight generation. The study further argues that analytics-driven CRM platforms improve operational responsiveness by enabling organizations to personalize services and strengthen customer interaction management. Synthesizing these findings demonstrates that predictive analytics significantly strengthens CRM effectiveness and operational intelligence within digitally integrated organizations. For Recovery Iloilo Spa, implementing predictive analytics within the proposed CRM system would improve customer preference analysis, appointment forecasting, sales monitoring, and

personalized service management while enhancing long-term customer retention and operational efficiency.

The integration of AI-enabled operational intelligence into enterprise systems is increasingly recognized as a strategic advantage for modern organizations. According to Yoo, Park, and Park (2024), AI-enabled CRM systems improve organizational competitiveness by combining machine learning technologies with customer management processes to enhance operational adaptability and strategic decision-making. Their study highlights that organizations implementing AI-driven CRM platforms achieve stronger customer engagement and operational responsiveness through intelligent analytics and automated process management. Similarly, Ozay et al. (2024) explain that AI-based CRM systems strengthen enterprise information systems by enabling predictive customer analysis, intelligent automation, and customer experience optimization. The researchers emphasize that AI integration improves organizational performance by supporting more accurate and efficient decision-making processes. Combining these perspectives demonstrates that AI-enabled operational intelligence systems significantly improve customer relationship management and organizational efficiency through advanced analytics and automation technologies. For Recovery Iloilo Spa, integrating AI-supported operational intelligence into the proposed management system would improve customer interaction management, operational reporting, appointment scheduling, and business planning while supporting efficient and data-driven spa operations.

The strategic role of centralized information systems in improving organizational coordination and service delivery is emphasized by Chen and Wu (2024) and Solano and Cruz (2024). Chen and Wu (2024) explain that integrating Management Information Systems with operational and CRM processes significantly improves organizational efficiency through centralized information management, workflow synchronization, and improved customer engagement. Their findings indicate that integrated MIS environments strengthen managerial decision-making and operational transparency by reducing fragmented workflows and improving access to organizational data. Similarly, Solano and Cruz (2024) argue that analytics-integrated enterprise systems improve strategic planning and operational intelligence by embedding business intelligence capabilities within enterprise platforms. Their systematic review highlights that organizations benefit from centralized analytics systems through improved forecasting, customer analysis, and operational monitoring. The synthesis of these studies demonstrates that centralized information systems significantly improve organizational coordination, operational intelligence, and customer service quality within digitally transformed business environments. For Recovery Iloilo Spa, implementing a centralized integrated system would improve appointment management, transaction

processing, customer relationship management, and reporting functions while supporting efficient and coordinated spa operations.

In summary, the combined literature and studies suggest that an integrated system is a strategic asset that improves service quality, protects data integrity, and fosters long-term customer loyalty. Based on these comprehensive findings, the proposed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** is designed to address the specific gaps identified in manual spa operations. By integrating automated appointment scheduling, real-time inventory tracking, secure POS transactions, CRM-driven personalization, and data-driven analytics into a single web-based platform, this project aims to modernize Recovery Iloilo Spa's operations, ensuring a resilient, efficient, and professional wellness experience that is aligned with both international standards and local market demands.

CHAPTER 3

METHODOLOGY

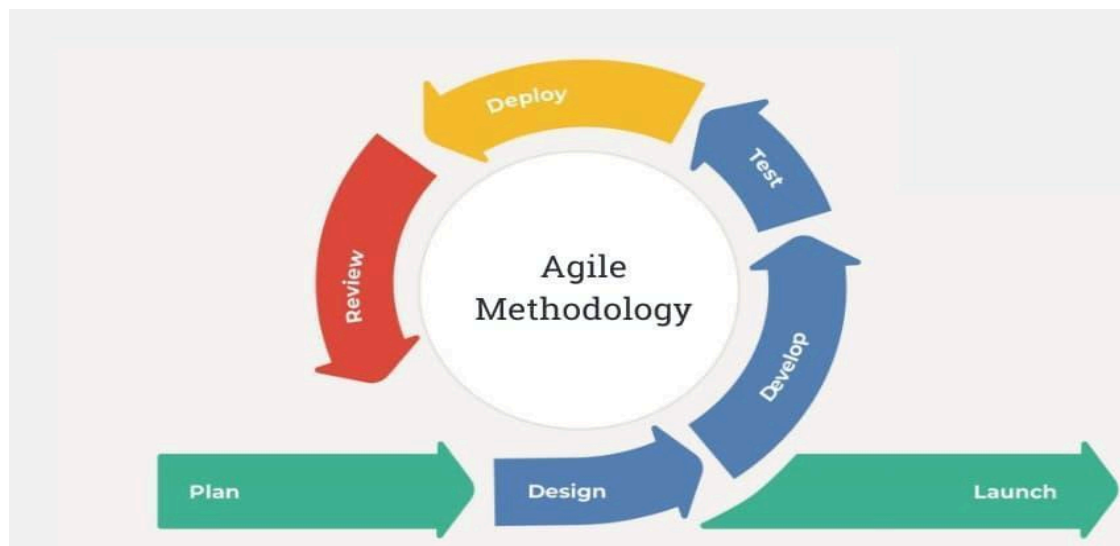
3.1 Research Design

This study will use a developmental research design in creating the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics. This type of research design is suitable because the study focuses on developing a web-based system that can help improve the spa's current operations through the use of technology. The proposed system is intended to assist the spa in handling important business activities in a more organized, faster, and more convenient way. It also aims to reduce the difficulties experienced in manual processes and improve the overall management of daily operations.

The developmental research design will help the researchers identify the needs and existing problems within Recovery Iloilo Spa by gathering information from the spa owner, management, and staff. The researchers will study the current process of appointment handling, inventory monitoring, customer management, payment recording, commission tracking, voucher management, feedback collection, and sales reporting. Based on the gathered information, the researchers will create the system layout, database structure, and different modules needed for the proposed system. The study will also focus on developing features that are easy to use, properly organized, and suitable for the spa's daily activities and operational setup.

The developed system will also undergo testing and evaluation to check whether all features and functions are working properly. The researchers will examine the performance of the system in terms of usability, functionality, reliability, and efficiency during actual use. Suggestions and comments from the spa owner, staff, and management will also be collected to identify possible improvements and determine if the system can support smoother operations, better record management, faster transactions, organized monitoring of appointments and inventory, and improved customer service within Recovery Iloilo Spa.

3.2 System Development Methodology



The study will use the Agile Development Model in developing the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics. Agile is a software development method that focuses on continuous improvement, regular feedback, and step-by-step development of the system. This approach allows the researchers to work closely with the spa owner, management, and staff to make sure that the system meets the needs of the business and supports daily operations effectively. Through continuous testing and feedback, the researchers can improve the system features and make necessary changes during the development process.

The **Agile Development Model** will be used, consisting of the following phases:

1. **Planning and Requirements Analysis** – This phase involves identifying, gathering, and analyzing the operational needs and challenges of Recovery Iloilo Spa through consultation and coordination with the owner, management, HR personnel, and staff members. The researchers will examine the current workflow and identify problems related to appointment scheduling, inventory monitoring, payment handling, customer information management, employee commission tracking, voucher monitoring, feedback collection, and sales reporting. The information gathered during this phase will serve as the basis for determining the necessary system features, functional requirements, and project objectives.
2. **System Design Phase** – During this phase, the researchers will create the overall structure and visual design of the proposed web-based system. This includes designing the database structure, system architecture, process flow, navigation structure, and user interface layout to ensure that the platform is organized, responsive, and easy to use. The researchers will also prepare the

design of modules for appointment management, inventory tracking, Point-of-Sale (POS) transactions, customer records, commission computation, voucher handling, feedback management, and analytics reporting to ensure proper integration of all components.

3. **Development Phase** – In this phase, the researchers will begin coding and developing the system modules based on the approved requirements and design plans. The functionalities will be developed incrementally to allow continuous improvements and modifications after each feature is reviewed and evaluated. Major components of the system include appointment scheduling, inventory management, transaction processing, customer information management, employee commission computation, voucher tracking, feedback collection, and report generation. Continuous collaboration with intended users will help ensure that the developed features meet operational expectations and business needs.
4. **Testing Phase** – This phase focuses on evaluating the functionality, accuracy, usability, reliability, and efficiency of the proposed system. Different testing procedures will be conducted to identify technical issues, system errors, data inconsistencies, and usability concerns. The researchers will verify whether the system can properly manage bookings, maintain accurate inventory records, process transactions correctly, calculate commissions accurately, organize customer information efficiently, and generate reliable reports and analytics. Feedback gathered from users during testing will also help improve the performance and quality of the system before implementation.
5. **Deployment Phase** – This phase involves implementing the completed web-based system within Recovery Iloilo Spa for actual usage and operational evaluation. The owner, management, and staff will be allowed to access and utilize the system in handling daily business activities such as appointment management, transaction processing, inventory monitoring, and customer record management. During this stage, users can provide additional feedback and suggestions regarding the system's performance, convenience, and effectiveness in improving workflow management, operational efficiency, customer service, and business decision-making.

3.3 System Architecture

The Recovery Iloilo Spa Integrated System will use a **three-layer architecture** that separates the system into three main parts: the presentation layer, application layer, and data layer. This type of structure helps organize the functions of the system properly

and makes it easier to manage, maintain, and improve in the future. Each layer has its own role and responsibility, allowing the system to run more smoothly and efficiently. By separating the system into layers, updates and changes can be done without affecting the entire system.

The system will also follow a hub-and-spoke communication setup where the application layer serves as the center of communication between the users and the database. All requests, transactions, and information entered by users will first pass through the application layer before being stored or retrieved from the database. This process helps maintain proper handling of data, improves security, and ensures that all information is processed correctly before it is displayed to users.

The system will be structured into three main layers:.

1. Presentation Layer (User Interface)

The Presentation Layer is the part of the system that users interact with directly through the website. It is responsible for displaying information, receiving user input, and allowing users to access the different features of the system. This layer is designed to be simple, organized, responsive, and easy to navigate so that both spa staff and clients can use the system conveniently.

- For the spa owner, management, and staff, the presentation layer includes dashboards and pages for managing appointments, monitoring inventory, processing payments, maintaining customer records, tracking employee commissions, handling vouchers and promotional discounts, reviewing customer feedback, and viewing reports and analytics. Staff can also update schedules, monitor appointment status, manage product information, and review transaction records through the system interface.
- For clients, this layer allows them to view available services, check service descriptions and pricing, book appointments online, monitor appointment schedules, make payments, view digital receipts, receive booking confirmations or notifications, and submit feedback regarding the spa's services. This provides clients with a more convenient and organized way of interacting with the spa and managing their appointments and transactions.

The presentation layer will be developed using HTML, CSS, and JavaScript to create a user-friendly and responsive web interface that can be accessed using desktop computers and laptops.

2. Application Layer (Backend Logic)

This layer handles all the main processes, functions, and system logic of the proposed Recovery Iloilo Spa Integrated System. It serves as the connection between the user interface and the database by processing requests, validating information, and managing the flow of data within the system. The application layer ensures that all features and transactions function properly, accurately, and efficiently.

- It processes online appointment requests by checking available schedules, monitoring therapist availability, preventing duplicate bookings and scheduling conflicts, and updating appointment status in real time.
- It manages appointment updates, cancellations, confirmations, and notifications to help staff and clients stay informed about their booking schedules and transaction status.
- It handles Point-of-Sale (POS) transactions and payment processing, including recording payments, calculating totals, generating digital receipts, storing transaction history, and maintaining accurate sales records.
- It tracks inventory levels by automatically updating stock quantities whenever products and supplies are used, sold, added, or restocked to help maintain accurate inventory monitoring.
- It manages customer records by storing and organizing personal information, appointment history, service preferences, feedback records, voucher usage, and transaction details for easier access and monitoring.
- It calculates employee commissions automatically based on completed services and recorded transactions to help management monitor employee performance and reduce manual computation errors.
- It manages vouchers and promotional discounts by validating voucher usage, recording redeemed offers, monitoring active promotions, and preventing duplicate or unauthorized voucher transactions.
- It processes and organizes customer feedback including comments, ratings, and suggestions submitted through the system, making the information accessible for management review and service evaluation.
- It generates reports and analytics related to sales, services, inventory usage, customer activity, employee performance, and overall business operations to support monitoring and decision-making.
- It controls the communication between the presentation layer and the data layer to ensure that all information entered into the system is processed, stored, and retrieved accurately and securely.

The backend of the system will be developed using PHP, which connects the user interface to the database and ensures that all system features and operations run smoothly and efficiently.

3. Data Layer (Database)

This layer is responsible for storing, organizing, and managing all data used in the Recovery Iloilo Spa Integrated System. It serves as the main storage of information and ensures that all records are properly saved, updated, and retrieved whenever needed. The database is designed to keep data accurate, secure, and well-organized to support smooth system operation and reliable reporting.

- It stores appointment records including booking details, schedules, assigned therapists, selected services, appointment status, and history of completed or cancelled bookings to ensure proper tracking of all client reservations.
- It keeps inventory records of spa products and supplies such as oils, lotions, towels, skincare items, and other materials, including stock quantities, usage records, restocking updates, and low-stock monitoring to help avoid shortages and overstocking.
- It records all payment and transaction data, including Point-of-Sale (POS) entries, total amounts, payment methods, discounts applied, receipts, and transaction history for accurate sales tracking and financial monitoring.
- It maintains customer information such as personal details, contact information, service history, preferred services, appointment records, and other relevant client data to support better customer service and management.
- It stores feedback records submitted by clients, including ratings, comments, suggestions, and service reviews that help management evaluate service quality and identify areas for improvement.
- It keeps records of vouchers and promotional usage, including issued vouchers, redeemed discounts, validity status, and usage history to ensure proper monitoring of promotions and prevent misuse.
- It stores employee-related records such as commission data, service assignments, and performance-related transaction history to support accurate commission computation and staff monitoring.
- It keeps system notifications and confirmation records such as booking confirmations, payment alerts, and system messages to ensure proper communication between users and the system.
- It stores generated reports and analytics data including sales summaries, service reports, inventory reports, and performance insights to support management in decision-making and business evaluation.

The database ensures that all information is **organized, secure, and easily accessible** for system operations and reporting.

3.4 Tools and Technologies Used

The development of the Recovery Iloilo Spa Integrated System will utilize the following tools and technologies to ensure functionality, reliability, and efficient system performance:

- **Frontend: HTML, CSS, JavaScript** – These technologies will be used to design and develop the user interface of the system. HTML will be used to structure the content of the web pages, CSS will be used for layout design and visual styling, and JavaScript will be used to add interactivity and improve user experience. These tools will help create a responsive and user-friendly interface for both spa staff and clients.
- **Backend: PHP** – PHP will be used as the server-side programming language responsible for handling system logic, processing user requests, managing transactions, and connecting the frontend to the database. It will support core functions such as appointment processing, inventory updates, payment handling, voucher management, and report generation.
- **Database: MariaDB** – MariaDB will be used to store, organize, and manage all system data. It will handle records such as appointments, customer information, inventory data, transactions, employee commissions, vouchers, feedback, and reports. It ensures data integrity, security, and fast retrieval of information when needed.
- **Development Tool: Visual Studio Code** – Visual Studio Code will be used as the primary code editor for developing the system. It provides features such as code highlighting, debugging support, extensions, and version control integration, which help improve development efficiency and organization of the project.

3.5 Data Gathering Techniques

We the researchers will use the following data gathering techniques to identify system requirements, understand current operational processes, and ensure that the proposed system meets the needs of Recovery Iloilo Spa:

- **Surveys and Questionnaires** – Structured questionnaires will be distributed to the spa owner, management, and staff to collect information regarding current

problems, system needs, and suggested improvements. This helps gather measurable feedback about appointment handling, inventory issues, payment processing, and customer service challenges.

- **Observation of Current Operations** – We researchers had observed the actual workflow of the spa, including how appointments are scheduled, how inventory is monitored, and how transactions are processed. This helps identify real-time issues such as delays, errors, and inefficiencies in manual processes.
- **Review of Existing Systems** – The researchers will study similar spa management systems or web-based booking systems to understand common features, system flow, and best practices that can be applied to the proposed system.
- **Document and Record Review** – Existing records such as appointment logs, sales records, inventory lists, and customer files will be reviewed to understand current data handling practices and identify gaps in the manual system that need improvement.

3.6 Testing Procedures

To ensure that the Recovery Iloilo Spa Integrated System works properly and meets the needs of the users, several testing methods will be applied during and after development. These tests will help identify errors, improve system performance, and make sure all features function as intended.

- **Unit Testing** – Each part or module of the system will be tested individually. This includes checking features such as appointment booking, inventory updates, payment processing, customer records, and reports to make sure each one works correctly on its own before combining them with other modules.
- **System Testing** – After all modules are integrated, the whole system will be tested as a complete platform. This process checks if all features work together properly, including data flow between the user interface, application processes, and database.
- **User Acceptance Testing (UAT)** – The system will be tested by the intended users such as the spa owner, staff, and management. This helps determine if the system meets their needs, is easy to use, and supports their daily operations effectively.

- **Performance Testing** – This will check how the system performs when handling multiple users, large amounts of data, and continuous transactions. It ensures that the system runs smoothly without delays or errors during normal usage.
- **Security Testing (Optional)** – This testing will check if the system is protected from unauthorized access and if user data such as customer records, transactions, and payments are stored securely.

3.7 Evaluation Criteria

The developed system will be evaluated based on the following criteria to determine its quality, performance, and usefulness in improving spa operations:

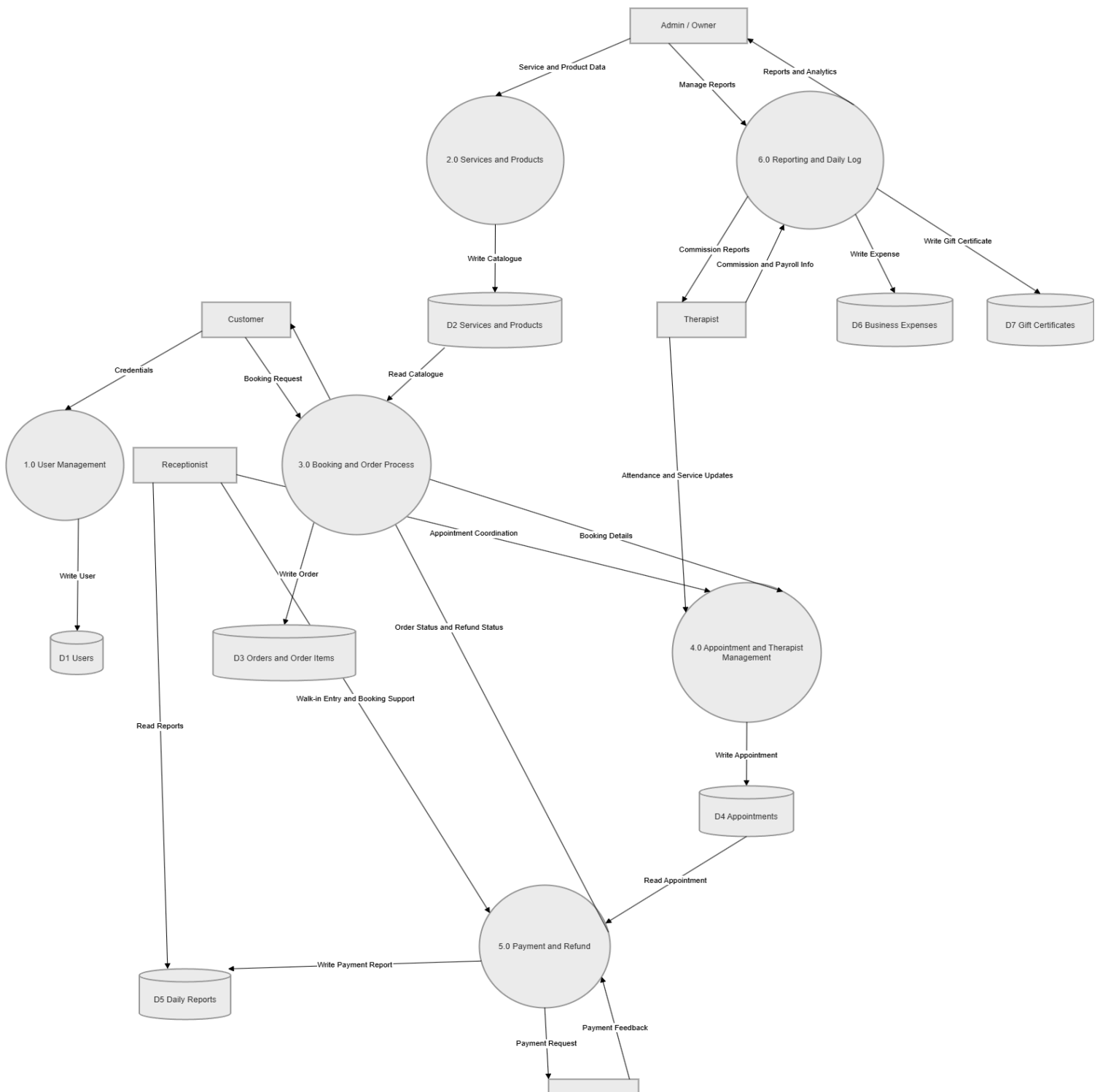
- **Functionality** – This criterion evaluates whether all features and functions of the system operate as intended. It checks if users can successfully perform tasks such as appointment booking, inventory management, payment processing, voucher handling, therapist assignment, and report generation. The system should complete these activities properly and produce the expected results without major issues.
- **Accuracy** – This criterion measures how correctly the system processes, stores, and displays information. It evaluates the correctness of appointment schedules, inventory records, transaction details, commission computations, customer information, and generated reports. Accurate information helps reduce errors and supports better daily operations.
- **Usability** – This criterion evaluates how easy the system is to understand and use for customers, therapists, receptionists, and administrators. It focuses on the clarity of menus, forms, buttons, and overall navigation throughout the system. A user-friendly system helps users complete tasks efficiently with minimal confusion.
- **Efficiency** – This criterion measures how quickly and smoothly the system performs different tasks. It evaluates the system's ability to process bookings, payments, record updates, report generation, and data retrieval within a reasonable amount of time. An efficient system helps improve productivity and reduces delays during daily operations.
- **Security** – This criterion evaluates how well the system protects sensitive information from unauthorized access and misuse. It includes account authentication, password protection, role-based access, and secure storage of

customer and business records. Proper security measures help maintain the confidentiality and safety of important data.

- **Feedback Integration** – This criterion evaluates how effectively the system collects, stores, and manages customer feedback and ratings. It checks whether customer comments and evaluations are properly recorded and made available for review by management. This information can help identify areas for improvement and support better service quality.
- **System Reliability** – This criterion measures the ability of the system to perform consistently during regular use. It evaluates whether the system can operate without frequent crashes, errors, or interruptions while handling daily transactions and activities. A reliable system helps ensure that users can complete their tasks without unnecessary disruptions.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 1)



1.0 User Management

- Customers and staff members provide their login or registration credentials to access the spa management system according to their assigned roles.
- The system validates the submitted information using D1: Users to ensure secure authentication, role-based access, and proper account management.
- Administrators can manage user accounts, staff roles, and therapist information while maintaining secure access to system records and transactions.

2.0 Services and Products Management

- Administrators manage spa services, wellness products, pricing, and category information within the system.
- The system stores and updates all catalogue data in D2: Services and Products, allowing customers to browse available services and products efficiently.
- The process ensures accurate inventory visibility, updated pricing, and proper service availability for bookings and purchases.

3.0 Booking and Order Process

- Customers submit booking requests for spa appointments and place orders for wellness products and services.
- The system retrieves catalogue information from D2: Services and Products and stores transaction records in D3: Orders and Order Items.
- Booking details are forwarded to the Appointment and Therapist Management process to organize schedules, assigned therapists, and service preparation while reducing scheduling conflicts and processing errors.

4.0 Appointment and Therapist Management

- The system manages customer appointments, therapist assignments, attendance records, and appointment coordination.
- Therapists provide attendance and service updates while receptionists assist with appointment scheduling and walk-in customer coordination.
- Appointment records are stored in D4: Appointments, allowing the system to monitor booking schedules, therapist availability, and service progress accurately.

5.0 Payment and Refund

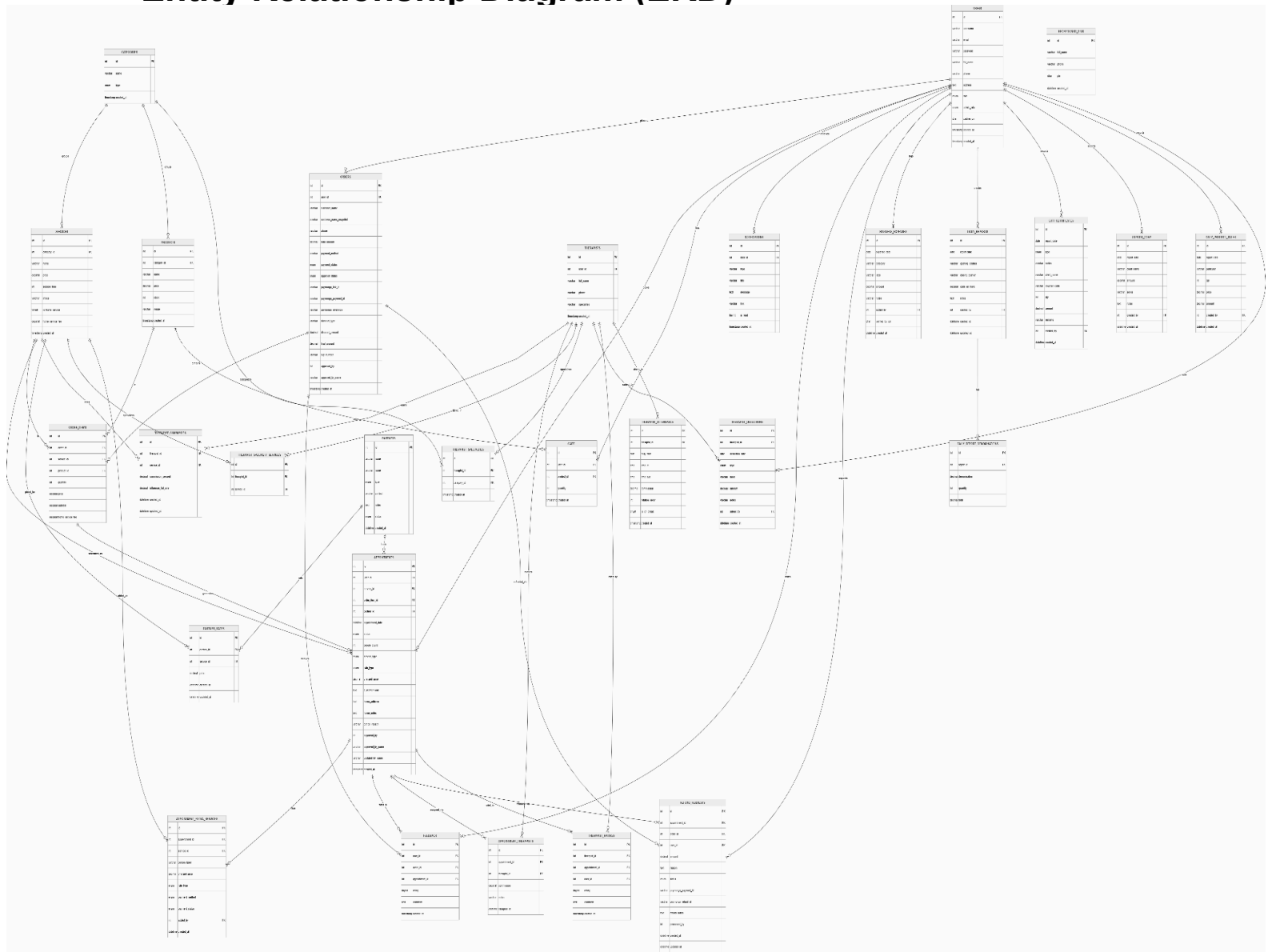
- Customers complete payments through onsite or online payment methods integrated with PayMongo.

- The system processes payment requests, receives payment feedback, and stores financial transaction summaries in D5: Daily Reports.
- Refund requests, payment verification, and order status updates are also handled within this process to ensure accurate financial recording and customer transaction tracking.

6.0 Reporting and Daily Log

- Administrators manage daily operational reports, business expenses, gift certificates, and financial summaries.
- The system retrieves payment and transaction data from D5: Daily Reports while storing expense records in D6: Business Expenses and promotional voucher records in D7: Gift Certificates.
- Analytics reports, commission summaries, payroll information, and business performance insights are generated to support management decision-making and operational monitoring.

Entity Relationship Diagram (ERD)



Entity Attributes

- users (id, username, email, password, full_name, phone, address, role, admin_role, cashier_pin, deleted_at, created_at)
- categories (id, name, type, created_at)
- services (id, category_id, name, price, session_time, image, is_home_service, home_service_fee, created_at)
- products (id, category_id, name, price, stock, image, created_at)
- cart (id, user_id, product_id, quantity, created_at)
- orders (id, user_id, customer_name, customer_name_snapshot, phone, total_amount, payment_method, payment_status, approval_status, paymongo_link_id, paymongo_payment_id, paymongo_reference, discount_type, discount_amount, final_amount, slip_number, approved_by, approved_by_name, created_at)
- order_items (id, order_id, service_id, product_id, quantity, price, subtotal, home_service_fee)
- appointments (id, user_id, service_id, order_item_id, partner_id, appointment_date, status, people_count, service_type, rate_type, charged_price, customer_note, home_address, home_notes, cancel_reason, approved_by, approved_by_name, updated_by_name, created_at)
- appointment_extra_services (id, appointment_id, service_id, person_label, charged_price, rate_type, payment_method, payment_status, added_by, created_at)
- appointment_therapists (id, appointment_id, therapist_id, commission, notes, assigned_at)
- feedback (id, user_id, order_id, appointment_id, rating, comment, created_at)
- notifications (id, user_id, type, title, message, link, is_read, created_at)
- refund_requests (id, appointment_id, order_id, user_id, amount, reason, status, paymongo_payment_id, paymongo_refund_id, refund_notes, processed_by, created_at, updated_at)
- therapists (id, user_id, full_name, phone, specialties, created_at)
- therapist_commission (id, therapist_id, service_id, commission_percent, influencer_flat_rate, created_at, updated_at)
- therapist_specialties (id, therapist_id, category_id, created_at)
- therapist_specialty_services (id, therapist_id, service_id)
- therapist_deductions (id, therapist_id, deduction_date, type, label, amount, notes, added_by, created_at)
- therapist_ratings (id, therapist_id, appointment_id, user_id, rating, comment, created_at)
- therapist_attendance (id, therapist_id, duty_date, time_in, time_out, commission, rotation_order, is_on_break, created_at)

- partners (id, name, type, contact, notes, status, created_at)
- partner_rates (id, partner_id, service_id, price, created_at, updated_at)
- receptionist_pins (id, full_name, phone, pin, created_at)
- business_expenses (id, expense_date, category, label, amount, notes, added_by, verified_by_pin, created_at)
- daily_reports (id, report_date, opening_cashier, closing_cashier, cash_on_hand, notes, created_by, created_at, updated_at)
- daily_report_denominations (id, report_id, denomination, quantity, total)
- gift_certificates (id, report_date, type, series, client_name, voucher_code, qty, amount, remarks, created_by, created_at)
- unpaids_corp (id, report_date, client_name, amount, series, notes, created_by, created_at)
- daily_product_sales (id, report_date, particular, qty, price, amount, created_by, created_at)

Relationships

- A **User** creates an account in the system to access different features based on their assigned role such as admin, receptionist, cashier, therapist, or client. Users can manage appointments, transactions, reports, and other system activities depending on their permissions.
- A **User** can schedule multiple **Appointments** for spa services. Appointment records store booking details such as selected services, appointment dates, number of persons, service type, home service information, customer notes, payment details, and appointment status.
- A **User** can place multiple **Orders** for spa services and wellness products. Each order contains payment details, discounts, approval information, transaction references, and final payment amounts for proper transaction monitoring.
- A **User** can add and manage selected items inside the **Cart** before proceeding to checkout and payment confirmation. The cart temporarily stores selected products and quantities for easier transaction handling.
- A **User** receives **Notifications** related to appointment confirmations, payment updates, refund processing, booking reminders, approval updates, and other important system activities.
- A **User** can submit **Feedback** regarding completed appointments and orders by providing ratings, comments, and suggestions about the services received and their overall experience.
- **Categories** organize both **Services** and **Products** to make records easier to manage, classify, and display within the system. One category may contain several services or products depending on their type.

- **Services** are connected to **Appointments** because every appointment requires a selected spa service. Services are also connected to **Order_Items** to record purchased or booked services included in transactions.
- **Services** are linked to **Therapist_Commission** records to calculate therapist earnings and commission percentages based on completed spa sessions and assigned services.
- **Services** are also connected to **Partner_Rates** to support special pricing arrangements for partnered companies, organizations, or promotional collaborations.
- **Products** are linked to the **Cart** and **Order_Items** to track selected products, purchased items, stock movement, and product sales transactions within the system.
- **Orders** contain multiple **Order_Items** representing the specific products and services included in a customer transaction. Each order item records quantities, pricing, service fees, and subtotal amounts for accurate billing.
- **Order_Items** may generate corresponding **Appointments** whenever spa services are included in the transaction. This connection helps link payment records with scheduled spa sessions.
- **Appointments** can include additional treatments or add-on services through **Appointment_Extra_Services**. These records store extra services selected by clients, pricing details, payment information, and service assignments.
- **Appointments** are assigned to therapists through **Appointment_Therapists** to manage therapist scheduling, assigned sessions, commission amounts, and work monitoring.
- **Appointments** are connected to **Feedback** and **Therapist_Ratings** to evaluate customer satisfaction, therapist performance, and overall service quality after completed sessions.
- **Appointments** may also generate **Refund_Requests** whenever customers request cancellations, payment refunds, or transaction adjustments. Refund records store refund amounts, reasons, payment references, processing details, and refund status.
- **Therapists** handle and perform scheduled spa services connected to appointments. Therapist records store personal details, specialties, and assigned services within the spa operations.
- **Therapists** receive earnings through **Therapist_Commission** records, which calculate commission percentages or fixed rates based on completed appointments and services rendered.
- **Therapists** may also receive deductions recorded under **Therapist_Deductions** for penalties, cash shortages, or other adjustments related to their commission or operational responsibilities.

- **Therapists** maintain attendance records through **Therapist_Attendance** to monitor duty schedules, working hours, break status, rotation order, and daily attendance performance.
- **Therapists** specialize in specific service categories through **Therapist_Specialties**, allowing management to identify therapist skills and assigned spa service areas more effectively.
- **Therapists** are also connected to specific spa services through **Therapist_Specialty_Services** to identify which treatments and procedures they are qualified to perform.
- **Partners** are connected to **Appointments** and **Partner_Rates** to support corporate clients, partnered businesses, promotional agreements, and discounted service pricing arrangements.
- **Daily_Reports** store overall daily operational summaries including cash records, opening and closing cashiers, daily notes, and generated reports related to spa activities and transactions.
- **Daily_Reports** contain multiple **Daily_Report_Denominations** used for recording cash breakdowns and denomination counts during cashier balancing and financial monitoring.
- **Business_Expenses** store operational expenses such as supplies, utilities, maintenance costs, and other business-related expenditures recorded by authorized users for financial tracking purposes.
- **Gift_Certificates** store issued vouchers, promotional certificates, client information, quantities, and redemption details for monitoring gift certificate transactions and promotional activities.
- **Unpays_Corp** stores records of unpaid corporate transactions, balances, billing details, and payment monitoring related to partnered companies or corporate clients.
- **Daily_Product_Sales** records daily sold products, quantities, prices, and total sales amounts to support inventory monitoring, sales tracking, and report generation within the system.

Use Case Diagram

Actors:

- Customer (User) - The Customer is the main user of the system who can access various services provided by Recovery Iloilo Spa. Customers can browse available spa services and products, schedule appointments, place orders, make payments, and manage their personal information through the system. They can

also monitor booking and payment status, receive notifications, submit feedback, rate therapists based on their experience, and request refunds or appointment cancellations when needed.

- Administrator (Staff and Management) - The Administrator is responsible for managing and overseeing the daily operations of Recovery Iloilo Spa through the system. This role handles customer accounts, appointments, therapist assignments, products, inventory, payments, reports, business expenses, gift certificates, partner records, and other important operational activities. Administrators also monitor transactions, generate reports, manage system information, and ensure that all records are accurate and updated.
- Therapist - The Therapist is responsible for providing spa treatments and wellness services to customers based on scheduled appointments. Through the system, therapists can view assigned bookings, check appointment details, monitor attendance records, review commissions and deductions, and receive notifications related to their work schedules. They can also view customer ratings and feedback to help improve service quality and customer satisfaction.
- Partner Establishment - The Partner Establishment refers to companies, organizations, or business partners that have an agreement with Recovery Iloilo Spa for special services and promotional arrangements. These partners may receive customized service rates, corporate booking privileges, and other benefits based on the partnership agreement. They can also monitor related appointments, service transactions, and partnership records maintained within the system.

Use Cases:

- Register and Login Account – Allows customers to create an account and securely access the system using their registered credentials. During registration, customers provide the required information needed to create their profile. After logging in, they can access booking, ordering, payment, and other available system features.
- Manage User Profile Information – Allows customers to view and update their personal information within the system. They can modify details such as their name, contact number, address, email, and account information when necessary. Keeping profile information updated helps maintain accurate customer records.
- Browse Spa Services and Products – Allows customers to view the spa services and products offered by Recovery Iloilo Spa. Customers can check service

descriptions, prices, and available options before making a decision. This helps them choose services and products that meet their needs.

- View Categories and Pricing – Allows customers to browse different categories of services and products available in the spa. The system displays corresponding prices to help customers compare available options. This makes it easier for them to plan their bookings and purchases.
- Add Products to Cart – Allows customers to select products and place them in their shopping cart. The cart temporarily stores chosen items before checkout and payment. This gives customers time to review their selected products before completing the transaction.
- Manage Cart and Checkout – Allows customers to review all items added to their cart before confirming their purchase. They can update quantities, remove unwanted items, and verify order details. Once satisfied, customers can proceed to checkout and complete the transaction.
- Place Orders for Products and Services – Allows customers to purchase spa products and confirm service transactions through the system. The system records order information and stores transaction details for future reference. This helps ensure that purchases are properly documented and monitored.
- Book Spa Appointments – Allows customers to schedule appointments for their preferred spa services. They can choose available dates and services based on their needs. The system records booking information and helps organize appointment schedules.
- Select Home Service or In-Store Service – Allows customers to choose where they want to receive the service. They may select an appointment at the spa branch or request a home service if available. This provides convenience and flexibility for different customer preferences.
- Add Extra Appointment Services – Allows customers to request additional services that can be added to an existing appointment. These extra services are recorded together with the main booking details. This allows customers to enjoy multiple treatments during a single appointment.
- Apply Gift Certificates or Discounts – Allows customers to use gift certificates, vouchers, or promotional discounts during transactions. The system automatically applies eligible discounts to the total amount due. This helps customers save money while availing services or products.
- Track Order and Appointment Status – Allows customers to monitor the current status of their orders and appointments. They can check whether transactions are pending, approved, completed, or cancelled. This keeps customers informed about their activities within the system.
- View Payment and Refund Status – Allows customers to check the status of payments made through the system. They can also monitor refund requests and

view updates regarding refund processing. This provides transparency for financial transactions.

- Receive System Notifications – Allows customers to receive important updates and announcements from the system. Notifications may include appointment confirmations, payment updates, reminders, and promotional information. This helps customers stay informed about their transactions and bookings.
- View Transaction History – Allows customers to access records of their previous appointments, purchases, and payments. The system stores completed transactions for future viewing and reference. This helps customers review their past activities whenever needed.
- Submit Service Feedback and Ratings – Allows customers to provide comments, suggestions, and ratings regarding the services they received. Their feedback helps the spa understand customer experiences and identify areas for improvement. This also helps management evaluate overall service quality.
- Rate Assigned Therapists – Allows customers to evaluate therapists based on the service they provided during appointments. Customers can submit ratings and comments regarding their experience. These evaluations help monitor therapist performance and customer satisfaction.
- Request Refunds or Appointment Cancellation – Allows customers to submit requests for appointment cancellation or transaction refunds when necessary. The system records the request details, including the reason provided by the customer. Management can then review and process the request accordingly.

For Administrator (Staff and Management):

- Login Account – Allows administrators to securely access the system using their assigned username and password. The login process helps protect sensitive information from unauthorized users. After successful login, administrators can access all management and operational functions available in the system.
- Manage User Accounts and Roles – Allows administrators to create, update, activate, and deactivate customer and staff accounts. They can also assign specific roles based on user responsibilities within the system. This helps ensure that users can only access features related to their assigned role.
- Manage Therapist Accounts and Assignments – Allows administrators to maintain therapist information, contact details, specialties, and work records. They can assign therapists to appointments based on availability and qualifications. This helps organize workloads and ensure proper service delivery.

- Manage Partner Establishments – Allows administrators to add, update, and maintain records of partner companies and organizations. Information such as contact details, partnership status, and service agreements can be stored in the system. This helps manage business partnerships more effectively.
- Manage Categories, Services, and Product Inventory – Allows administrators to organize service categories and maintain information about available products and treatments. They can add new items, update details, and monitor inventory levels. This helps keep service and product information accurate and updated.
- Update Service Pricing and Partner Rates – Allows administrators to modify service prices and special rates offered to partner establishments. Pricing information can be adjusted whenever necessary to reflect business requirements. This helps ensure accurate charges for customers and partners.
- Process Customer Bookings and Appointments – Allows administrators to review and manage appointment requests submitted by customers. They can confirm, update, reschedule, or cancel bookings when necessary. This helps maintain an organized appointment schedule and improve service coordination.
- Assign Therapists to Appointments – Allows administrators to assign available therapists to scheduled appointments. The system helps match therapists based on service requirements and availability. This ensures that appointments are properly handled and completed on time.
- Manage Orders and Payment Status – Allows administrators to review customer orders and monitor payment records. They can check whether transactions are pending, paid, approved, or completed. This helps maintain accurate transaction monitoring and record keeping.
- Approve or Reject Transactions – Allows administrators to review transactions that require verification before processing. They can approve valid transactions or reject those with incomplete or incorrect information. This helps maintain accuracy and prevent transaction errors.
- Handle Refund Requests and Customer Concerns – Allows administrators to review refund requests submitted by customers. They can evaluate the reason for the request and take the necessary action based on company policies. This helps resolve customer concerns and maintain customer satisfaction.
- Monitor Therapist Attendance and Deductions – Allows administrators to track therapist attendance records, duty schedules, and working hours. They can also record deductions related to company policies or other approved reasons. This helps maintain accurate employee monitoring and documentation.
- Manage Therapist Commissions – Allows administrators to configure commission rates for therapists based on services provided. The system records earnings generated from completed appointments. This helps ensure that commission records are accurate and easy to monitor.

- **Manage Gift Certificates and Corporate Unpaid Records** – Allows administrators to record and monitor gift certificates issued by the spa. They can also track unpaid balances from corporate clients and partner organizations. This helps maintain organized records of promotional and corporate transactions.
- **Record Business Expenses and Product Sales** – Allows administrators to record operational expenses and daily product sales. These records include spending related to business operations and income generated from product purchases. This helps support financial monitoring and reporting.
- **Generate Daily Financial and Sales Reports** – Allows administrators to generate reports showing daily sales, payments, expenses, and other financial information. The reports provide a summary of business activities within a selected period. This helps management review business performance and monitor operations.
- **View Business Analytics Dashboard** – Allows administrators to access charts, summaries, and reports generated by the system. Information related to sales, appointments, services, and customer activity can be viewed in one location. This helps management understand business trends and make informed decisions.
- **View Therapist Ratings and Feedback** – Allows administrators to review ratings and comments submitted by customers after completed services. These records help identify therapist strengths and areas that may need improvement. This supports service quality monitoring and employee evaluation.
- **Manage Receptionist Security Pins** – Allows administrators to create, update, and manage security pins used by reception staff. These pins may be required for specific transactions and verification processes. This helps improve security and control access to sensitive activities.
- **Send Notifications and Updates to Customers** – Allows administrators to send important announcements and updates directly to customers. Notifications may include appointment reminders, payment confirmations, promotional offers, and service updates. This helps improve communication between the spa and its customers.

For Therapist:

- **Login Account** – Allows therapists to securely access the system using their assigned account credentials. The login process helps protect therapist information and prevents unauthorized access. After logging in, therapists can access schedules, appointments, attendance records, commissions, and other related functions.
- **View Assigned Appointments** – Allows therapists to view appointments assigned to them by the administrator. The information includes customer details, service

schedules, appointment dates, and service locations. This helps therapists prepare for their upcoming appointments and manage their daily workload.

- View Appointment Details – Allows therapists to review complete information about assigned appointments. They can view selected services, additional treatments, customer requests, notes, and home service details when applicable. This helps therapists understand customer needs before providing the service.
- Update Appointment Progress and Status – Allows therapists to update the current status of appointments within the system. They can indicate whether a service is ongoing, completed, or cancelled based on the actual situation. This helps keep records updated and allows management to monitor appointment progress.
- View Service Commissions and Deductions – Allows therapists to monitor commissions earned from completed services. They can also review deduction records that may affect their earnings. This helps therapists stay informed about their income and work performance.
- Record Attendance and Duty Schedule – Allows therapists to record attendance and monitor their working schedules. The system stores information about duty dates, working hours, and attendance records. This helps maintain accurate employee records and supports workforce monitoring.
- View Assigned Specialties and Services – Allows therapists to view the services and categories assigned to them based on their skills and qualifications. This helps ensure that therapists only perform treatments that match their expertise. It also helps administrators assign appointments more effectively.
- Receive Customer Ratings and Feedback – Allows therapists to view ratings and comments submitted by customers after completed services. Customer feedback provides information about service quality and customer satisfaction. This helps therapists identify areas where they can further improve their performance.
- Receive System Notifications – Allows therapists to receive important updates and announcements from the system. Notifications may include new appointments, schedule changes, attendance reminders, and other work-related information. This helps therapists stay informed about their responsibilities and daily activities.

For Partner Establishment:

- View Partner Service Appointments – Allows partner establishments to view appointments associated with their organization or partnership agreement. They can monitor booking schedules, appointment dates, and services requested by

partnered clients. This helps them stay updated on activities related to their partnership with the spa.

- **Manage Partner Service Rates** – Allows partner establishments to view and coordinate special service rates provided under the partnership agreement. They can review agreed pricing for services offered to their clients or members. This helps ensure that partnered customers receive the correct rates and benefits.
- **Coordinate Appointment Schedules with Administrators** – Allows partner establishments to communicate and coordinate appointment schedules with spa administrators. This includes arranging corporate bookings, group appointments, and other partnership-related services. This helps avoid scheduling issues and ensures that appointments are properly organized.
- **Monitor Service Transactions and Status Updates** – Allows partner establishments to track the progress of appointments and services connected to their partnership. They can view whether bookings are pending, confirmed, completed, cancelled, or rescheduled. This helps them stay informed about the status of partnered transactions.
- **View Partner Service Records** – Allows partner establishments to review records of services provided under their partnership agreement. These records may include appointment details, service history, and transaction information. This helps maintain organized documentation and supports monitoring of partnership activities.
- **Receive Notifications and Updates** – Allows partner establishments to receive important announcements and updates from Recovery Iloilo Spa. Notifications may include appointment confirmations, schedule changes, service updates, and partnership-related information. This helps maintain clear communication between the spa and its business partners.
- **View Partnership Information** – Allows partner establishments to access information related to their partnership with Recovery Iloilo Spa. This includes available services, agreed rates, partnership status, and other relevant details. This helps both parties follow the terms and conditions of their business arrangement.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in application development, database management, systems analysis and design, and emerging technologies such as machine learning. The project follows outcomes-based education principles and integrates ethical con

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